

UNIT- I

1

Web Essentials and Mark - up Language - HTML

Syllabus

*The Internet, basic internet protocols, the World Wide Web, HTTP Request message, HTTP response message, web clients, web servers. **HTML** : Introduction, history and versions.*

***HTML elements** : headings, paragraphs, line break, colors and fonts, links, frames, lists, tables, images and forms, Difference between HTML and HTML5. **CSS** : Introduction to Style Sheet, CSS features, CSS core syntax, Style sheets and HTML, Style rule cascading and inheritance, text properties. Bootstrap.*

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Part I : Web Essentials

1.1 Basic Concepts

1.1.1 What is Web Technology ?

- Web technology is a technology that acts as an interface between web client and web server. It includes markup languages, programming interfaces, online presentation tools, Java applets, video editing tools and so on.
- The commonly used client side scripting languages are HTML, JavaScript, CSS, XML and so on.
- The commonly used server side technologies are ASP, JSP, Java Servlets, PHP and so on.

1.1.2 Concept of Internet

Definition : Internet is global system in which millions of computers are connected together. It is basically a network of networks.

- Using internet many people can share resources and can communicate with each other. To have internet service one must go to the service providers. That means your computer must be connected to the Internet Service Providers (ISP) through phone-line modem or DSL.
- There are some privately owned internet service providers from which we can hire the internet services.

1.1.3 Concept of World Wide Web

Definition : World Wide Web (WWW) is collections of software and corresponding protocols used to access the resources over the network.

- The world wide web is a **information system** in which various documents containing information are interlinked together. User can access this information or write the information via computers. This information is typically stored on the web pages and through web browsers we can access these web pages.
- The web pages may contain the information in the form of text, audio, video, images and graphics. We can navigate between the web pages using hyperlinks.
- The concept of WWW was introduced by **Sir Tim Berners-Lee** the contractor at the **European Organization for Nuclear Research (CERN)**, Switzerland in 1980. He built a personal database of people and software models and used hypertext so that each new page of information was linked to an existing page.

1.1.4 Internet or WWW ?

- The term internet and WWW is often used interchangeably, but these are two different terms.
- The **internet** is collection of computers and other devices (such as printers, scanners etc.) connected together whereas World Wide Web (WWW) is collection of software and corresponding protocols used to access the resources over the network.
- The world wide web contains huge amount of documents, images and other resources which can be accessed using the hyperlinks.
- Thus people use internet through the Web.

1.2 Basic Internet Protocols

Various protocols used in internetworking are -

1) File Transfer Protocol (FTP)

- The file transfer protocol sets the rules for transferring files between computers.
- When user wants to download a file from the server FTP is used.
- FTP uses **two connections** between client and server. One connection is used for actual **data transfer** and other is used for **control information** (used for commands). This separation of data and commands makes the FTP more efficient.

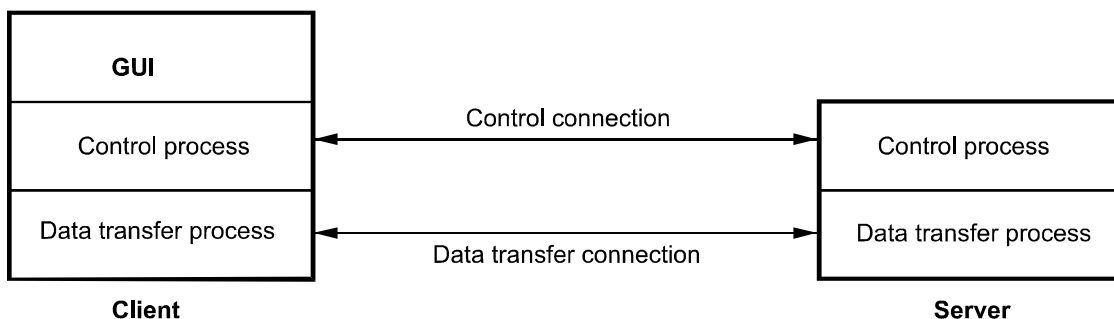


Fig. 1.2.1 Two connections used in FTP processing

- When client makes a request for particular file download then using the data transfer connection actual data gets transmitted from server to the client.
- At the same time server keeps track of how much data is sent so far and how much is remaining. This tracking can be done using the control transfer connection.
- Hence during the file downloading/uploading we can see a message about how many bytes are getting transferred and how much time is remaining.

2) HTTP

- The **Hyper Text Transfer Protocol (HTTP)** is a **request/response** protocol.
- It is a **communication protocol** used to transfer the information on local area network and World Wide Web (WWW).
- It is the network protocol used to deliver virtually all files and other data (collectively called resources) on the World Wide Web, whether they're HTML files, image files, query results, or anything else. Usually, HTTP takes place through TCP/IP sockets.
- It is also called as a **stateless protocol** because this protocol is not able to maintain the previous conversation/information.

3) SNMP

- **Simple Network Management Protocol (SNMP)** is a protocol which enables network administrators to manage network devices and to diagnose network problems.
- The network management system is based on two main elements : a **supervisor** and **agents**.
 - The supervisor is the terminal at which the network administrator requests for network management.
 - The agents are found at the level of each interface connecting the managed devices to the network. With the help of these agents information on the different objects (such as switch, hub, routers) can be collected.

4) SMTP

- **Simple Mail Transfer Protocol (SMTP)** is a simple protocol which is extensively used for transfer of **e-mails to remote servers**.
- It is an asynchronous protocol, because it allows delayed delivery of message.
- With the help of mail transfer agent and user agent the SMTP sends and receives the emails.

5) POP3

- **Post Office Protocol version 3 (POP3)** is used by local email clients (such as Microsoft Outlook Express).
- The POP3 protocol works only at the receivers end and has no work at the sender's end.

- The POP protocol has two parts, a client POP i.e. receiver's POP and a server POP i.e. receiver's email server. The client i.e. the receiver opens TCP connection with receiver's POP server. This client must be authenticated first by using the user name and password. Then the client can receive the emails from the mailbox.

6) TCP

- The transmission control protocol is used for,
 1. Safe delivery of data
 2. Error detection
 3. Assurance of the correct sequencing of data being received.
- This protocol is called **connection oriented protocol** because before sending the data this protocol requires that two computers have established connections.
- The TCP allows the transmission of arbitrary amount of data by breaking it into stream of separate **IP packets**.
- These IP packets are numbered so that it could be reassembled properly at arrivals. Along with the data an **acknowledgement** is also sent/received in order to know whether the **reliable connection** has occurred or not.

7) UDP

- The user datagram protocol is a **connectionless** protocol without any error detection facility.
- This protocol is used for simply transmission of data.
- The UDP is known as an **unreliable protocol** however this is much faster than TCP.

8) IP

- Internet Protocol (IP) is a network layer protocol which consists of addressing information.
- Using this information the communication between uniquely addressed computers is possible.

1.3 The Web Site Design Issues

SPPU : Dec.-18, May-19, Marks 5

- Following design goals are for the web design -

1. Simplicity :

- It is a general tendency of web designers to provide lot of animations, huge amount of information, extreme visuals and so on. This makes the web design enormous and it should be avoided. The web application must be moderate and simple.

2. Identity :

- Web design must be based on the nature of the web application. It is driven by the objective of the web application, category of user using it. A web engineer must work to establish an identity for the web application through the design.

3. Consistency :

- The contents of the web application should be constructed consistently. For example : Text formatting, font style should be the same over all the text document of the web application. Similarly, the graphics design, color scheme and style must be identical over all the web pages of the web application. Architectural design should produce the templates using which the consistent hypermedia structure can be formed. Interface design should define the consistent modes of interaction, navigation and content display. Navigation mechanism must be used consistently across web application elements.

4. Robustness :

- The users always expect robust contents and functions of the web application. That means any required functionality should not be missing at all, If any function or content is missing or insufficient then that web application will fail.

5. Navigability :

- The navigation should be simple and consistent. The design of navigations should intuitive and predictable in nature. That means any novice user should be in a position to make use of navigation links without any help.

6. Visual appeal :

- The web applications are most visual and most dynamic and aesthetic in nature. There are various factors that contribute to visual appeal. These factors are -
- Look and feel of the content, interface layout, color co-ordination, the balance of text, graphics and other media, navigation mechanism and so on.

7. Compatibility :

- The web application can be used in variety of environment and configurations such as different browsers, internet connection types, operating systems and various browsers.

1.3.1 Browser Compatibility Issues

- One of the greatest challenge faced by the authors of web document is designing the web pages which can be displayed properly by multiple browsers.
- Basically, every browser contains a program called **rendering engine**. This program interprets the markup tags of your web document and display the result in the browser. This logic of interpreting the tags varies from browser to browser. As a result, conflicting results are displayed on different browsers.
- The responsibility of a web designer is to test the web page of as many browsers as possible to ensure that the work appears correctly as per the design.
- To settle the browser compatibility issues following guideline must be adopted –
 - o *Follow W3C standards* : Make use of scripting languages such as HTML, CSS, JavaScript correctly.
 - o *Validate the code* : Test the syntax of the code. Make the web document free from any kind of coding errors.
 - o *Know your audience* : The web design must be accessible, readable and unambiguous for the target audience.
 - o *Test your work on multiple browsers and devices* : Repeatedly test and try to execute the web document on multiple browsers and devices for ensuring its correctness.

1.3.2 Bandwidth

- The term bandwidth means amount of data that can be carried from one point to another in a given period of time. The bandwidth is generally used to define user's internet connection speed. It is expressed in terms of bits per seconds (bps).
- The web pages get slowly displayed on the browser as they contain large and detailed graphics and flash animations. Normally user never prefer to view the web pages that are displayed slowly on the browser window.
- The web designer **must test** the execution of web pages at **variety of connection speed**.
- The iphones, or other cellular handsets that display the web pages have **lower bandwidth** than the home or business computers.

1.3.3 Cache

- The term cache means the **temporary storage area**.

- The computer on which all the web pages are stored is called **web server**. When we type URL address in the browser, it connects to appropriate web server and requests the file specified by the user. The server sends the file to the user and the user's browser can download it.
- The first time when user visit the web site, entire contents, including images, stylesheets and so on get downloaded on user's hard disk. The next time when user visits the web site, the user's browser downloads and parses the web page from the site. The browser checks if any specified image is already present on the computer's hard drive in the cache.
- Cache is the browser's temporary storage area for web pages and images.
- Thus the browser always try to load images from the cache instead of downloading it from the web server.
- The browser cache is useful for displaying the frequently visited web pages more efficiently.
- The cookie is a small text file that server embeds in clients web browser. Each time when user makes a request for the web page, it sends the cookies as well. The browser retrieves the values from cookies.

Review Question

1. List and discuss the different design issues in web development.

SPPU : Dec.-18, May-19, Marks 5

1.4 HTTP Request Message and Response Message

SPPU : May-18, March-19, Marks 5

- The basic feature of HTTP protocol is that it follows the **request response** model.
- The client makes a request for desired web page by giving the URL in the address bar. This **request** is submitted to the web server and then web server gives the **response** to the web browser by returning the required web page.

1. HTTP Request Message Structure

The basic structure of request message is given by following general form -

```
<Start line>

<Header fields>

<Blank Line>

<Message Body>
```

<start Line>

The start line consists of three parts which are separated by a single space. These parts are -

i) Request method ii) Request-URI iii) HTTP version

i) Request method : Various methods used for making the request are enlisted in the following table.

HTTP method	Description
GET	The GET method is used to retrieve information from a specified URI and is assumed to be a safe, repeatable operation by browsers.
POST	The POST method is used to request the server for desired web page and the request made is accepted as a new subordinate of the resource identified. The POST method is used for operations that have side effects and cannot be safely repeated. For example, transferring money from one bank account to another has side effects and should not be repeated without explicit approval by the user.
HEAD	The HEAD method is identical to GET. The only difference is that the server should not return a message-body in the response. The meta-information contained in the HTTP headers in response to a HEAD request should be similar to the information sent in response to a GET request.
OPTION	This method supports for the specified URL. It can be used to check the functionality of a web server by requesting '*' instead of a specific resource.
PUT	This method uploads a representation of the specified resource.
DELETE	This method is useful in deleting the specified resource.
TRACE	When request is made using TRACE method the server echoes back the received request so that a client can see what intermediate servers are adding or changing in the request.

(ii) Request-URI :

- The **Uniform Resource Identifier (URI)** is a string used to identify the names or resources on the Internet.
- The URI is a combination of URL and URN.
- The URL stands for **Uniform Resource Locator** and URN stands for **Uniform Resource Name**.
- The web address denotes the URL and specific name of the place or a person or item denotes the URN. For example

urn : ISBN 978-81-8431-123-2

specifies the address of some book.

- If the URI is written in the form of http : then it is both an URI and URL but there are some other URI which can also be used as URL. For example

URL	Intended server
ftp://ftp.mywebsite.com/index.txt	File can be located on FTP server
telnet://mywebsite.org	Telnet server
mailto:myself@mywebsite.org	Mail box
http://www.mywebsite.com	Web server

(iii) HTTP Version : The first HTTP version was HTTP/0.9 but the official version of HTTP was HTTP/1.1

<Header Field> and <Message Body>

- The host header field is associated with the http request.
- The header fields are in the form of field name and field value.
- Thus typical structure of http request is given by following example -

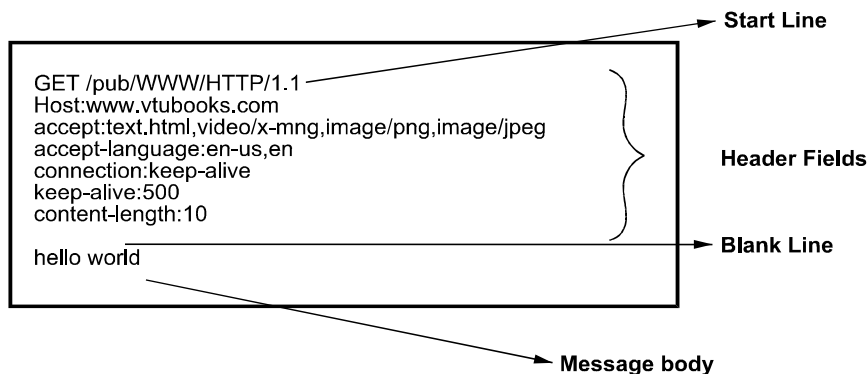
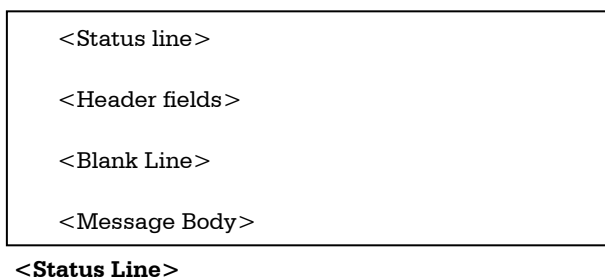


Fig. 1.4.1 HTTP request message structure

2. HTTP Response Message Structure

The basic structure of response message is given by following general form -

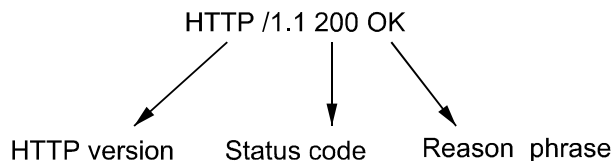


Status line is similar to the start line in the request message. It consists of three fields.

HTTP version	Status code	reason phrase
--------------	-------------	---------------

The HTTP version denotes the HTTP version such as HTTP/1.1. The **status code** is a numeric code indicating the type of response. The **reason phrase** is in the text string form and presents the information about the status code.

For example -



Following table explains some commonly used status codes.

Status code	Reason phrase	Description
200	OK	This is a standard response for successful request.
201	Created	It shows that the request is fulfilled and a new resource is being created.
202	Accepted	When the request is accepted for processing but is not processed yet is denoted by this status code.
301	Moved permanently	The URI for requested resource is moved at some another location.
401	Unauthorized	The requested resource is protected by some password and the user has not provided any password.
403	Forbidden	The requested resource is present on the server but the server is not able to respond it.
404	Not Found	The requested resource is not present currently but may be available in future.
500	Internal Server Error	It is a generic error message that appears due to software internal failure.

<Header Fields>

The **header field** in response message is similar to that of request message.

<Message Body>

The message body consists of response message.

For example

```
HTTP/1.1 200 OK
Date: Sat, 30 Mar 2019 07:59:01 GMT
Server: Apache/2.0.50 (Unix) mod_perl/1.99_10 Perl/v5.8.4
mod_ssl/2.0.50 OpenSSL/0.9.7d DAV/2 PHP/4.3.8
Last-Modified: Mon, 23 Feb 2009 08:32:41 GMT
Accept-Ranges: bytes
Content-Length: 2010
Content-Type: text/html
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN">
<html>...</html>
```

The response header fields are enlisted in following table.

Header field	Description
Date	It represents the date and time at which the response is generated.
Server	The name of the server which is responding
Last-Modified	The date and time at which the response is last modified.
Accept-ranges	It specifies the unit which is used by the client to accept the range request. For example if there is a large document and only a single web page is currently needed then this specifies the Accept-range.

Example 1.4.1 *List and discuss any three HTTP commands.*

SPPU : May-18, Marks 5

Solution : HTTP protocol defines a set of request methods. A client can use one of these request methods to send a request message to an HTTP server.

Method	Purpose
GET	Client can use the GET request to get a web resource from the server.
HEAD	Client can use HEAD request to get the header that a GET request would have obtained.
POST	It is used to post data to the web server.
PUT	It asks the user to store data.
DELETE	It asks the server to delete the data.

Example 1.4.2 *Describe the steps involved when a web browser requests for and obtains a web page from a web server.*

SPPU : March-19, Marks 5

Solution : The steps in web browser and web server communication are :

Step 1 : The user submits the request for web page via web browser.

Step 2 : The web server software routes the request to the server.

Step 3 : The web server decodes the request and then processes it.

Step 4 : The web server then encodes the response as HTML web page. Then the server submits back the response to web browser.

Step 5 : The web browser updates the display for displaying the requested web page.

1.5 Features of HTTP Protocol

1. It is a **communication** protocol used between web browser and web server.
2. This protocol is based on **request-response** messaging. That means client makes the request of desired web page and then the server responds it by sending the requested resource.
3. It is a **stateless protocol**. That means HTTP protocol can not remember the previous user's information nor it remembers the number of times the user has visited particular website.
4. The request-response message consists of plain text in fairly **readable form**.
5. The HTTP protocol has a **cache control**. This is an advanced feature of HTTP. Most of the web browsers automatically store (Cache) the recently visited web pages. This is very useful feature because if the user requests the same web page that has been visited already then it can be displayed from the cache memory instead of requesting the web server and bringing it from there.

1.6 Web Clients

- Web client is a kind of software that runs on the clients machine.
- This software sends the HTTP request to the server and then processes the HTTP response.
- Various forms of web client software are -
 - 1) Web browsers
 - 2) Browsers running on mobile phones.
 - 3) Software robots which crawl on the web for retrieving the information.
 - 4) User agents which assists the user in browsing the web.

- **Web browser** is a kind of software which is basically used to use resources on the web.
- Over the networks, two computers communicate with each other. In this communication, when request is made by one computer then that computer is called a **client** and when the request gets served by another computer then that computer is called server. Thus exchange of information takes place via Client-Server communication.
- When user wants some web document then he makes the request for it using the web browser.
- The browsers are the programs that are running on the clients' machines. The request then gets served by the server and the requested page is then returned to the client. It is getting displayed to the client on the web browser.
- The **commonly used web browsers** are (i) Internet explorer (ii) Mozilla Firefox (iii) Netscape Navigator (iv) Opera (v) Google Chrome (vi) Safari.
- Web browser supports variety of **protocols** but the most commonly used protocol on the web browser is Hyper Text Transfer Protocol (HTTP). This protocol is typically used when browser communicates with the server.



1.6.1 Functions Defined by Web Browser

Various functions of web browser are -

1. Reformats the URL and send a valid HTTP request.
2. When user gives the address of particular web site it is in the form of domain name. The web browser converts the DNS to corresponding IP address.

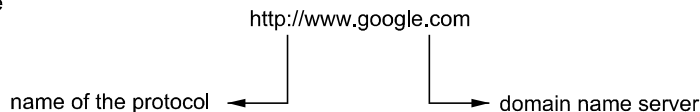
3. The web browser establishes a TCP connection with the Web browser while processing the user's request.
4. The web browsers send the HTTP request to the web server.
5. The web server processes the HTTP request sent by the web browser and returns the desired web page to the client machine. The web browser on the client's machine displays this web page in appropriate format.

1.6.2 URL

- The **Uniform Resource Locator (URL)** is unique address for the file that has to be accessed over the internet.
- When we want to access some website we enter it's URL in the address bar of the web browser.
- For example if we want to access **www.google.com** then we must specify its URL in the address bar as shown.



Here



- However any other file such as some text file or image file or some HTML file can also be specified. The URL contains name of the protocol such as **http://**
- The URL may contain the name of the protocol as such as ftp. For example **ftp://ftp.funet.fi/pub/standards/RFC/rfc2166.txt**
- The protocol identifier and the resource name are separated by a colon and two forward slashes. The syntax of writing URL is as given below,
protocol://username@hostname/path/filename
- Sometimes instead of domain name servers IP addresses can also be use. For example,
http://192.168.0.1
- But use of IP address as URL is not preferred because human can not remember numbers very easily but they can remember names easily.

Absolute and Relative URL

- The absolute URL is a URL which directly point to a file. It exactly specifies exact location of a file or directory on the internet. Each absolute URL is unique. For example -
http://www.vtubooks.com/home.aspx
- The relative URL points to the file or a directory in relation to the present directory. For example
- Consider the **absolute address** which refers an image mother.jpg
http://www.mywebsite.com/myphotos/mother.jpg
- For the above given absolute address the **relative address** will be -
../myphotos/mother.jpg
- That means from the current URL the directory **myphotos** will be searched for the image **mother.jpg**. The two periods.. instruct the server to move up one directory which is the root directory, then enter myphotos directory (**/myphotos**) and finally point at **mother.jpg**. Thus using relative URL writing of long path name can be avoided.

1.7 Web Servers

SPPU : April-18, Dec.-18, Marks 5

- Web server is a special type of server to which the web browser submits the request of web page which is desired by the client.
- There are some popularly used web servers such as Apache and IIS from Microsoft.

Functions of web server

Various functions of web server are -

1. The web servers **accepts the requests** from the web browsers.
2. The user **request is processed** by the web server.
3. The web servers respond to the users by **providing the services** which they demand for over the web browsers.
4. The web servers **serve the web based applications**.
5. The DNS translate the **domain names into the IP addresses**.
6. The servers **verify** given address exists, **find necessary files** ,run appropriate scripts exchange cookies if necessary and returns back to the browser.
7. Some servers actively participate in **session handling techniques**.

Working Principle of Web Server

- When user submits a request for a web page, he/she is actually demanding for a page present on the web server.
- When web browser submits the request for a web page, the web server responds this request by sending back the requested page to the web browser of the client's machine.

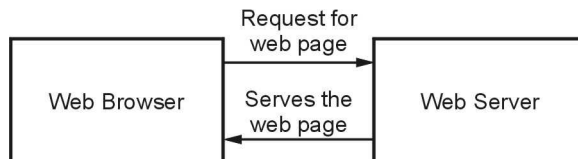


Fig. 1.7.1

Step 1 : Web client requests for the desired web page by providing the IP address of the website.

Step 2 : The web server locates the desired web page on the website and responds by sending back the requested page. If the page doesn't exist, it will send back the appropriate error page.

Step 3 : The web browser receives the page and renders it as required.

1.7.1 Apache

- It is an excellent server because of its two important features : **Reliability** and **Efficiency**.
- Secondly it is more popular because it is an **open source software**. That means it is freely available to anybody. Apache web server is best suitable for UNIX systems but it can also be used on Windows box.
- The apache web server can be configured as per the requirements using the file **httpd.conf**. This file is present in the Apache software package.

1.7.2 IIS

- The **Internet Information Services** or Internet Information Server is a kind of web server provided by Microsoft.
- This server is most popular on Windows platform.
- Following are some differences between Apache and IIS servers.

Sr. No.	Apache web server	IIS web server
1.	Apache web server is useful on both Unix based systems and on Windows platform.	IIS web server is used on Windows platform.
2.	It is an open source product.	It is a vendor specific product and can be used on windows products only.
3.	The Apache web server can be controlled by editing the configuration file httpd.conf	For IIS server, the behaviour is controlled by modifying the window based management programs called IIS snap in . We can access IIS snap-in through the Control-Panel->Administrative Tools.

Example 1.7.1 Define the terms :

i) Website ii) Web page iii) Web server iv) URL and v) Home page.

SPPU : April-18, Marks 5

Solution :

- i) **Website** : It is a set of related web pages located under a single domain name.
- ii) **Web page** : It is hypertext document connected to world wide web.
- iii) **Web server** : Web server is a server software that delivers content or services to end users over internet.
- iv) **URL** : The Uniform Resource Locator (URL) is unique address for the file that has to be accessed over the internet. When we want to access some website we enter it's URL in the address bar of the web browser.
- v) **Home page** : It is the introductory web page of web site that acts as a start-up page.

Example 1.7.2 What are the technologies used in traditional web programming ? Also identify location of each technology used in this model.

SPPU : April-18, Marks 5

Solution : The server side technologies and client side technologies are used in traditional web programming. The server side technologies work at the server level and client side technologies are used at the client level. Following table illustrates the comparative description of these technologies.

Sr. No.	Server side scripting	Client side scripting
1.	The server side scripting is used to create the web pages that provide some services.	The client side scripting is used to create the web pages as a request or response to server. These pages are displayed to the user on web browser.
2.	These scripts generally run on web servers.	These scripts generally run on web browser.
3.	A user's request is fulfilled by running a script directly on the web server to generate dynamic HTML pages. This HTML is then sent to the client browser.	The processing of these scripts takes place on the end users computer. The source code is transferred from the web server to the users computer over the internet and run directly in the browser.
4.	Uses : Processing of user request, accessing to databases.	Uses : Making interactive web pages, for interacting with temporary storages such as cookies or local storage, sending request to server and getting the response and displaying that response in web browser.
5.	Example : PHP, ASP.NET, nearly all the programming languages including C++, Java and C#	Example : HTML, CSS, JavaScript (primarily)

Review Question

1. Write the difference between client side scripting and server side scripting.

SPPU : Dec.-18, Marks 4

1.8 Client Sever Communication

- In this architecture, there are three components.
 - 1) Client PC or Web client
 - 2) An application server or web server
 - 3) A database server.

Working

- Step 1 :** The client PC or web client submits the request for desired web page to the web server.
- Step 2 :** The work of server is distributed among application server and database servers. Application server possesses the required communication functions.
- Step 3 :** The data required by this business logic is present on database server. The required data is returned to application servers
- Step 4 :** The web server or application server prepares the response page and sends it to the web client.

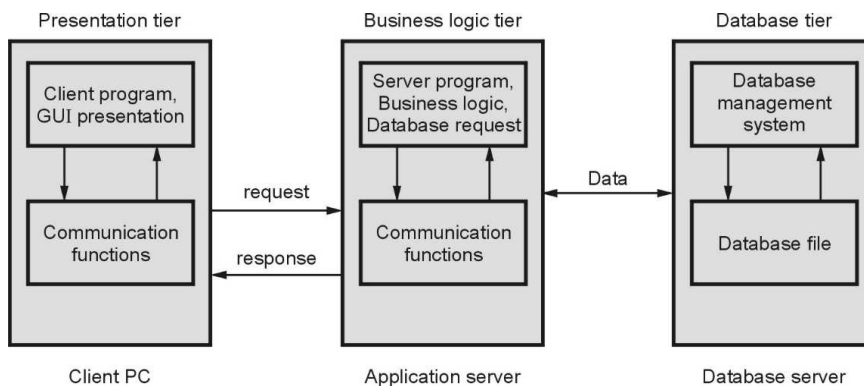


Fig. 1.8.1 Client server communication

Advantages :

1. High performance and persistent objects take part in communication.
2. This architecture is scalable. That means it can support increased number of users and resources.
3. High degree of flexibility in deployment platform and configurations.
4. Code/data reusability can be achieved.
5. Improved data integrity
6. Improved security as clients can not directly access the database.
7. The maintenance and modifications can be done effectively without disturbing other modules in the architecture.

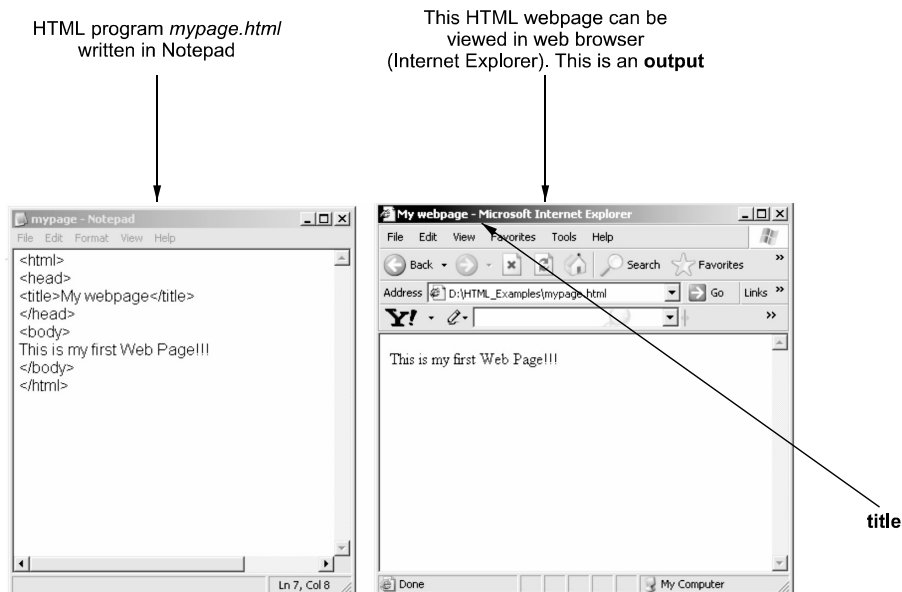
Disadvantages :

1. The complexity gets increased.
2. Cost for network maintenance and deployment is increased.

Part II : HTML

1.9 Introduction

- HTML stands for HyperText Markup Language. **Hypertext** is simply a piece of text that works as a link.
- **Markup Language** is language of writing layout information within documents.
- Basically an HTML document is a plain text file.
- It contains rich text. The rich text means text with tags.
- HTML is **not a case sensitive** language.
- Any HTML program can be written in simple Notepad or WordPad text editors. The extension to this program should be either **html** or **htm**.
- This program then can be opened in some web browser and the corresponding web page can be viewed. Let us create our first web page using HTML.



HTML Document

```
<html>
<head>
<title>My webpage</title>
</head>
<body>
This is my first Web Page!!!
</body>
</html>
```

Script Explanation

- 1) Note that the program consists of some strings enclosed within **angular brackets**. Such strings are called **tags**.
- 2) The HTML program should be written within **<html>** and **</html>**. The **<html>** indicates the start of html program and **</html>** denotes end of html program. Use of slash (/) in the angular bracket indicates end of that particular tag.
- 3) Any HTML program has two prominent parts : **head and body**. The head part acts as a header of a file and contains some information like setting the title of web page.

Comments in HTML

- The comment in HTML can be denoted as follows -

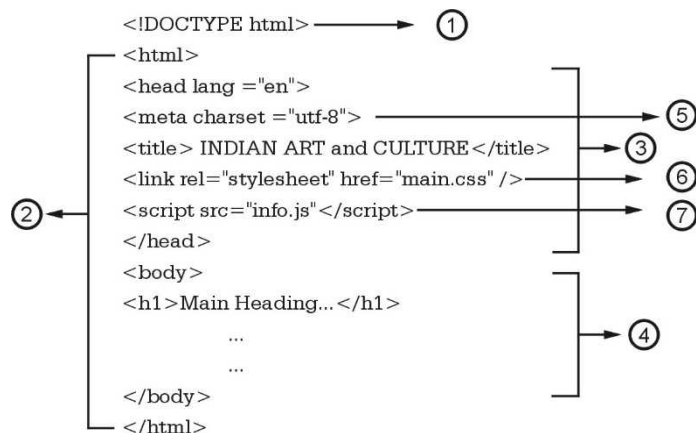
```
<!--It is a comment statement -->
```
- There should **not be a space** between angular bracket and exclamation mark. This comment is beginning with **<!--** and ending with **-->**. Also note one thing that there should not be any **--** inside the comment.

1.10 History and Versions

- HTML stands for **HyperText Marks up Language**. It is a markup language.
- In 1991 Tim Berners Lee invented HTML.
- In 1991 there is a recommendation for HTML 4.01. Thus From 1991 to 1999, HTML developed from version 1 to version 4.
- In year 2000, the World Wide Web Consortium (W3C) recommended XHTML 1.0. The XHTML syntax was strict, and the developers were forced to write valid and "well-formed" code.
- In 2004, WHATWG (Web Hypertext Application Technology Working Group) was formed. The WHATWG wanted to develop HTML, consistent with how the web was used, while being backward compatible with older versions of HTML.
- In 2004 - 2006, the WHATWG gained support by the major browser vendors.
- In 2008, the first **HTML5** public draft was released.

1.11 Structure of HTML

- The structure of HTML document can be illustrated by following sample HTML document.

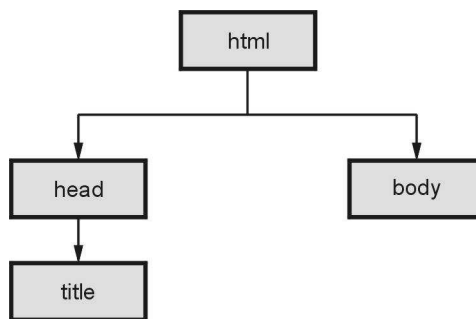


- Doctype** : The Doctype stands for Document type which tells the browser what type of document (HTML) it is about to process.
- HTML** : The `<html>` element is sometimes called the root element as it contains all the other HTML elements in the document.
- Head** : HTML pages are divided into two sections : The head and the body, which correspond to the `<head>` and `<body>` elements. The head contains descriptive elements about the document, such as its title, any style sheets or JavaScript files it uses.
- Body** : The body contains content (both HTML elements and regular text) that will be displayed by the browser.
- Character encoding** : The `<meta>` element specifies that the character encoding for the document is **UTF-8**. Character encoding refers to which character set standard is being used to encode the characters in the document. UTF-8 is a more complete variable-width encoding system that can encode all 110,000 characters in the unicode character set.
- External style sheet** : The external stylesheet such as css can be used with HTML document. We can specify the name of the CSS stylesheet.
- External scripts** : Sometimes external JavaScript files can be associated with HTML document.

1.12 HTML Elements

SPPU : April-18, May-18, Marks 5

- In all HTML/XHTML documents the root element is **<html>** which has two children : head and body.
- Any text contained within the head element does not appear directly in the **client area** of web browser. The head element is used for providing certain information to web browser. The **<title>** element is used to give title to the web page. Note that this **title** does not appear in client area. Rather it appears at the top of web browser.
- The **<body>** element contains the information which is to be displayed in the client area of web browser.
- The body element can contain several fundamental elements such as **<p>**, **<h1>**, **<div>**, **<a>**, **<hr>**, **
** and so on. The elements within the body part is for deciding the layout of your web page.

**Fig. 1.12.1 Element tree**

1.12.1 Heading

- There are header tags which help to display the text as some header.
- The header tag is denoted by h1, h2 and so on upto h6.
- Following HTML document along with its output is itself self explanatory.

HTML Document [headerDemo.html]

```
<!DOCTYPE html>
<html>
  <head>
  </head>
  <body>
    <h1> This is a first header </h1>
    <h2> This is a second header </h2>
    <h3> This is a third header </h3>
    <h4> This is a forth header </h4>
    <h5> This is a fifth header </h5>
    <h6> This is a sixth header </h6>
  </body>
</html>
```

Output**1.12.2 Paragraphs**

Following are the tags that are used for creating paragraphs.

Tag	Meaning
<p>	This tag should be put at the end of every paragraph.
<pre>	This tag is used to preserve the white spaces and lines in the text.
<div>	This tag is used to make division of sections in the XHTML document.

For example**HTML Document[TextFormat.html]**

```
<!DOCTYPE html>
```

```
<html >
```

```
  <head>
```

```
  </head>
```

```
  <body>
```

```
    <p>
```

Once upon a time, there was a king who kept a monkey as a pet.

The monkey served the king by all the possible ways.

```
  </p>
```

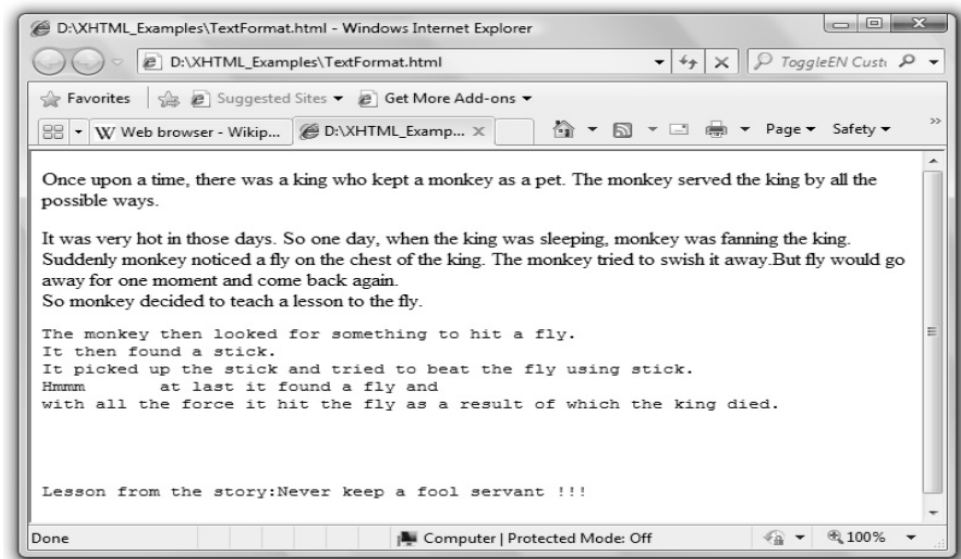
```
  <div>
```

It was very hot in those days. So one day, when the king was sleeping, monkey was fanning the king. Suddenly monkey noticed a fly on the chest of the king. The monkey tried to swish it away. But fly would go away for one moment and come back again.

```
</div>
<div> So monkey decided to teach a lesson to the fly.</div>
<pre>
The monkey then looked for something to hit a fly.
It then found a stick.
It picked up the stick and tried to beat the fly using stick.
Hmmm    at last it found a fly and
with all the force it hit the fly as a result of which the king died.

Lesson from the story:Never keep a fool servant !!!
</pre>
</body>
</html>
```

Output



1.12.3 Line Break

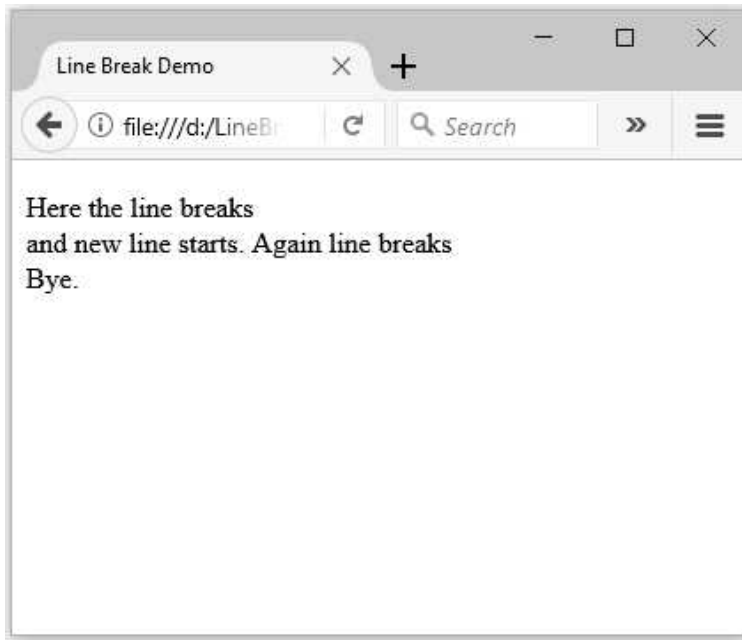
For the line break the **
** tag is used. This is a single line break tag. Generally this tag is kept at the end of every line.

For example

```
<!DOCTYPE html>
<html>
  <head>
    <title>Line Break Demo</title>
  </head>
  <body>
    <p>
```

```
Here the line breaks <br/> and new line starts.  
Again line breaks<br/> Bye.  
</p>  
</body>  
</html>
```

Output



1.12.4 Setting Font Style

The font style can be of various types such as boldface, italics, and strikethrough.

Tag	Purpose
 	Displays the text in bold
<i> </i>	Displays the text in italics
 	Displays the text in bold
<strike> </strike>	Displays the text with strike

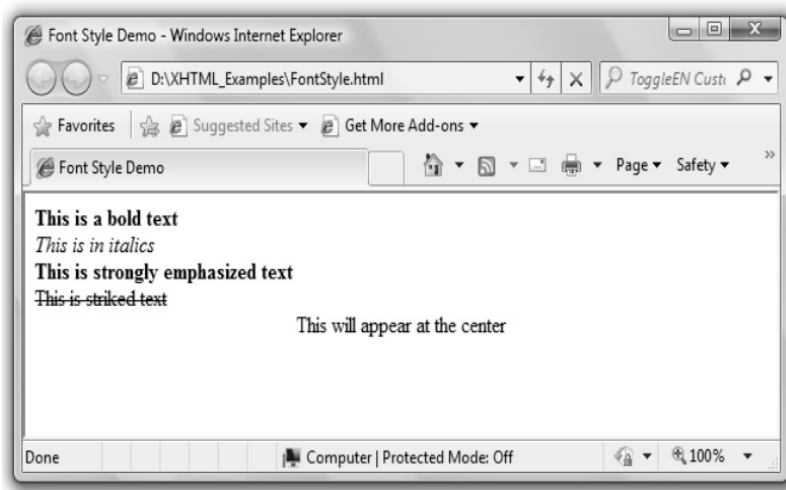
Following HTML document illustrates this -

HTML Document [FontStyle.html]

```
<!DOCTYPE html>  
<html>  
  <head>  
    <title>Font Style Demo</title>  
  </head>  
  <body>  
    <b> This is a bold text </b> <br/>
```

```
<i> This is in italics</i> <br/>
<strong> This is strongly emphasized text</strong> <br/>
<strike> This is striked text </strike>
<center> This will appear at the center</center> <br/>
</body>
</html>
```

Output



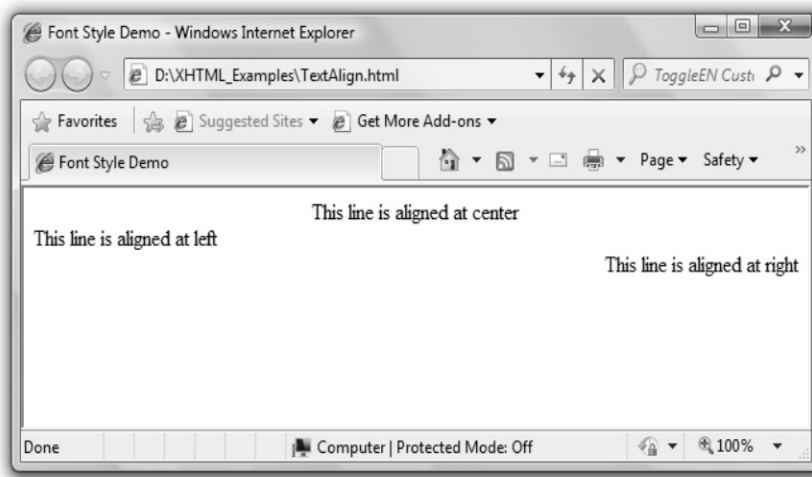
1.12.5 Text Alignment

We can align the text at left, right or at the center using a `<div>` tag. Here is a HTML program which shows the text aligned left, right and centre.

HTML Document[TextAlign.html]

```
<!DOCTYPE html >
<html>
  <head>
    <title>Font Style Demo</title>
  </head>
  <body>
    <div align="center">This line is aligned at center</div>
    <div align="left">This line is aligned at left</div>
    <div align="right">This line is aligned at right</div>
  </body>
</html>
```

Output



1.12.6 Setting the Font and Color

We can set the font, size and color of the text in the web page. The tag **<basefont>** is used for this purpose. The optional attributes used with **<basefont>** tag are as given below :

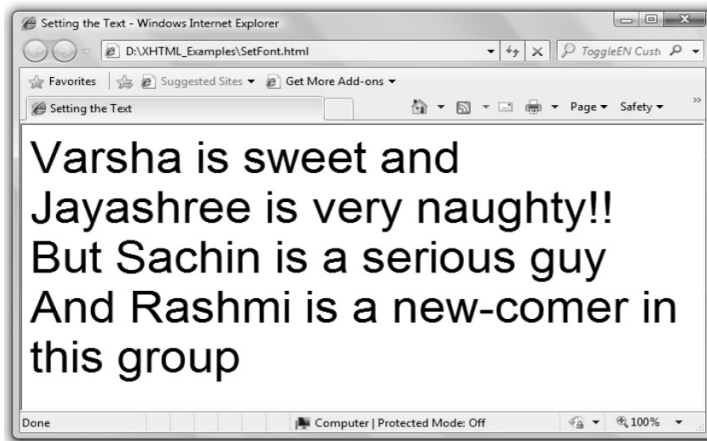
Attribute	Value	Purpose
color	Color value such as "red", "yellow" and so on.	It displays the text with specified color.
face	Font-family such as "arial", "verdana", "sans-serif" and so on.	It displays the text with specific font family.
size	Number	The text of specified size will be displayed using this attribute value.

For example -

HTML Document[SetFont.html]

```
<!DOCTYPE html>
<html >
  <head>
    <title>Setting the Text </title>
  </head>
  <body>
    <basefont face="arial" size="10">
    Varsha is sweet and <br/>
    Jayashree is very naughty!!<br/>
    But Sachin is a serious guy<br/>
    And Rashmi is a new-comer in this group
  </body>
</html>
```

Output



The **<basefont>** tag is supported in Internet Explorer 9 and not by FireFox, Chrome, Opera and so on.

Most of the web designers prefer to have their web pages white in color. HTML allows the web developer to use **color** attribute for coloring the text or for setting the background color.

First of all we will learn how to display a text colourful on the webpage.

HTML Document[ColorFont.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title>coloring the text</title>
  </head>
  <body>
    <basefont face="arial" size="5" color="blue">
    Great people are simple and <br>
    <font face="Times new roman" color="red" size="8">
    Simple people are great!!!<br>
    <font face="GoudyHandtooled BT" color="green" size="14">
    Great people rule the world.
  </body>
</html>
```

Output**Script Explanation :**

- 1) In the above HTML document, we have used **font face** tag to set the font.
- 2) Using the attributes **color** and **size** we can specify values color and size in double quotes.

We can set the **background color** of the web page using the attribute **bgcolor**. Following program sets the background color of the web page to red.

HTML Document[BgColor.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title>Coloring the background</title>
  </head>
  <body bgcolor="#FFFF00">
    <h3>This is document has colored background</h3>
  </body>
</html>
```

Script Explanation :

- 1) In above document we have specified background color using the **bgcolor** attribute.
- 2) The color can be specified by either using the color name in double quotes or using the hex.code as given in above document.
- 3) The first two digits of hex code represents the red value, then next two digits specify the green value and final two digits specify the blue value. Hence the corresponding output will be as follows -

Output

Note that the background color is specified by the hexadecimal value. Following table shows the hex and corresponding decimal values -

Decimal value	Hexadecimal value
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D
14	E
15	F

In HTML, the colors are specified by beginning with # and the FF denotes 255. In this color code first two digits specify the amount of red color then next two digits specify the green color and the last two digits specify blue color.

FF 00 00
 red colour

Hence is used to set the background color red. Each pair of digits specifies 0 to 255 color values. Thus hexadecimal color coding helps us to specify.

$$256 * 256 * 256 = 16777216 \text{ colors.}$$

The color can also be specified by its name as follows -

```
<html>
<body bgcolor=yellow>
</html>
```

Following is a list of colors that can be specified for setting the background colors.

Aqua	Navy
Black	Olive
Blue	Purple
Fuchsia	Red
Gray	Silver
Green	Teal
Lime	White
Maroon	Yellow

1.12.7 Special Character

- In HTML certain symbols are treated as special. For example - < i.e. less than symbol is considered as special symbol because it is used as the beginning of the tag.
- Hence we cannot directly use certain symbols such as <, >, ", © and so on.
- To print these symbols we use the units which are called as **reference**. This reference always begins with ampersand (&) sign. Such reference is called as **entity reference**.
- Following table enlists some characters and entity references -

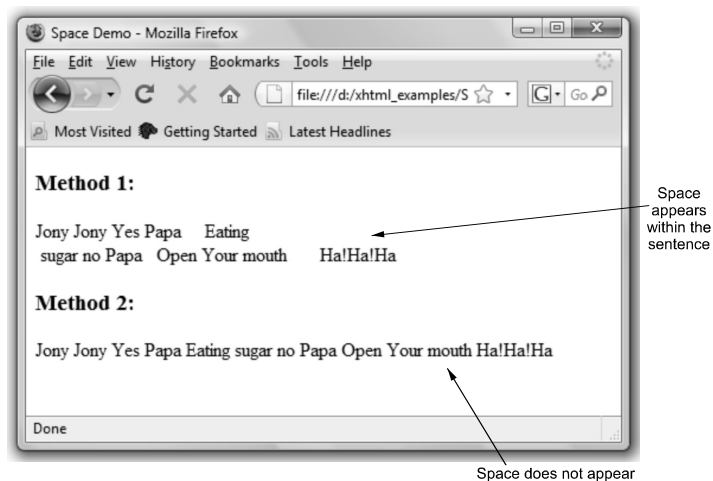
Character	Meaning	Entity Reference
<	less than	<
>	greater than	>
&	ampersand	&
"	double quote	"
'	apostrophes	'
©	copyright	©

- Another type of entity reference that is used frequently is ** **; that is **non breaking space character**.
- This entity reference is useful for defining the space between two string and informing browser for not performing the word wrapper between the strings.
- Following is a sample script that makes use of ** **.

HTML Document[SpaceDemo.html]

[illegible]

Output



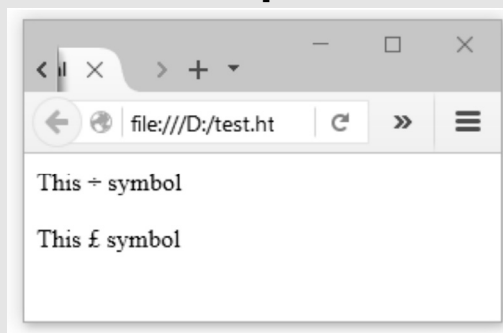
Example 1.12.1 How divide and pound symbol can be put on HTML document ?

SPPU : May-18, Marks 5

Solution : With the help of entity references denoted using & these two symbols can be put in HTML document. The code can be

```
<html>
<head>
</head>
<body>
<p>This &divide symbol</p>
<p>This &pound symbol</p>
</body>
</html>
```

Output



Review Question

1. List and describe any five HTML tags.

SPPU : April-18, Marks 5

1.13 Links

- There is a common practice to specify the web link in the web page. The link acts as a pointer to some web page or some resource.
- Use of hyperlink in the web page allows that page to link logically with other page.
- We can use hyperlinks by using a tag **<a>** and by specifying the URL for **href**.
- The value assigned to **href** specifies the target of the link.
- The **<a>** means beginning of the web link and **** means end of the web link.
- The most important attribute of the **<a>** element is the href attribute, which indicates the link's destination.
- By default, links will appear as follows in all browsers :
 - 1) An unvisited link is underlined and blue.
 - 2) A visited link is underlined and purple.
 - 3) An active link is underlined and red.
- Following are the steps to be followed to specify web link in the web page.

Step 1 : The beginning of web link can be specified by the tag **<a href = " "**.

Inside the double quotes mention the URL of desired link.

Step 2 : Write some text that should act as a hyperlink.

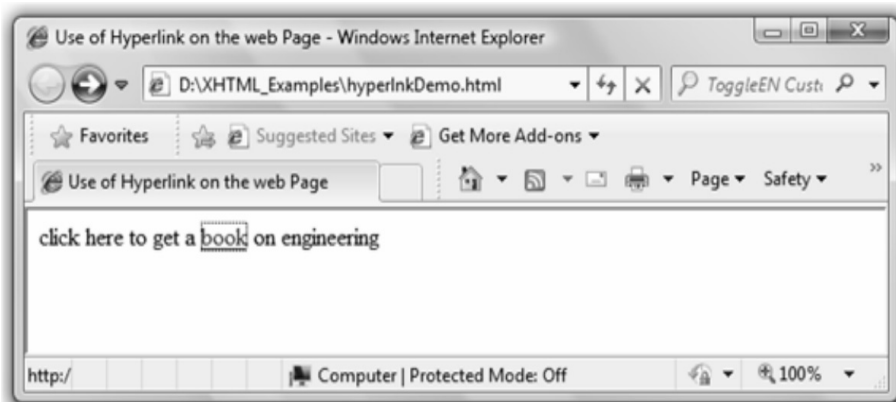
Step 3 : End the web link `</a>`

Here is a sample program which implements the above given idea -

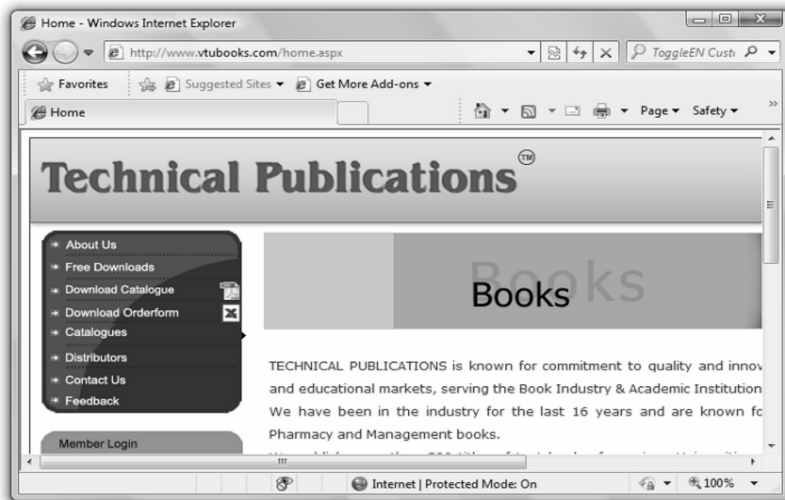
HTML Document [HyperLinkDemo.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title> Use of Hyperlink on the web Page </title>
  </head>
  <body>
    click here to get a
    <a href="http://www.vtubooks.com">book</a> on engineering
  </body>
</html>
```

Output



If you click on the hyperlink book then you will get the following output.



The target attribute

- If we want to get that link opened in another window we can mention **_target** property. Various targets can be -
- **_self** loads the page into the current window.
- **_blank** loads the page into a new separate browser window.

Example 1.13.1 *Develop a HTML page to use image as a link to another page.*

Solution :

Step 1 : Write the HTML document containing an hyperlink to an image. When we click on the image the html page referred by the **<a href>** tag must get opened.

XHTML Document[ImgLink.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title> Image Demo </title>
  </head>
  <body>
    <h1> Indian Heritage </h1>
    </p>
    <a href="TajMahal.html" target = "_blank">
    <img src= "TajMahal.jpg" alt= "Taj Mahal!!" />
    </a>
    <br/>History of Taj Mahal
    <br/>
  </body>
</html>
```

Step 2 : Write the HTML document(TajMahal.html) mentioned by the **<a href>** tag in Step 1 as follows.

HTML Document[TajMahal.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title> Information on TajMahal </title>
  </head>
  <body>
    <h1> Indian Heritage </h1>
    <p>
    Taj Mahal is one of the wonders of the World. It is located in Agra,India.
    <br/>
```

```

Taj Mahal is built by <strong>Shah Jahan</strong> in memory of his wife
<strong>Mumtaz Mahal</strong>
<br/>
It is considered to be the excellent example of Mughal Architecture
</p>
<br/>
</body>
</html>

```

Step 3 : Open some web browser and specify the file name **ImgLink.html** in the address bar and you will get following output.

Output



1.13.1 Playing an Audio File

Using HTML script we can play the audio files. The simplest way to play audio file is the use of hyperlinks. Following script shows how simple it is to a play a sound file using HTML

```

<html>
<body>
<h2>Playing The Audio</h2>
<form>
  <a href="chimes.wav">Click here to Play the sound</a>
</form>
</body>
</html>

```

The above script executes in almost all the web browsers such as Internet explorer, Mozilla Firefox, Opera and so on. But the above HTML document launches the Media player software and then the audio file is played within it. For playing the audio files using hyperlinks it is necessary to get the audio file downloaded first.



1.13.2 Uses of Links

- Following are the uses of the hyperlinks for a web document -
 1. One can link logically related documents together using the links in the web page.
 2. Use of link enhances the readability of the web document.
 3. User can click on the link and can learn more about a subtopic and then can return to the main topic. This navigation within the web pages is possible due to the links.

1.14 Frames

- HTML frames allow us to present documents in **multiple views**.
- Using multiple views we can keep certain information visible and at the same time other views are scrolled or replaced.
- For example, within the same window, one frame can display a company information, a second frame can be a navigation menu and a third frame may display selected document that can be scrolled through or replaced by navigating in the second frame.
- Various frames can be set in one browser window.
- To set the frames in the browser window we use **frame set**.
- **For example -**

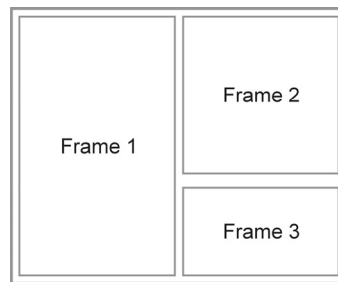


Fig. 1.14.1 Frames

```
<frameset cols="150,*">
```

will allow us to divide the window into two columns (i.e. in two vertical frames). One frame occupying the size of 150 pixels and the other occupies the remaining portion of the window. Here * means any number of pixels.

- Similarly

```
<frameset rows="*,120">
```

will divide the window into two rows (i.e. in two horizontal frames). The second part of horizontal frame will be of 120 pixels and upper horizontal frame will occupy remaining portion of the window.

- Similarly we can also specify the frameset in percentage form. For example

```
<frameset rows="30%,70%">
```

Using frameset we can divide the rows and columns in any number of frames. For example

```
<frameset rows = "20%,30%,50%" cols = "30%,*">
```

This will create a frames in the browser's window as follows -



Fig. 1.14.2 Frames

- In every layout frame we can load the desired html page by using **frame src**. For example

```
<frame src="D:\\html_examples\\bulleted1.html" name="Left_Vertical">
```

By this statement we are loading the web page **bulleted1.html** in the specific frame and the frame is named as **Left_Vertical**.

Attributes in frameset tag

Attribute	Value	Purpose
cols	pixels % *	It specifies the number and size of columns in a frameset
rows	pixels % *	It specifies the number and size of rows in a frameset.

Attributes of frame tag

The <frame> tag has no end tag. The <frame> tag defines one frame within a <frameset> tag. Various attributes of frame tag are,

Attribute	Value	Purpose
frameborder	0 or 1	Value 1 specifies that the frame is displayed with border and 0 indicates that there is no border.
name	Some name	It specifies name of the frame
Nosize		Due to this attribute we cannot resize the particular frame.
scrolling	yes, no or auto	It specifies whether or not to display the scrollbar along with the frame.
src	URL	It specifies the name of the Document to be displayed within the frame.

Example of browser containing frame

Step 1 : Create a main HTML document which will display three HTML documents in three vertical frames.

FrameSet.html

```
<!DOCTYPE html>
<html>
<frameset cols="25%,*,50%">
  <frame src="frame1.html">
  <frame src="frame2.html">
  <frame src="frame3.html">
</frameset>
</html>
```

Step 2 : Create frame1.html, frame2.html and frame3.html files as follows –

Frame1.html

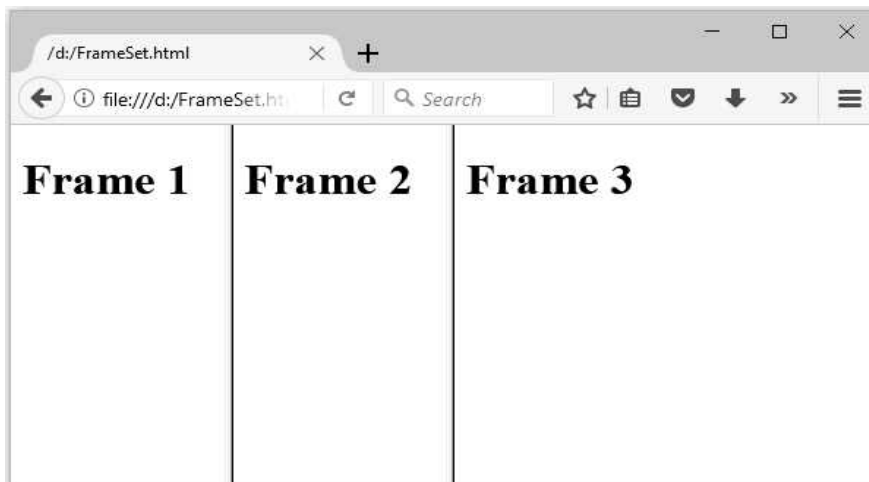
```
<!DOCTYPE html>
<html>
<body>
  <h1> Frame 1 </h1>
</body>
</html>
```

Frame2.html

```
<!DOCTYPE html>
<html>
<body>
  <h1> Frame 2 </h1>
</body>
</html>
```

Frame3.html

```
<!DOCTYPE html>
<html>
<body>
  <h1> Frame 3 </h1>
</body>
</html>
```

Output

Script explanation : The above HTML document, the `<frameset>` tag is used to define frameset. The `col` attribute is used to create the three column frames.

1.14.1 Frames with Scrollbars

```
<!DOCTYPE html>
<html>
<frameset cols="25%,*,50%">
  <frame src="frame1.html" scrolling=no>
  <frame src="frame2.html" scrolling=auto>
  <frame src="frame3.html">
</frameset>
</html>
```

Output

To set the scroll bar we should assign the value **auto** to the **scrolling** parameter and if we do not want the scroll bar accompanying the frame then we must assign the value **no** to the scrolling parameter.

Example 1.14.1 Create XHTML document that has two frames. The left frame displays the contents.html and the right frame displays the cars.html where second frame is the target of link from the first frame. [Note : Contents.html is a list of links for the cars description].

Solution :

Step 1 : We will create the basic html file that contains the left and right frames. The HTML document is as follows :

CarDemo.html

```
<!DOCTYPE html >
<html>
  <head>
    <title>My Frames Page</title>
  </head>
  <frameset cols="50%,50%">
    <frame src="contents.html" name="Left_Vertical" noresize />
    <frame src="Description.html" name="Right_Vertical" scrolling=no/>
  </frameset>
</html>
```

Step 2 : Following is a HTML document that displays the contents that are loaded at first. The left frame HTML document is as follows –

contents.html

```
<!DOCTYPE html >
<html >
  <head>
    <title>My Frames Page</title>
  </head>
  <body>
    <h3>
      <a href="Innova.html" target="Right_Vertical">Toyota Innova</a>
    </h3>
    <h3>
      <a href="Scorpio.html" target="Right_Vertical">Mahindra Scorpio</a>
    </h3>
    <h3>
      <a href="Etios.html" target="Right_Vertical">Toyota Etios Liva</a>
    </h3>
  </body>
</html>
```

The right frame HTML document is as follows -

Description.html

```
<!DOCTYPE html >
<html >
  <head>
    <title>My Frames Page</title>
  </head>
  <body>
    <center>
      <h3> WELCOME </h3>
      This page displays the details of a car you have chosen.
    </center>
  </body>
</html>
```

Step 3 : Following are the three html documents each containing the description of each car.

Innova.html

```
<!DOCTYPE html >
<html>
  <head>
    <title>My Frames Page</title>
  </head>
  <body>
    <h2> TOYOTA INNOVA</h2>
    <ul type="disc">
      <li>Price: Rs.9,35,731-Rs.14,82,852</li>
      <li><td>Mileage: </td><td>10Kmpl(City) and 13 Kmpl(hwy)</li>
    </ul>
  </body>
</html>
```

Scorpio.html

```
<!DOCTYPE html>
<html >
  <head>
    <title>My Frames Page</title>
  </head>
  <body>
    <h2> MAHINDRA SCORPIO</h2>
    <ul type="disc">
      <li>Price: </td><td>Rs.7,49,583-Rs.12,85,479</li>
      <li>Mileage: </td><td>12Kmpl(City) and 16 Kmpl(hwy)</li>
```

```

</ul>
</body>
</html>

```

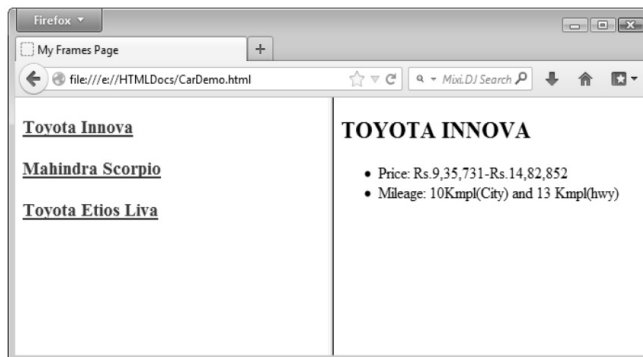
Etios.html

```

<!DOCTYPE html>
<html >
  <head>
    <title>My Frames Page</title>
  </head>
  <body>
    <h2> TOYOTA ETIOS LIVA</h2>
    <ul type="disc">
      <li> Price: </td><td>Rs.4,46,276-Rs.6,86,426</li>
      <li> Mileage: </td><td>17Kmpl(City) and 19 Kmpl(hwy)</li>
    </ul>
  </body>
</html>

```

Step 4 : Open the suitable web browser and invoke the CarDemo.html(created in Step 1), the output will be as follows -



Example 1.14.2 Create a HTML document for a company home page and explain.

Solution : For creation of this web page the frames are used.

Step 1 : The main page is created as follows -

Main.html

```

<?xml version = "1.0" encoding = "utf-8"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Frameset//EN"
  "http://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <frameset rows="20%,80%">
    <frame src="logo.html">
    <frameset cols="30%,70%">

```

```
<frame src="menu.html">
<frame src="contents.html">
</frameset>
</frameset>
</html>
```

Step 2 : In the main page a layout of the company home page is created. Various html files that are called in this HTML document are as given below -

Menu.html

```
<html>
<head><title>LOGO</title></head>
<body bgcolor="aqua">
<h3><a href="about.html"> About us</a></h3>
<h3><a href="download1.html"> Free Downloads</a></h3>
<h3><a href="download2.html"> Download Catalogue</a></h3>
<h3><a href="download3.html">Download Orderform</a></h3>
<h3><a href="catalogues.html">Catalogues</a></h3>
<h3><a href="distributors.html">Distributors</a></h3>
<h3><a href="Contacts.html">Contact Us</a></h3>
<h3><a href="feedback.html">Feedback</a></h3>
</body>
</html>
```

Logo.html

```
<html>
<head><title>LOGO</title></head>
<body bgcolor="magenta">
<h1> Technical Publications</h1>
</body>
</html>
```

Contents.html

```
<html>
<head><title>LOGO</title></head>
<body bgcolor="khaki">
<center>
<h2>BOOKS</h2>
</center>
<p>TECHNICAL PUBLICATIONS is known for commitment to quality and innovation. We are Leaders in our chosen scholarly and educational markets, serving the book Industry & Academic Institutions.
</p>
```

<p>We have been in the industry for the last 18 years and are known for the quality and scholarly publications in Engineering books. We publish more than 1000 titles of text books, for various Universities across the India.

</p>

<p>We are specialized in Engineering text books and have been publishing titles for various Engineering branches such as Electrical, Electronics, Computer Science, Information Technology, Mechanical etc. and other management books.

</p>

</body>

</html>

Step 3 : Open the suitable web browser and type the address for **main.html**. The output will be as follows -



1.15 Lists

- List is nothing but the collections of items or elements. There are two types of lists - unordered **lists** and **ordered lists**.
- The unordered list is useful for simply listing the items but if we want the items in some specific sequence then the ordered lists are used. Let us discuss how to use these types lists in the HTML document.

1.15.1 Unordered List

Following HTML document makes use of unordered list.

HTML Document [UnordLstDemo.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title> Use of Unordered List </title>
  </head>
  <body>
    <h2>All About Computer ...</h2>
```

Following are some popular operating systems used in computer

```
<ul type="disc">
  <li>DOS</li>
  <li>Windows 98</li>
  <li>Windows XP</li>
  <li>Windows Professional</li>
  <li>Windows Vista</li>
  <li>Unix</li>
</ul>
```

Following are some core subjects on computer science

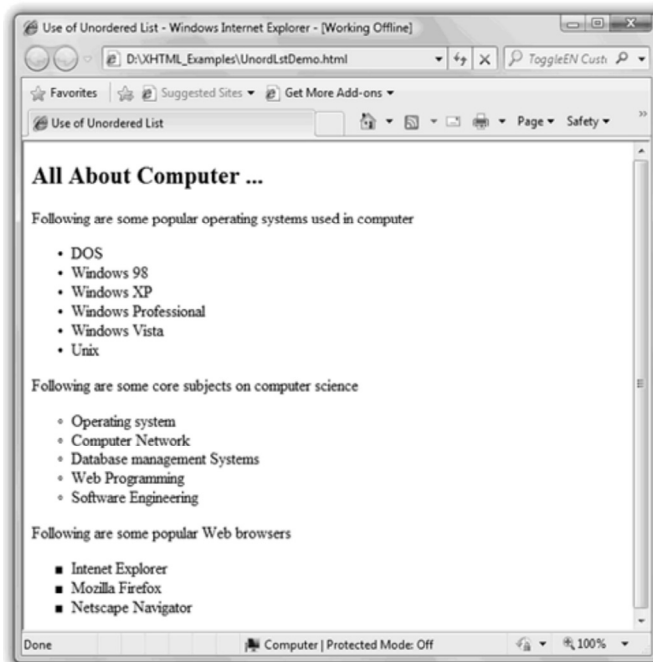
```
<ul type="circle">
  <li>Operating system</li>
  <li>Computer Network</li>
  <li>Database management Systems</li>
  <li>Web Programming</li>
  <li>Software Engineering</li>
</ul>
```

Following are some popular Web browsers

```
<ul type="square">
  <li>Intenet Explorer</li>
  <li>Mozilla Firefox</li>
  <li>Netscape Navigator</li>
</ul>
</body>
```

```
</html>
```

Output



Script explanation :

In above script we have used the tag `<ul>` for specifying the unordered list. The list items can be mentioned using the tag `<li>` Due to which the list items can appear in bulleted form. The style of the bullet can be as given below -

tags	meaning
<code></code> and <code></code>	Beginning and end of the bulleted list
<code></code> and <code></code>	Displays the bulleted text on separate line.
<code><ul type="circle"></code>	Displays the circular bullets.
<code><ul type="disc"></code>	Displays the solid round bullets.
<code><ul type="square"></code>	Displays the squared bullets.

1.15.2 Ordered List

The ordered list is a list of items which must follows some specific sequence. We can number the text using `<ol>` tag. Following table shows various styles by which the list can be numbered.

Tags	Meaning
and	Beginning and end of the numbered list
and	Displays the numbered text on separate line.
<ol type="A">	Displays the list in following manner
	A.
	B.
	C.
	...
<ol type="i">	Displays the list in following manner
	i.
	ii.
	iii.
	...
<ol type="I">	Displays the list in following manner
	I.
	II.
	III.
	...
<ol type="1">	Displays the list in following manner
	1.
	2.
	3.
	...

Here is one illustrative program -

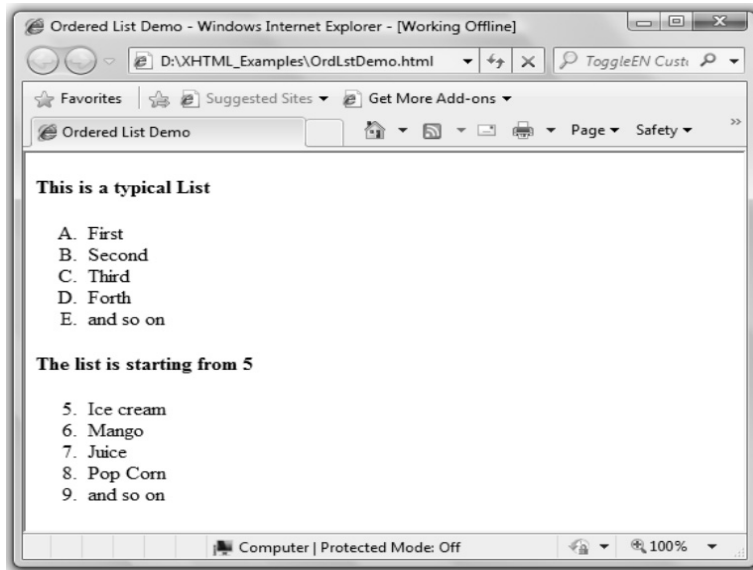
HTML Document [OrdLstDemo.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title> Ordered List Demo </title>
  </head>
  <body>
```

```
<h4>This is a typical List</h4>
<ol type="A">
<li>First</li>
<li>Second</li>
<li>Third</li>
<li>Forth</li>
<li> and so on</li>
</ol>
<h4>The list is starting from 5</h4>
<ol start="5">
<li>Ice cream</li>
<li>Mango</li>
<li>Juice</li>
<li>Pop Corn</li>
<li> and so on</li>
</ol>
</body>
</html>
```

We can have the nested ordered list. Here is an example of it.

Output

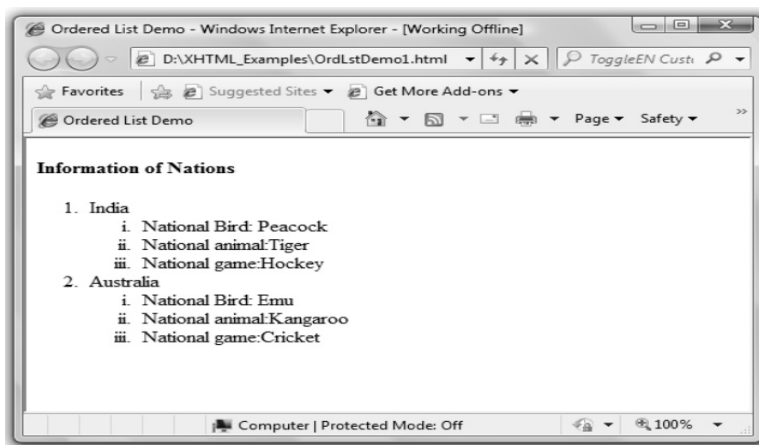


HTML Document [OrdLstDemo1.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title> Ordered List Demo </title>
  </head>
```

```
<body>
  <h4>Information of Nations</h4>
  <ol type="1">
    <li>India
      <ol type="i">
        <li>National Bird: Peacock</li>
        <li>National animal:Tiger</li>
        <li>National game:Hockey</li>
      </ol>
    </li>
    <li>Australia
      <ol type="i">
        <li>National Bird: Emu</li>
        <li>National animal:Kangaroo</li>
        <li>National game:Cricket</li>
      </ol>
    </li>
  </ol>
</body>
</html>
```

Output



1.16 Tables

- For systematic arrangement of information we often require tabular structure. HTML allows us to create tables on the web pages.
- Many web designers are using invisible tables on the web pages.
- The **biggest advantage of using tables** on the web pages is that the information gets arranged systematically.

- The table is a matrix of rows and columns and the area formed due to intersection of a row and a column is called cell.

1.16.1 Basic Table Structure

- To create a table on the web page the table beginning tag is `<table>` and `</table>` tag is used for ending the table.
- Within `<table> ... </table>` tag we can create rows and columns.
- The rows are created using `<tr> </tr>` and columns are created using `<td> </td>`
- For example -

<table>

<tr>	One <td>	Two <td>	Three <td>
<tr>	Four <td>	Five <td>	Six <td>
<tr>	Seven <td>	Eight <td>	Nine <td>

HTML Document[TabDemo.html]

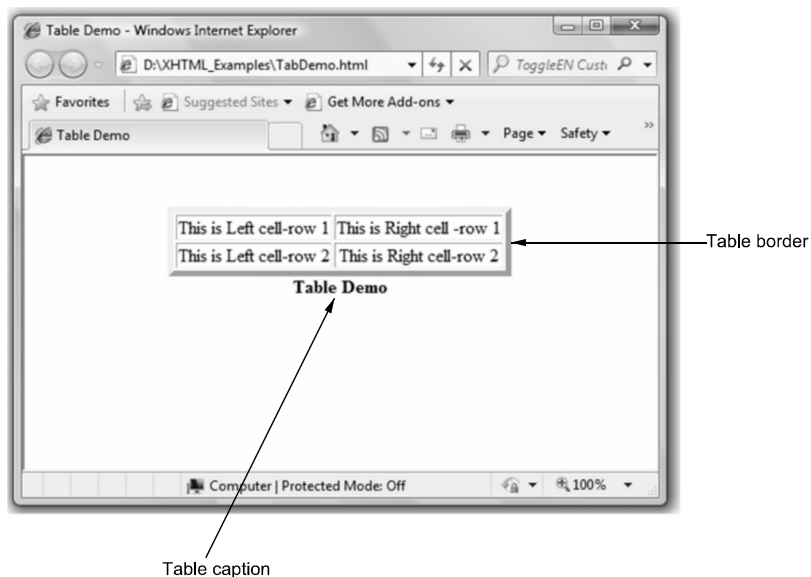
```

<!DOCTYPE html>
<html>
  <head>
    <title> Table Demo </title>
  </head>
  <body>
    <br/><br/>
    <center>
      <table border="5">
        <caption align="bottom">
          <b> Table Demo </b>
        </caption>
        <tr>
          <td>This is Left cell-row 1</td>
          <td>This is Right cell -row 1</td>
        </tr>

```

```
<tr>
    <td>This is Left cell-row 2</td>
    <td>This is Right cell-row 2</td>
</tr>
</table>
</center>
</body>
</html>
```

Output



Script Explanation

- 1) In above program, the parameter border = "5" is set in order to set the table border. You can give any value to set the desired border.
- 2) The caption to the table is given by the parameter caption, we can set this caption either at the top or at the bottom by using the parameter align.
- 3) Then using the tag <tr> the table row can be set. The <td> tag is used to create columns from left to right.
- 4) Thus in the above program we are filling up the table values from top to bottom and from left to right. Just refer the output provided along with the program for clear understanding of look and feel of the table.

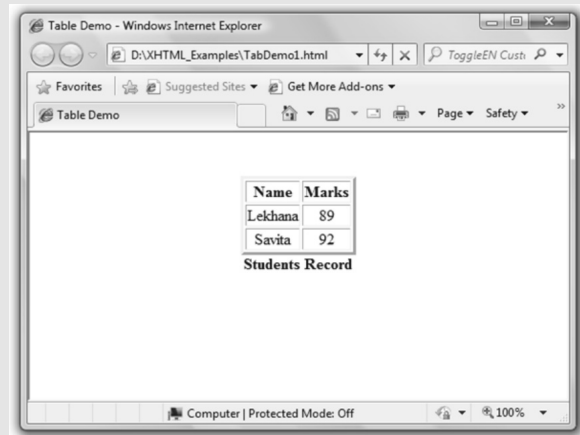
Example 1.16.1 Write HTML document to create a table with header to each column.

Solution : We can set the **header** to each column of the table by **<th>** tag. Here is a simple HTML program that simply adds the header to each column.

HTML Document [TabDemo1.html]

```
<!DOCTYPE html >
<html >
  <head>
    <title> Table Demo </title>
  </head>
  <body>
    <br/><br/>
    <center>
      <table border="3">
        <caption align="bottom">
          <b> Students Record </b>
        </caption>
        <th>Name</th>
        <th>Marks</th>
        <tr align=center>
          <td>Lekhana</td>
          <td>89</td>
        </tr>
        <tr align=center>
          <td>Savita</td>
          <td>92</td> </tr>
        </tr>
      </table>
    </center>
  </html>
```

Output



The data in each row can be aligned centrally by using **<tr align=center>** Other types of alignment can be left or right by **<tr align=left>** or **<tr align=right>** respectively.

Example 1.16.2 Discuss the process of dividing the table into columns with an example. How would you give headers to the table columns ?

Solution : The process of dividing the table into column is as follows :

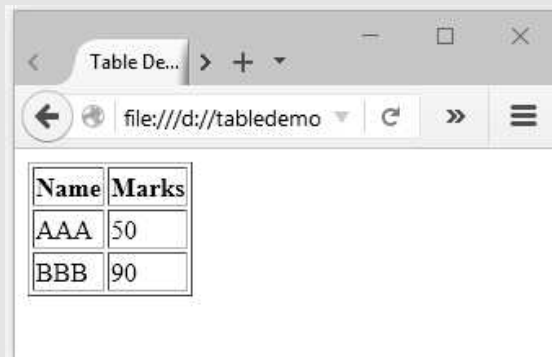
1. Create a table using the tag **<table>**.
2. Set the border to the table using attribute **border="some value"**.
3. Create rows using the tag **<tr>** and for each row create the columns using **<td>** **</td>** tags.
4. The header for the table column can be given using the tag **<th>** **</th>**.

Example 1.16.3 Following HTML document divides the table in columns and gives the header to the column.

Solution :

```
<!DOCTYPE html>
<html>
<head>
<title>Table Demo</title>
</head>
<body>
  <table border = "1">
    <th>Name</th>
    <th>Marks</th>
    <tr>
      <td>AAA</td>
      <td>50</td>
    </tr>
    <tr>
      <td>BBB</td>
      <td>90</td>
    </tr>
  </table>
</body>
</html>
```

Output



1.16.2 RowSpan and ColSpan

- Sometimes we may require adding sub-columns or sub-rows to categorize the information properly. In such a situation colspan and rowspan can be used.
- The **rowspan** is used to extend the **row vertically** and **colspan** is used to extend the **column horizontally**.

For example -

When **rowspan** = 2 then the **row can be extended vertically by two cells**.

rowspan {	First	Second	<td rowspan=2>First
		Third	<td>Second</td><td>Third</td>

When **colspan**=2 then the **column can be extended horizontally by two cells**.

First	Second	<td>First <td>Second
Third		<td colspan=2>Third

colspan

Example 1.16.4 Explain the structure of HTML. Write code to display following table.

Sr. No.	College Name			
	Section A		Section B	
	X	Y	X	Y
1				
2				
3				

Solution : ImageDemo.html

```
<html>
  <head>
    <title>My Table</title>

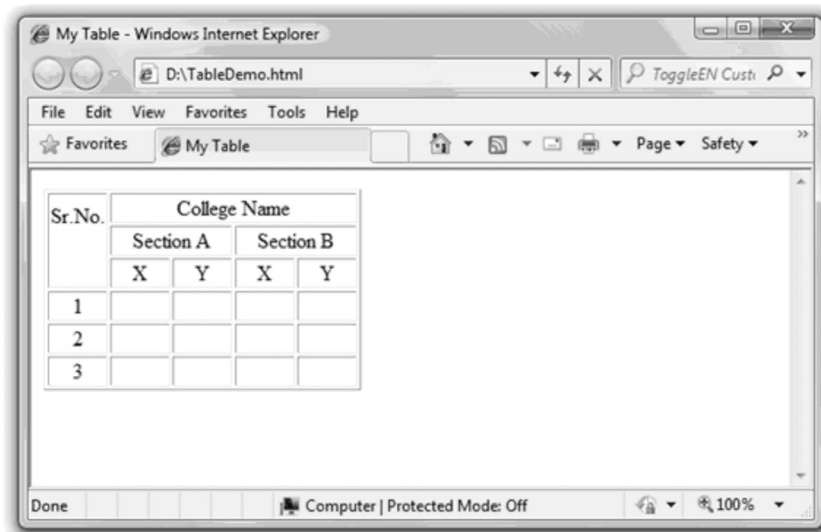
  </head>
  <body>
    <table border=1>
      <tr>
        <td rowspan="3">
          <p align=center>Sr.No.</p>
          <p>&nbsp;</td>

          <td colspan="4">
            <p align="center">College Name</p>
          </td>
        </tr>
        <tr>
          <td width="40%" colspan="2" align="center">Section A</td>
          <td width="40%" colspan="2" align="center">Section B</td>
        </tr>
        <tr>
          <td align="center">X</td>
          <td align="center">Y</td>
          <td align="center">X</td>
          <td align="center">Y</td>
        </tr>
        <tr>
          <td align="center">1</td>
          <td align="center">&nbsp;</td>
          <td align="center">&nbsp;</td>
          <td align="center">&nbsp;</td>
          <td align="center">&nbsp;</td>
        </tr>
      </tr>
    </table>
  </body>
</html>
```

```

        <td align="center">2</td>
        <td align="center">&nbsp;</td>
        <td align="center">&nbsp;</td>
        <td align="center">&nbsp;</td>
        <td align="center">&nbsp;</td>
    </tr>
    <tr>
        <td align="center">3</td>
        <td align="center">&nbsp;</td>
        <td align="center">&nbsp;</td>
        <td align="center">&nbsp;</td>
        <td align="center">&nbsp;</td>
    </tr>
</table>
</body>
</html>

```

Output

Example 1.16.5 Write HTML code to draw table given below :

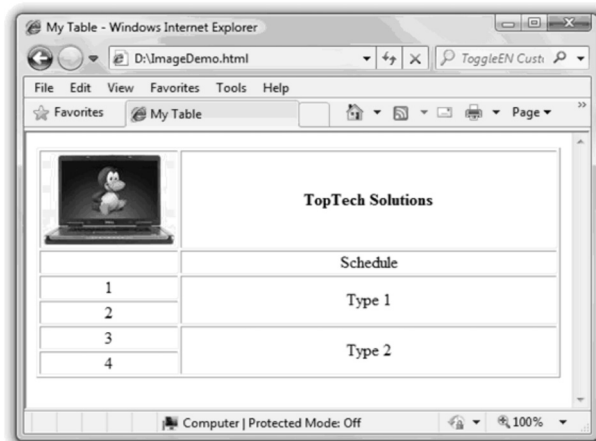
Image (20%)	Company Name (80%)
	Schedule
1	Type 1
2	
3	Type 2
4	

Solution : ImageDemo.html

```

<html>
  <head>
    <title>My Table</title>
  </head>
  <body>
    <table border=1>
      <tr>
        <td width="20%" align=center></td>
        <th width="80%" align=center>TopTech Solutions</th>
      </tr>
      <tr>
        <td>&nbsp;</td>
        <td align=center>Schedule</td>
      </tr>
      <tr>
        <td align=center>1</td>
        <td rowspan="2" align=center>Type 1</td>
      </tr>
      <tr>
        <td align=center>2</td>
      </tr>
      <tr>
        <td align=center>3</td>
        <td rowspan="2" align=center>Type 2</td>
      </tr>
      <tr>
        <td align=center>4</td>
      </tr>
    </table>
  </body>
</html>

```

Output

Example 1.16.6 Write HTML code to draw table given below.

Items		Price
Shirt	Trouser	₹. 1000/-
₹. 400/-	₹. 600/-	

Solution :

```
<html>
<head>
<title>Item-Price List</title>
</head>
<body>
<table border=1>
<tr>
<th colspan="2"><center>Items</center></th>
<th>Price</th>
</tr>
<tr>
<th><center>Shirt</center></th>
<th><center>Trouser</center></th>
<td rowspan="2"><center>Rs.<br/>1000/-</center></td>
</tr>
<tr>
<td><center>Rs.400/-</td>
<td><center>Rs.600/-</td>
</tr>
</table>
</body>
</html>
```

Output



Example 1.16.7 Write HTML code which includes table, Hyperlink, character formatting ordered and unordered list to display your resume.

Solution : resume.html

```
<html>
<head>
<title>RESUME APPLICATION</title>
```

```
</head>
<body>
<center><h2>RESUME</h2></center>
<a href="my_photo.jpg">PHOTO</a>
<br/><br/>
<strong>Name: </strong> Mr.Sachin Patil<br/>
<strong>Address: </strong> D10,SunEmpire,SunCityRd,Pune<br/>
<strong>Sex: </strong>Male<br/>
<strong>Phone: </strong>9229229221<br/>
<strong>Email: </strong>puntambekar_a@rediffmail.com
<br/><br/>
<strong>Educational Qualification:</strong>
<table border=2>
<tr>
<th>Examination</th>
<th>Year Of Passing</th>
<th>Board/University</th>
<th>Marks obtained</th>
</tr>
<tr>
<td>S.S.C</td><td>1991</td><td>Pune</td><td>93%</td>
</tr>
<tr>
<td>H.S.C</td><td>1993</td><td>Pune</td><td>90%</td>
</tr>
<tr>
<td>B.E.(Computer)</td><td>1997</td><td>Pune</td><td>72%</td>
</tr>
<tr>
<td>M.Tech(Computer)</td><td>2001</td><td>IIT(Kanpur)</td><td>A grade</td>
</tr>
</table>
<strong>
<br/><br/>
```

Programming Languages known :

```
</strong>
<ul>
<li>Cobol
<li>C
<li>C++
<li>Visual C++
```

```
<li>Visual Basic
```

```
<li>Java
```

```
</ul>
```

```
<br/> <br/>
```

```
<div align=left>Date:</div>
```

```
<div align=right>Place:</div>
```

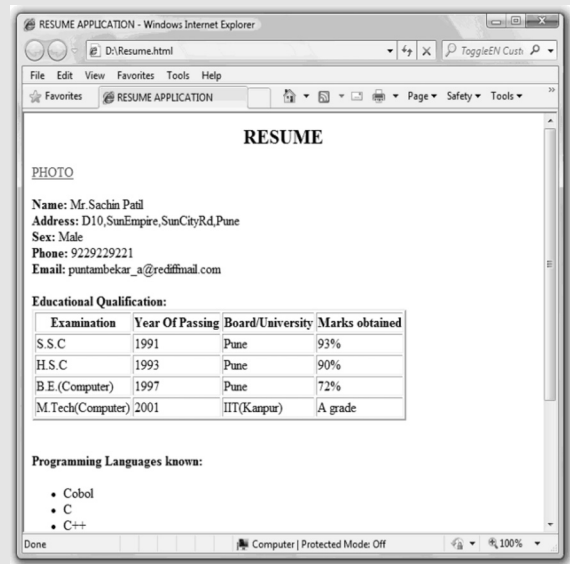
```
<br/>
```

```
<div align=left>Signature</div>
```

```
</body>
```

```
</html>
```

Output



Example 1.16.8 Create the following table using HTML tags.

Name of Train	Place	Destination	Time Arrival Departure		Fare
Rajdhani	Mumbai	Delhi	07.30	08.45	Rs. 989.00
Madras Mail	Mumbai	Madras	09.00	10.15	Rs. 450.00
Konkan Exp.	Mumbai	Manglore	13.30	14.45	Rs. 756.00
Deccan Exp.	Mumbai	Pune	16.00	17.30	Rs. 345.00

Solution :

```
<html>
<head>
<title> Train Time Table </title>
</head>
<body>
<table border="1">
<tr>
<th rowspan=2>Name of Train</th>
<th rowspan=2>Place</th>
<th rowspan=2>Destination</th>
<th colspan=2>Time</th>
<th rowspan=2>Fare</th>
</tr>
```

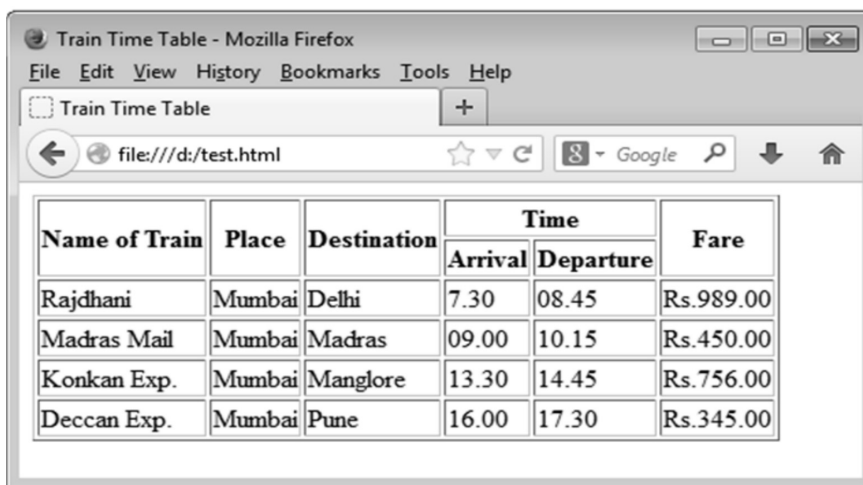
```

<tr><th>Arrival</th><th>Departure</th></tr>
<tr>
<td>Rajdhani</td><td>Mumbai</td><td>Delhi</td>
<td>7.30</td><td>08.45</td>
<td>Rs.989.00</td>
</tr>
<tr>
<td>Madras Mail</td><td>Mumbai</td><td>Madras</td>
<td>09.00</td><td>10.15</td>
<td>Rs.450.00</td>
</tr>
<tr>
<td>Konkan Exp.</td><td>Mumbai</td><td>Manglore</td>
<td>13.30</td><td>14.45</td>
<td>Rs.756.00</td>
</tr>
<tr>
<td>Deccan Exp.</td><td>Mumbai</td><td>Pune</td>
<td>16.00</td><td>17.30</td>
<td>Rs.345.00</td>
</tr>

</table>
</body>
</html>

```

Output



Name of Train	Place	Destination	Time		Fare
			Arrival	Departure	
Rajdhani	Mumbai	Delhi	7.30	08.45	Rs.989.00
Madras Mail	Mumbai	Madras	09.00	10.15	Rs.450.00
Konkan Exp.	Mumbai	Manglore	13.30	14.45	Rs.756.00
Deccan Exp.	Mumbai	Pune	16.00	17.30	Rs.345.00

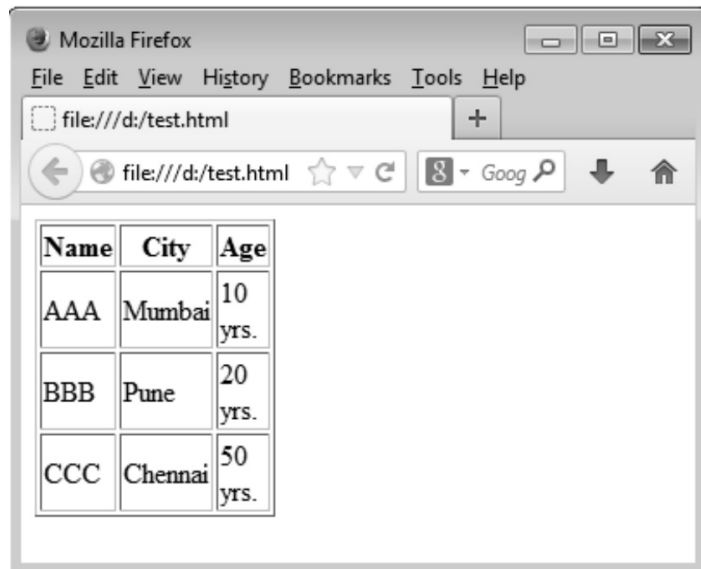
Example 1.16.9 Differentiate between `<table width="400" height="200">` and `<table width="100%" and height="50%">`.

Solution : i) `<table width="400" height="200">` denotes that the table width is 400 pixels and height is 200 pixels. It denotes the **absolute value**.

For example

```
<html>
<head>
</head>
<body>
  <table border="1" width="100" height="50">
    <tr>
      <th>Name</th>
      <th>City</th>
      <th>Age</th>
    </tr>
    <tr>
      <td>AAA</td> <td>Mumbai</td> <td>10 yrs.</td>
    </tr>
    <tr>
      <td>BBB</td> <td>Pune</td> <td>20 yrs.</td>
    </tr>
    <tr>
      <td>CCC</td> <td>Chennai</td> <td>50 yrs.</td>
    </tr>
  </table>
</body>
</html>
```

Output



The screenshot shows a Mozilla Firefox browser window with the address bar displaying 'file:///d:/test.html'. The rendered HTML table is displayed in the main content area. The table has three columns: Name, City, and Age. The rows contain the following data: AAA, Mumbai, 10 yrs.; BBB, Pune, 20 yrs.; and CCC, Chennai, 50 yrs.

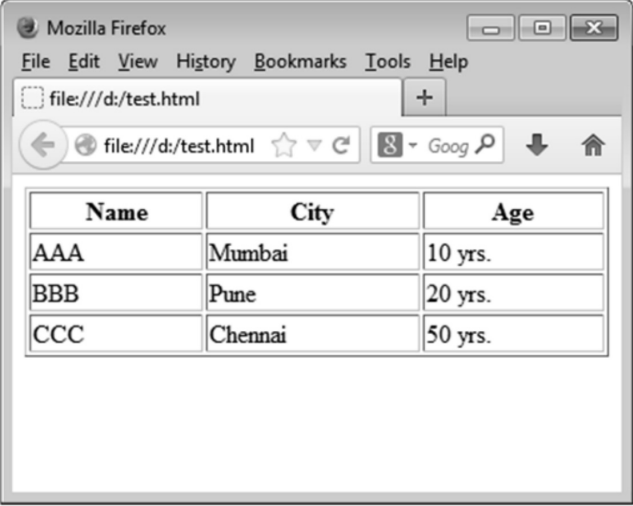
Name	City	Age
AAA	Mumbai	10 yrs.
BBB	Pune	20 yrs.
CCC	Chennai	50 yrs.

- ii) On the other hand `<table width="100%" and height="50%">` denotes the width of the table is 100% of the screen and height is 50% of the browser screen. It denotes the **relative value**.

For example

```
<html>
<head>
</head>
<body>
  <table border="1" width="100%" height="50%">
    <tr>
      <th>Name</th>
      <th>City</th>
      <th>Age</th>
    </tr>
    <tr>
      <td>AAA</td><td>Mumbai</td><td>10 yrs.</td>
    </tr>
    <tr>
      <td>BBB</td><td>Pune</td><td>20 yrs.</td>
    </tr>
    <tr>
      <td>CCC</td><td>Chennai</td><td>50 yrs.</td>
    </tr>
  </table>
</body>
</html>
```

Output



The screenshot shows a Mozilla Firefox browser window with the address bar displaying 'file:///d:/test.html'. The browser content area displays a table with three columns: Name, City, and Age. The table contains three rows of data: AAA, Mumbai, 10 yrs.; BBB, Pune, 20 yrs.; and CCC, Chennai, 50 yrs.

Name	City	Age
AAA	Mumbai	10 yrs.
BBB	Pune	20 yrs.
CCC	Chennai	50 yrs.

1.16.3 Additional Table Elements

There are some additional tables that can add additional meaning and accessibility to the table. For example -

1. The <caption> element :

- The <caption> tag defines a table caption.
- The <caption> tag must be inserted immediately after the <table> tag.
- For example –

```
<!DOCTYPE html>
<html>
<head>
<style>
table,th,td{
    border:1px solid black;
}
</style>
</head>
<body>
<table>
    <caption>Student Information</caption>
    <tr>
        <th>RollNo</th>
        <th>Name</th>
    </tr>
    <tr>
        <td>1</td>
        <td>AAA</td>
    </tr>
    <tr>
        <td>2</td>
        <td>BBB</td>
    </tr>
    <tr>
        <td>3</td>
        <td>CCC</td>
    </tr>
</table>
</body>
</html>
```

Output

RollNo	Name
1	AAA
2	BBB
3	CCC

Table
Caption

2. The <col> and <colgroup> element :

- The <col> tag specifies column properties for each column within a <colgroup> element.

- The **<col>** tag is useful for **applying styles to entire columns**, instead of repeating the styles for each cell, for each row.
- For example –

```

<!DOCTYPE html>
<html>
<head>
<style>
table, th, td {
    border: 1px solid black;
}
</style>
</head>
<body>

<table>
<colgroup>
    <col span="2" style="background-color:pink">
    <col span="2" style="background-color:aqua">
</colgroup>
<tr>
    <th>RollNo</th>
    <th>Name</th>
    <th>City</th>
    <th>Marks</th>
</tr>
<tr>
    <td>1</td>
    <td>AAA</td>
    <td>Pune</td>
    <td>78</td>
</tr>
<tr>
    <td>2</td>
    <td>BBB</td>
    <td>Mumbai</td>
    <td>88</td>
</tr>
<tr>
    <td>3</td>
    <td>CCC</td>

```

Output

RollNo	Name	City	Marks
1	AAA	Pune	78
2	BBB	Mumbai	88
3	CCC	Bangalore	97
4	DDD	Hyderabad	84

```
<td>Bangalore</td>
  <td>97</td>
</tr>
<tr>
  <td>4</td>
  <td>DDD</td>
  <td>Hyderabad</td>
  <td>84</td>
</tr>
</table>
</body>
</html>
```

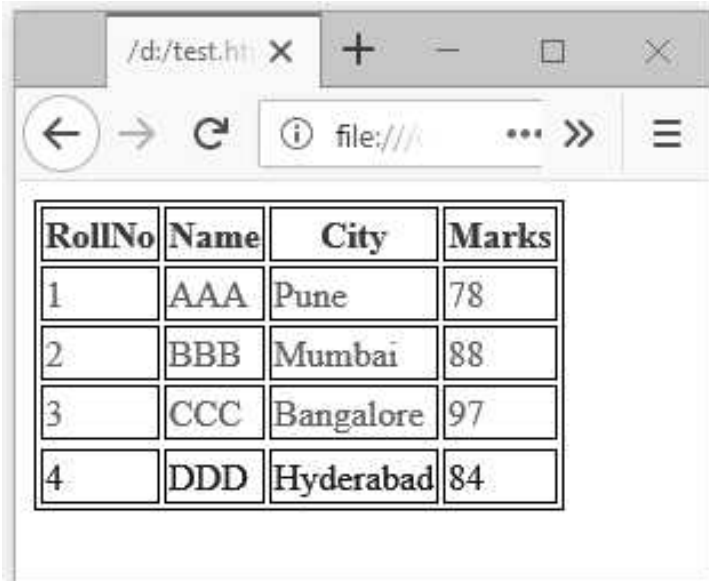
3. The <tbody>, <thead> and <tfoot> elements

- The <tbody> tag is used to group the body content in an HTML table.
- The <tbody> element is used in conjunction with the <thead> and <tfoot> elements to specify each part of a table such as body, header, and footer.
- For example –

```
<!DOCTYPE html>
<html>
<head>
<style>
table, th, td {
  border: 1px solid black;
}
thead {color:red;}
tbody {color:green;}
tfoot {color:blue;}
</style>
</head>
<body>

<table>
<thead>
<tr>
<th>RollNo</th>
  <th>Name</th>
  <th>City</th>
```

```
<th>Marks</th>
</tr>
</thead>
<tbody>
<tr>
<td>1</td>
<td>AAA</td>
<td>Pune</td>
<td>78</td>
</tr>
<tr>
<td>2</td>
<td>BBB</td>
<td>Mumbai</td>
<td>88</td>
</tr>
<tr>
<td>3</td>
<td>CCC</td>
<td>Bangalore</td>
<td>97</td>
</tr>
<tr>
<td>4</td>
<td>DDD</td>
<td>Hyderabad</td>
<td>84</td>
</tr>
</tbody>
<tfoot>
<td>4</td>
<td>DDD</td>
<td>Hyderabad</td>
<td>84</td>
</tr>
</tfoot>
</table>
</body>
</html>
```

Output

RollNo	Name	City	Marks
1	AAA	Pune	78
2	BBB	Mumbai	88
3	CCC	Bangalore	97
4	DDD	Hyderabad	84

1.17 Images

- We can display images as a link. We have to use the `<img src = " " >` tag for this purpose.
- Within the double quotes image file should be mentioned. Normally .gif or .jpg files are used to display the images on the web browser.
- **Example**
- Following HTML Script shows how to insert an image in the web page.

HTML Document [imagedemo.html]

```
<!DOCTYPE html>
<html>
<head>
  <title> Image Demo </title>
</head>
<body>
  <h1> Indian Heritage </h1>
  <p>
    In India, People celebrate the festival of light <br/>
    Which is called as <strong>Diwali</strong>
  </p>
  <img src= "lamp1.jpg" alt= "Light!!" />
  <br/>
  Diwali suggests us to enlighten our life with hope,health and wealth
  <br/>
</body>
</html>
```

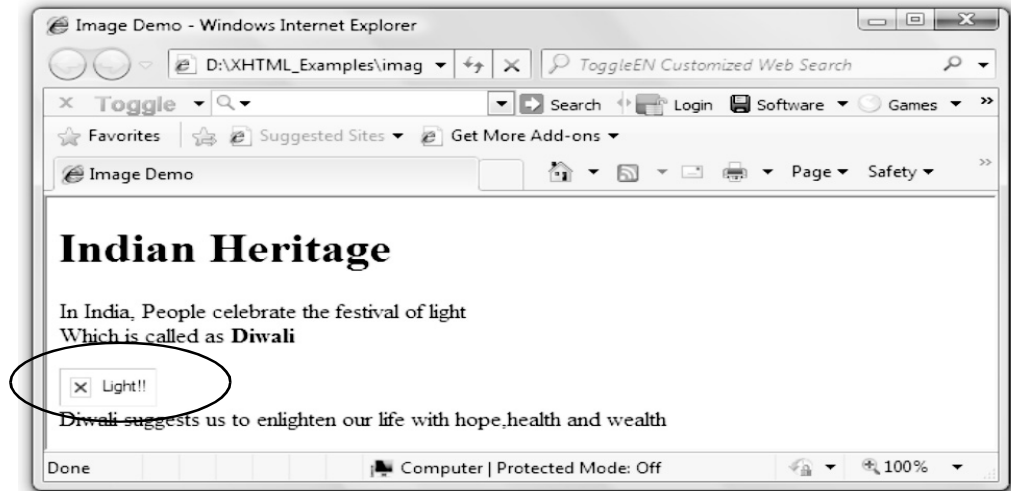
Output



Script explanation : In above script we have used `<img />` tag in order to display an image.

The image tag has two attributes -

- The **src** suggests us to give the **image file name**. If the desired image is not present in the current directory of web page then the complete path where the image is located must be given.
- The **alt** attribute displays the text when it is not possible to display the image. For instance - in above given **imagedemo.html** script if, the **lamp1.jpg** is not present then we will get following output.



Types of Images

- There are two types of images that are more commonly used -
 1. **Graphic Interchange Format (GIF)** : It uses 8-bit color representation for pixels. Hence a single pixel can have 256 different colors. This file have the extension **.gif**.
 2. **Joint Photographic Expert Group (JPEG)** : This format uses 24-bit color representation for pixel. Hence a JPEG can have 16 million different colors. The file have the extension **.jpg** or **.jpeg**.

The JPEG image have more color information than a GIF image. Secondly JPEG can be smaller than GIF image. Hence normally JPEG image is preferred. But the drawback of JPEG is that it does not support transparency.

1.18 Forms

- Form is a typical layout on the web page by which a user can interact with the web page.

- Typical component of forms are **text, text area, checkboxes, radio buttons and push buttons**.
- HTML allows us to place these form components on the web page and send the desired information to the destination server.
- All these form contents appear in the <form> tag. The form has an attribute action which gets executed when user clicks a button on the form.
- Various attributes of form are –

Attribute	Description
action	It specifies the url where the form should be submitted.
method	It specifies the HTTP methods such as GET, POST
name	This attribute denotes the name of the form.
target	It specifies the target of the address in the action attribute.

Let us learn various form components with the help of simple HTML documents.

1.18.1 Text

- **Text** is typically required to place one line text. For example if you want to enter some name then it is always preferred to have Text field on the form.
- The text field can be set using

```
<input type="text" size="30" name="username" value=" " >
```

The input type is **text** and the **value** of this text field is " " That means the blank text field is displayed initially and we can enter the text of our choice into it. There is **size** parameter which allows us to enter some size of the text field.

- Some other parameters or attributes can be
 - **maxlength** that allows us to enter the text of some maximum length.
 - **name** indicates name of the text field.
 - **align** denotes the alignment of the text in the text field. The alignment can be left, right, bottom and top.

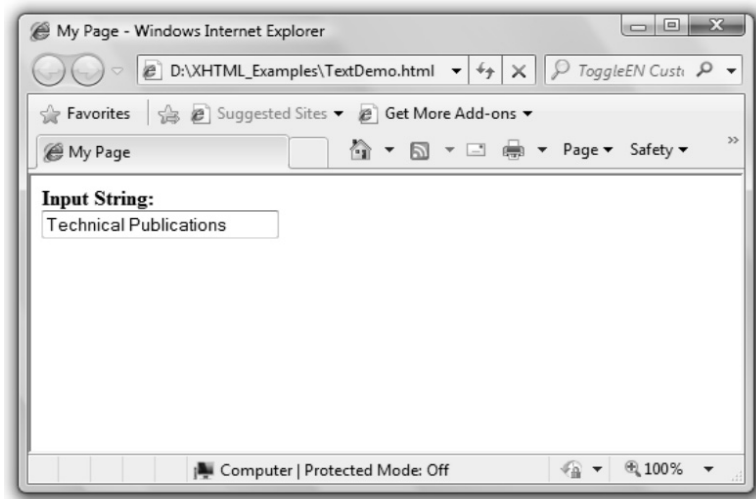
Example code

HTML Document [TextDemo.html]

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml11.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
```

```
<title>My Page</title>
</head>
<body>
<form>
  <b>Input String:</b><br/><input type="text" size="25" value="">
</form>
</body>
</html>
```

Output



Script explanation : In above document

- 1) We have the label “Input String” just before the <input> tag. We can also specify the label by using the <label> tag as follows -

```
<label>Input String:<br/><input type="text" size="25" value=""></label>
```
- 2) Thus the label gets bound to the text box. This aspect is always beneficial for a web programmer because using label control we can focus on the corresponding text box contents.
- 3) Initially the text box field is blank. We can type some text inside this text box.

Example 1.18.1 *How will you create password field in a HTML form ?*

Solution : The password field is typically created on the form. Hence following code can be written to create it -

```
<form name="form1">
  Password:<input type="password"/>
</form>
```

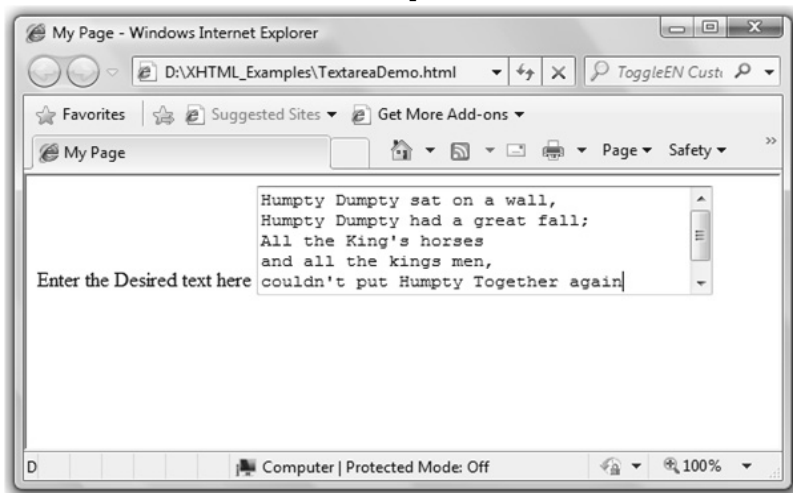
1.18.2 Text Area

Text field is a form component which allows us to enter single line text, what if we want to have multiple line text? Then you must use textarea component.

HTML Document [TextareaDemo.html]

```
<!DOCTYPE html>
<html>
<head>
  <title>My Page</title>
</head>
<body>
  <form>
    Enter the Desired text here
    <textarea cols="40" rows="5" name="myname">
    </textarea>
  </form>
</body>
</html>
```

Output



Various parameters that can be set for the text area can be,

- **row** denotes total number of rows in the text area.
- **col** specifies total number of columns in the text area.
- **name** denotes the name of the text area which can be utilised for handling that component for some specific purpose.

- **wrap** can be **virtual** or **physical**. If the **wrap** is virtual then the line breaks get disappeared when the text is actually submitted to the server. But if the wrap is assigned to the **physical** then the line breaks (if any) appear as it is in the text.

1.18.3 Checkbox

- It is the simplest component which is used particularly when we want to make some selection from several options.
- For having the checkbox we have to specify the input type as **checkbox**. For example

```
<input type="checkbox" name="option1" value="mango"
checked="checked">Mango<br/>
```

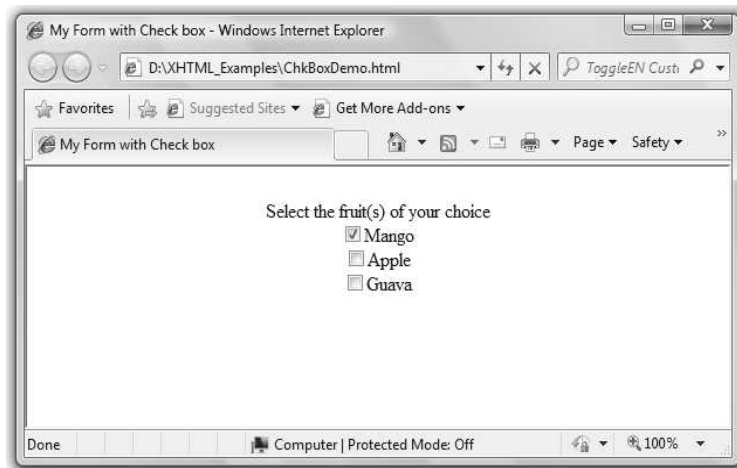
- If we want to get the checkbox displayed as checked then set **checked="checked"**

Example Code

HTML Document[ChkBoxDemo.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Form with Check box</title>
  </head>
  <body>
    <form name ="checkboxForm">
      <div align="center"> <br>
      Select the fruit(s) of your choice<br/>
      <input type="checkbox" name="option1" value="mango"
checked="checked">Mango<br/>
      <input type="checkbox" name="option2" value="apple">Apple<br/>
      <input type="checkbox" name="option3" value="guava">Guava<br/>
    </div>
    </form>
  </body>
</html>
```

Output



Script explanation

- 1) In the above program to set some checkbox in checked state we can mention the attribute **checked** as **checked**.
- 2) We can set the **value** attribute as " " but this then the checkbox will not get associated with any value. The **Mango**, **Apple** and **Guava** are the labels of the checkboxes.

1.18.4 Radio Button

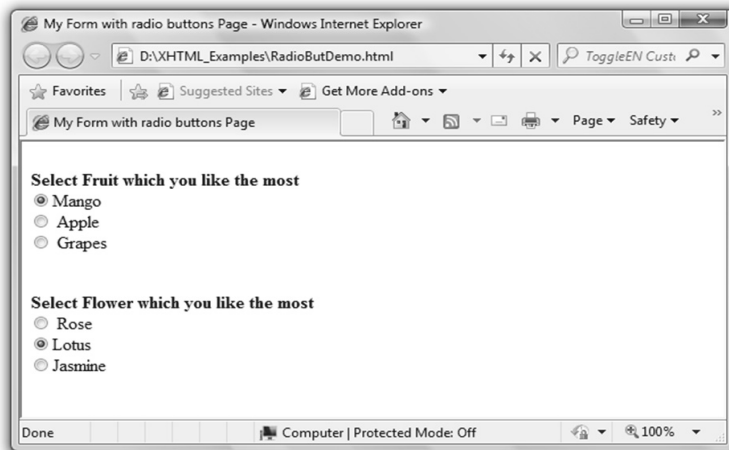
- This form component is also use to indicate the selection from several choices.
- Using **input type="radio"** we can place radio button on the web page.
- This component allows us to make only one selection at a time.
- We can create a group of some radio button component.
- Following HTML document displays the radio buttons for two different groups.

HTML Document[RadioButDemo.html]

```
<!DOCTYPE html>
<html >
  <head>
    <title>My Form with radio buttons Page</title>
  </head>
  <body>
    <form name="myform">
      <div align="left"> <br>
      <b>Select Fruit which you like the most</b> <br>
      <input type="radio" name="group1" value="Mango">Mango<br>
      <input type="radio" name="group1" value="Apple" checked> Apple<br>
      <input type="radio" name="group1" value="Grapes"> Grapes
```

```
<br/><br/><br/>
<b>Select Flower which you like the most</b><br/>
<input type="radio" name="group2" value="Rose"> Rose<br/>
<input type="radio" name="group2" value="Lotus">Lotus<br/>
<input type="radio" name="group2" value="Jasmine" checked>Jasmine<br/>
</div>
</form>
</body>
</html>
```

Output



Example 1.18.2 *What is the difference between group of checkbox buttons and group of radio buttons ?*

Solution :

- The checkbox and radio buttons are used for making the selection from a group of choices.
- When a user selects (checks) a checkbox, its value gets assigned as the current value of the checkbox group's control name.
- A checkbox group's control name may get paired with several current values if the user selects more than one checkbox.
- Radio buttons work just like checkboxes except they are typically set up to be mutually exclusive of one another, i.e. when one is selected, all the others are automatically deselected.

1.18.5 Button

- We can create the button using `<input type="button">` tag.
- There are two types of buttons that can be created in HTML. One is called **submit** button and the another one is **reset** button.

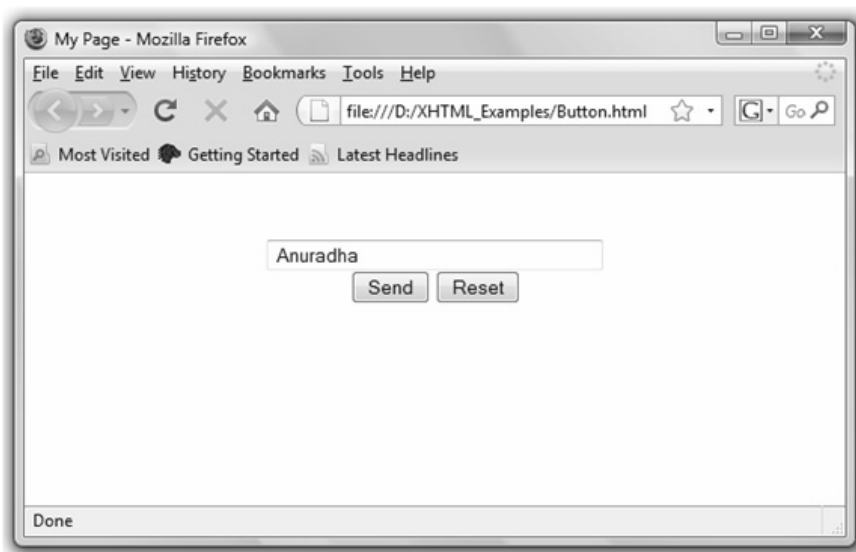
- Various parameters of submit button are :
 - 1) **name** denotes the name of the submit button.
 - 2) **value** is for writing some text on the text on the button.
 - 3) **align** specifies alignment of the button.

Example code

HTML Document[Button.html]

```
<!DOCTYPE html>
<html >
  <head>
    <title> My Page </title>
  </head>
  <body>
    <form name="myform" action="http://www.localhost.com/cgi-bin/hello.cgi" method="POST">
      <div align="center">
        <br/><br/>
        <input type="text" size="35" value=" ">
        <br><input type="submit" value="Send">
        <input type="reset" value="Reset"><br>
      </div>
    </form>
  </body>
</html>
```

Output



Script explanation

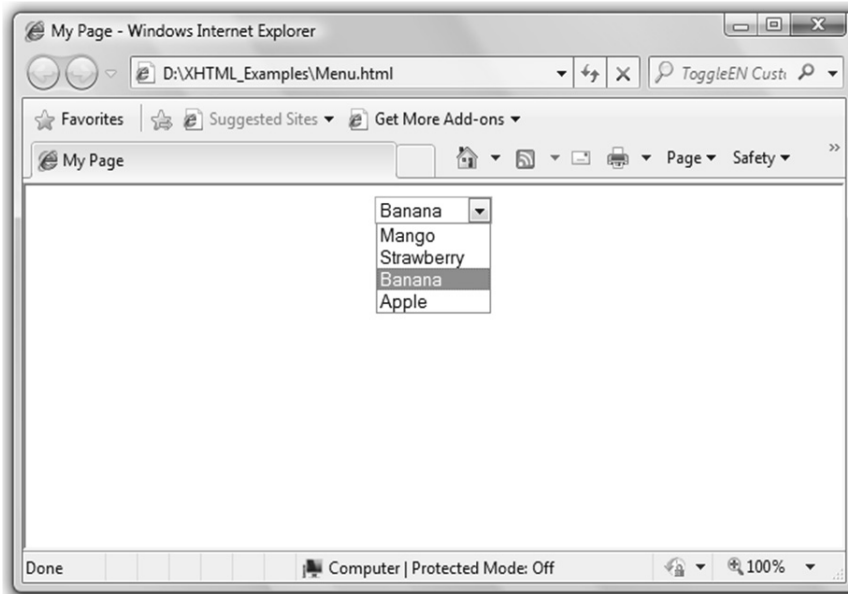
- 1) In above HTML document, we have used the form whose name is “**myform**”.
- 2) There are two attributes associated with the **form** tag and those are **action** and **method**. The **action** parameter indicates the address and the cgi script where the contents should go and **method** parameter is for the methods for submitting the data. The **method** can be **get** or **post**. Thus by specifying the action and method for a form we can send the desired data at desired location.

1.18.6 Menus

- HTML allows us to have pop down menu on the web page so that the desired selection can be made.
- The parameter **select** is for the menu component and **option** parameter is for setting the values to the options of drop down menu.
- We can make some specific option selected by **selected value = .**
- In the following HTML document we have created one drop down menu in which various fruits are enlisted. By default “Banana” is set as selected.

HTML Document [Menu.html]

```
<!DOCTYPE html>
<html >
  <head>
    <title> My Page </title>
  </head>
  <body>
    <form name="myform">
      <div align="center">
        <select name="My_Menu">
          <option value="Mango">Mango</option>
          <option value="Strawberry">Strawberry</option>
          <option selected value="Banana">Banana </option>
          <option value="Apple">Apple</option>
        </select>
      </div>
    </form>
  </body>
</html>
```

Output

Example 1.18.3 Create a HTML document that has the form with the following controls

- a) A text box to collect the customer name.
- b) Four checkboxes, one for the following items :
 - i. Four HTML textbooks for ₹1000
 - ii. Eight XML textbooks for ₹2000
 - iii. Four JavaBeans books for ₹2500
 - iv. Eight UML textbooks for ₹1500
- c) A collection of radio buttons that are labelled as follows –
 - i. Cash
 - ii. Cheque/DD
 - iii. Credit card.

Solution :

```
<!DOCTYPE html>
<html >
<body>
<center><h2>REGISTRATION FORM</h2></center>
<hr/>
<form>
<table>
<tr>
<td>Name:</td>
<td><input type="text" size="25" value=""></td>
</tr>
<tr>
<td>Select the desired Items:</td>
<td>
```

```

<select name="f">
<option value="4 HTML Books(Rs. 1000)">4 HTML Books(Rs. 1000)</option>
<option value="8 XML Books(Rs. 2000)">8 XML Books(Rs. 2000)</option>
<option value="4 JavaBeans Books(Rs.2500)">4 JavaBeans Books(Rs.2500)</option>
<option value="8 UML Books(Rs.1500)">8 UML Books(Rs.1500)</option>
</select>
</td>
</tr>
<tr>
<td>
Mode of Payment:
</td>
</tr>
<tr>
<td><input type="radio" name="group1" value="Cash">Cash</td>
</tr>
<tr>
<td><input type="radio" name="group1" value="Cheque/DD">Cheque/DD</td>
</tr>
<tr>
<td><input type="radio" name="group1" value="Credit Card">Credit Card</td>
</tr>
<tr></tr><tr></tr><tr></tr>
<td><input type="submit" value="Submit"></td>
<td><input type="reset" value="Reset"></td>
</tr>
</table>
</form>
</body>

```

Example 1.18.4 Write a form to collect details of a user such as name, address, radio button to choose subject of book he wants to buy, dropdown to choose favorite author, comments for the last book he read.

Solution :

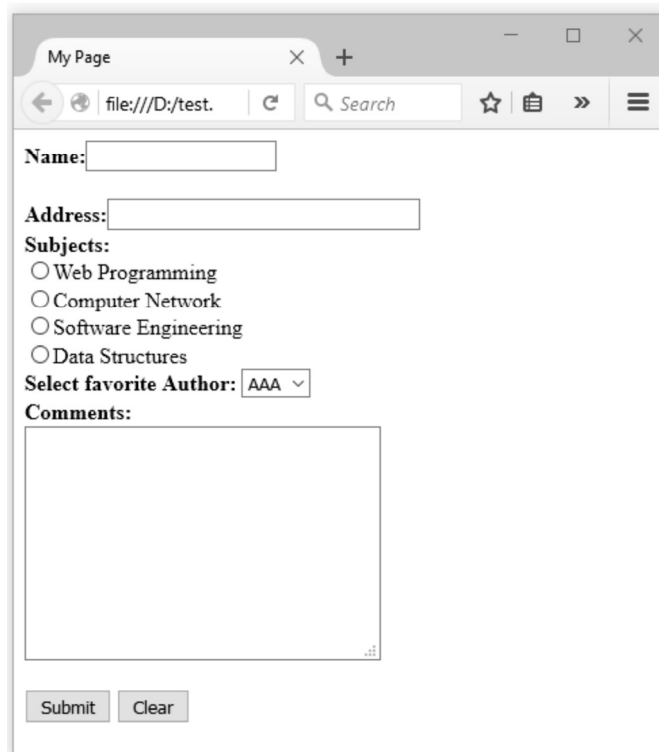
```

<!DOCTYPE html >
<html >
<head>
<title>My Page</title>
</head>
<body>
<form>
<b>Name:</b><input type="text" size="20" value=""><br/>
<b>Address:</b><input type="text" size="35" value=""><br/>
<b>Subjects:</b><br/>
<input type="radio" name="authors" value="Web Programming">Web Programming<br/>

```

```
<input type="radio" name="authors" value="Computer Network">Computer
Network<br/>
<input type="radio" name="authors" value="Software Engineering"> Software
Engineering<br/>
<input type="radio" name="authors" value="Data Structures">Data Structures<br/>
<b>Select favorite Author:</b>
<select name="MyMenu">
  <option value="AAA">AAA</option>
  <option value="BBB">BBB</option>
  <option value="CCC">CCC</option>
  <option value="DDD">DDD</option>
</select>
<br>
<b>Comments:</b><br>
<textarea cols="30" rows="10" name="comments">
</textarea>
<br><br>
<input type="submit" value="Submit"/>
<input type="reset" value="Clear"/>
</form>
</body>
</html>
```

Output

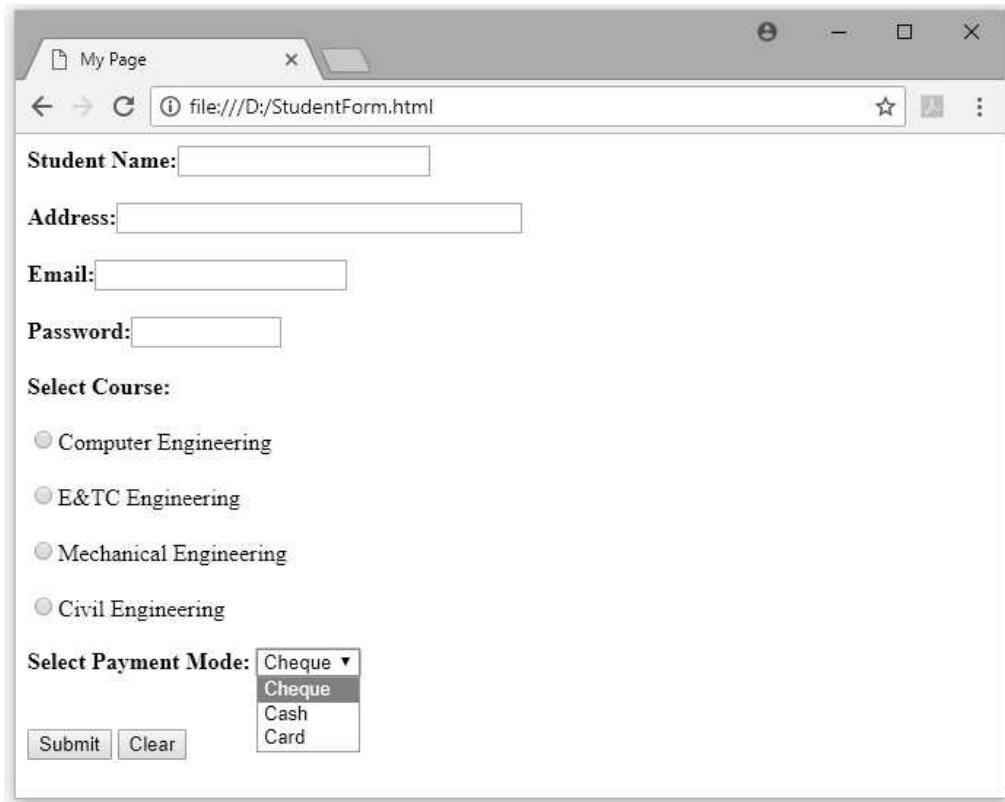


The screenshot shows a web browser window titled "My Page" with the address bar displaying "file:///D:/test.". The rendered form contains the following elements:

- Name:** A text input field.
- Address:** A text input field.
- Subjects:** A list of radio buttons with labels: "Web Programming", "Computer Network", "Software Engineering", and "Data Structures".
- Select favorite Author:** A dropdown menu currently showing "AAA".
- Comments:** A large text area for comments.
- Submit and Clear buttons:** Two buttons at the bottom of the form.

Example 1.18.5 *Design a simple HTML form for registration of a student.***Solution : HTML Form for Student Registration**

```
<!DOCTYPE html>
<html >
<head>
  <title>My Page</title>
</head>
<body>
  <form>
    <b>Student Name:</b><input type="text" size="20" value=""><br/><br/>
    <b>Address:</b><input type="text" size="35" value=""><br/><br/>
    <b>Email:</b><input type="text" size="20" value=""><br/><br/>
    <b>Password:</b><input type="password" size="10" value=""><br/><br/>
    <b>Select Course:</b><br/><br/>
    <input type="radio" name="courses" value="Computer Engineering">Computer
Engineering<br/><br/>
    <input type="radio" name="courses" value="E&TC Engineering">E&TC
Engineering<br/><br/>
    <input type="radio" name="courses" value="Mechanical Engineering">Mechanical
Engineering<br/><br/>
    <input type="radio" name="courses" value="Civil Engineering">Civil
Engineering<br/><br/>
    <b>Select Payment Mode:</b>
    <select name="MyMenu">
      <option value="Cheque">Cheque</option>
      <option value="Cash">Cash</option>
      <option value="Card">Card</option>
    </select>
    <br/><br/><br/>
    <input type="submit" value="Submit"/>
    <input type="reset" value="Clear"/>
  </form>
</body>
</html>
```

Output

The screenshot shows a web browser window with the title 'My Page' and the address bar displaying 'file:///D:/StudentForm.html'. The form contains the following elements:

- Student Name:** A text input field.
- Address:** A text input field.
- Email:** A text input field.
- Password:** A text input field.
- Select Course:** A group of four radio buttons with labels: 'Computer Engineering', 'E&TC Engineering', 'Mechanical Engineering', and 'Civil Engineering'.
- Select Payment Mode:** A dropdown menu currently showing 'Cheque'. The dropdown list is open, showing options: 'Cheque', 'Cash', and 'Card'.
- Submit** and **Clear** buttons at the bottom left.

1.19 HTML5**SPPU : May-19, Dec.-19, Marks 5**

- The HTML 5.0 is the fifth revision of HTML standard of World Wide Consortium (W3C) which is finalized in October 2014.
- The earlier version was HTML 4.0 which was released in 1997.
- The objective of this markup language version was - i) To provide support for latest multimedia and ii) To make the script readable and consistently understood by the developer.
- The HTML5 adds many new syntactic features. It includes the elements such as <video>, <audio> and <canvas> elements. It also has integration of Scalable Vector Graphics (SVG) contents and MathML for mathematical Formula.

1.19.1 Features of HTML 5.0

1. The XHTML doctype is just impossible to memorize (!DOCTYPE html PUBLIC "...). Hence a new HTML doctype is <!DOCTYPE html>.
2. There are new graphic elements such as <canvas> and <svg> in HTML5.0.

3. The support for multimedia elements such as <audio> and <video>
4. It has support for <header>, <footer>, <article>, and <section>
5. It has support for new form controls such as number, date, time, calendar, and range.
6. It has a rich set of Application Programming Interface (API) for geolocations, HTML Drag and Drop, Local Storage, Application cache and so on.

1.19.2 Difference between HTML and HTML5

Sr. No.	HTML	HTML5
1.	The DOCTYPE declaration is much longer such as <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml11.dtd">	The DOCTYPE declaration is simple <!DOCTYPE html>
2.	<audio> and <video> tags are not supported.	<audio> and <video> tags are supported for playing audio and video files.
3.	Finding out geographic location is impossible using HTML	The HTML5 supports the API for identifying the geographic location .
4.	It supports the tag such as <applet>, <big>, <center>, , <frame>, <strike>	The tags that are removed from HTML5 are <applet>, <big>, <center>, , <frame>, <strike>
5.	Does not allow JavaScript to run in browser. JavaScript runs in same thread as browser interface.	It allows JavaScript to run in background.
6.	There is no support for <canvas>	The <canvas> tag is for 2D drawing.
7.	It needs external plugins such as flash.	The need for external plugin is reduced .

Review Questions

1. Write at least any five differences between HTML and HTML5.

SPPU : May-19, Marks 5

2. Describe the following web technology : HTML 5.

SPPU : Dec.-19, Marks 3

Part III : CSS

1.20 Introduction to Style Sheet

- The CSS stands for Cascading Style Sheet.
- The Cascading Style Sheet is a markup language used in the web document for presentation purpose.
- The **primary intension of CSS** was to separate out the web content from the web presentation.
- Various elements such as text, font and color are used in CSS for presentation purpose. Thus CSS specification can be applied to bring the styles in the web document.

1.21 CSS Features

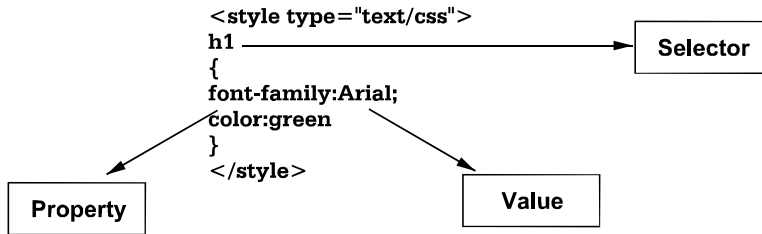
- By combining CSS with the html document, considerable amount of **flexibility** into the content submission can be achieved.
- Similarly, separating out the style from actual contents help in **managing** large-scale complex sites. Thus CSS facilitates publication of content in multiple presentation formats.
- If CSS is used, effectively then **global style sheet** can be applied to a web document. This helps in maintaining the **consistency in the web document**.
- If a **small change** needs to be done in the style of web content, then CSS makes it more convenient.
- All above mentioned advantages of CSS show the need for it in web development.

1.22 CSS Core Syntax

- The style specification is specified differently for each different level. For instance :
- For inline cascading style sheets the **style** appears in side the tag for defining the value.

```
<p style="font-size: 30pt ;font-family:Script">
```

- For the document cascading style sheet the **style** specification appear as the content of a style element within the header of a document.



The **type** attribute tells the browser that what it is reading is text which is affected by the Cascading Style Sheet. The type specification is necessary because there is one more specification used in JavaScript.

- The external style sheet makes use of **style** specification in the same manner as in document cascading style sheet.
- The most commonly used CSS properties are enlisted in the following table -

Property Type	Property
Fonts	font font-family font-size font-style font-weight @font-face
Text	letter-spacing line-height text-align text-decoration text-indent
Color and background	background background-color background-image background-position background-repeat color

Borders	border border-color border-width border-style border-top border-top-color border-top-width
Spacing	padding padding-bottom, padding-left, padding-right, padding-top margin margin-bottom, margin-left, margin-right, margin-top
Sizing	height max-height max-width min-height min-width width
Layout	bottom, left, right, top clear display float overflow position visibility z-index

Lists	list-style list-style-image list-style-type
-------	---

1.23 Selector

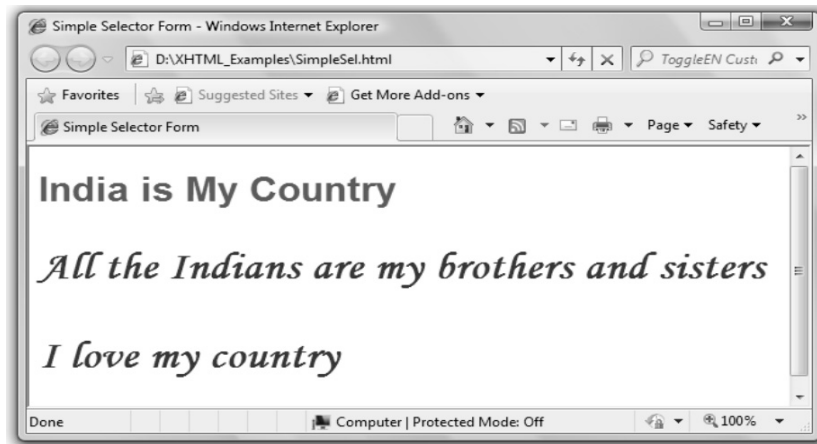
1.23.1 Simple Selector Form

- The simple selector form is a single element to which the property and value is applied.
- We can also apply property and value to group of elements.
- Following is an illustration for simple selector form.

HTML Document[SimpleSel.html]

```
<!DOCTYPE html>
<html >
<head>
  <title>Simple Selector Form</title>
  <style type="text/css">
    h1
    {
      font-family:Arial;
color:green;
    }
    h2,h3
    {
      font-family:Monotype Corsiva;
      color:red;
      font-size: 28pt;
    }
  </style>
</head>
<body>
  <h1>India is My Country</h1>
  <h2>All the Indians are my brothers and sisters</h2>
  <h3>I love my country</h3>
</body>
</html>
```

We can apply style to more than one selector

Output

Similarly the style can also be applied to the elements at specific positions.

For example :

```
body b p {font-size:18pt;}
```

Note that there are more than one element to which the style is applied and these elements are separated by the white spaces.

1.23.2 Class Selectors

- Using class selector we can assign different styles to the same element.
- These different styles appear on different occurrences of that element.

For example

HTML Document[ClassSel.html]

```
<!DOCTYPE html>
<html>
<head>
<title>Class Selector Form</title>
<style type="text/css">
  h1.RedText
  {
    font-family:Monotype Corsiva;
    color:red;
    font-size: 14pt;
  }
h1.BlueText
{
  font-family:Arial;
  color:blue;
  font-size: 10pt;
}
```

```
</style>
</head>
<body>
  <h1 class="RedText">India is My Country</h1>
  <h1 class="BlueText">All the Indians are my brothers and sisters</h1>
  <h3>I love my country</h3>
</body>
</html>
```

Output



Script explanation :

Note that in above given script we have used two different classes for the element h1. One class selector is for displaying the string in red color and another is for displaying the string in blue color. This definition is given in the head section. In the body section these class names appear in double quotes inside the h1 tag.

Example 1.23.1 Create a HTML document that displays a table of basketball scores at national games in which the team names have their respective team colors. The score of the leading/wining team should appear larger and in different font than the losing team. Use CSS.

Solution : Football.html

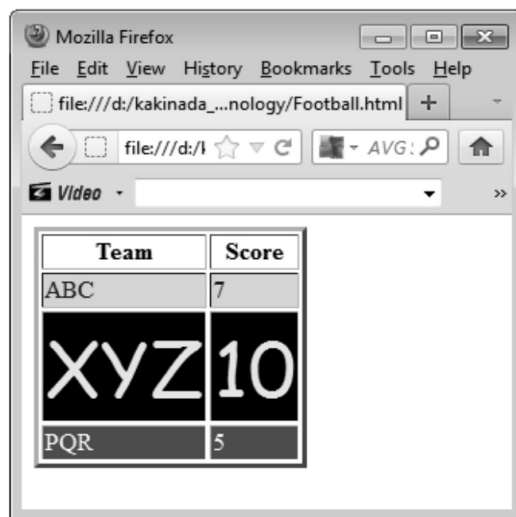
```
<!DOCTYPE html>
<html >
<head>
<style type="text/css">
tr.one
{
  background-color:pink;
  color:blue;
}
tr.two
{
```

```

font-family:cursive;
background-color:black;
color:yellow;
font-size:50px;
}
tr.three
{
background-color:green;
color:white;
}
</style>
</head>
<body>
<table border="3">
<th>Team</th>
<th>Score</th>
<tr class="one"><td>ABC</td><td>7</td>
</tr>
<tr class="two"><td>XYZ</td><td>10</td>
</tr>
<tr class="three"><td>PQR</td><td>5</td>
</tr>

</body>
</html>

```

Output**1.23.3 Generic Selectors**

- We define the class in generalised form.
- In the sense, that particular class can be applied to any tag.
- Here is the HTML document which makes use of such generic selector.

HTML Document [ClassSel1.html]

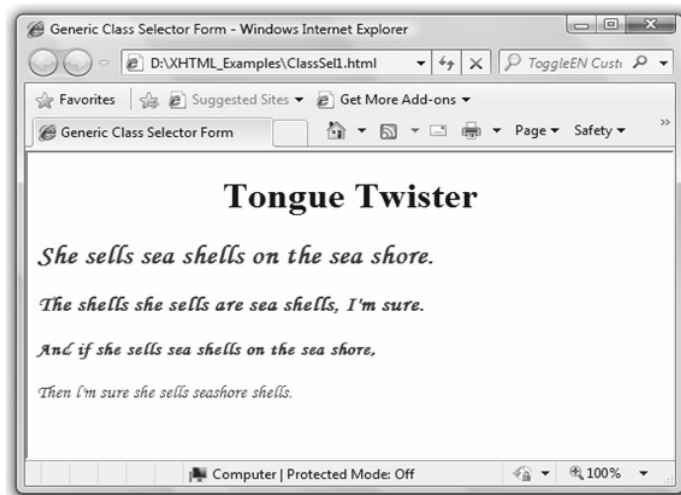
```

<!DOCTYPE html>
<html>
  <head>
    <title>Generic Class Selector Form</title>
    <style type="text/css">
      .RedText
      {
        font-family:Monotype Corsiva;
        color:red;

```

```
    }  
    </style>  
  
</head>  
  <body>  
    <center>  
      <h1> Tongue Twister</h1>  
    </center>  
      <h2 class="RedText">  
        She sells sea shells on the sea shore.  
      </h2>  
      <h3 class="RedText">  
        The shells she sells are sea shells, I'm sure.  
      </h3>  
      <h4 class="RedText">  
        And if she sells sea shells on the sea shore,  
      </h4>  
      <p class="RedText">  
        Then I'm sure she sells seashore shells.  
      </p>  
    </body>  
</html>
```

Output



Note that the class selector must be preceded by a dot operator.

1.23.4 Id Selectors

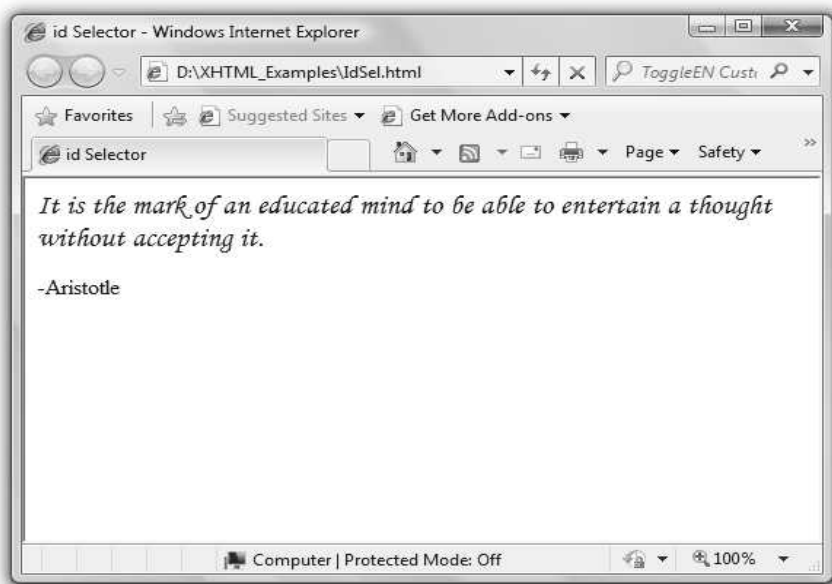
- The id selector is similar to the class selector but the only difference between the two is that class selector can be applied to more than one elements where as using the id selector the style can be applied to the one specific element.
- The syntax of using id selector is as follows -

```
#name_of_id {property:value list;}
```

- Following HTML document makes use of id selector.

HTML Document[IdSel.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title>id Selector</title>
    <style type="text/css">
      #top
      {
        font-family:Monotype Corsiva;
        color:blue;
        font-size:16pt;
      }
    </style>
  </head>
  <body>
    <div id="top">
      It is the mark of an educated mind to be able to
      entertain a thought without accepting it.
    </div>
    <p>
      -Aristotle
    </p>
  </body>
</html>
```

Output**1.23.5 Universal Selectors**

This selector is denoted by * (asterisk). This selector can be applied to all the elements in the document.

HTML Document[UniverSel.html]

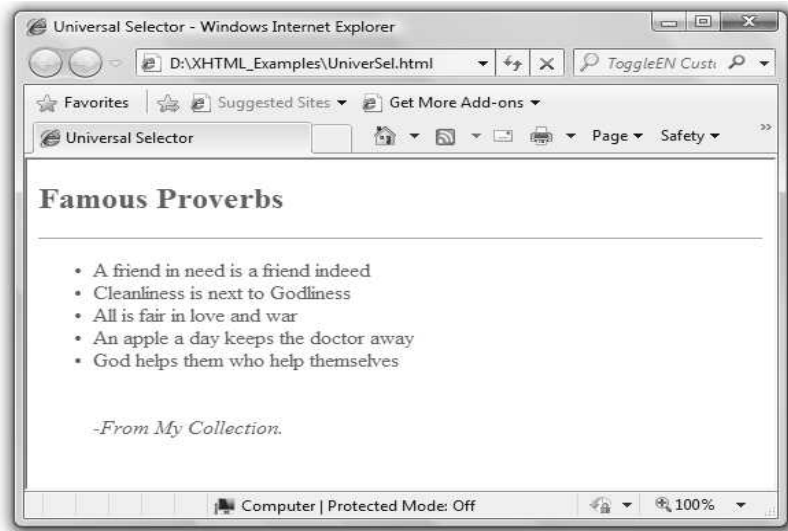
```
<!DOCTYPE html>
<html >
  <head>
    <title>Universal Selector</title>
    <style type="text/css">
      * {
        color:green;
      }
    </style>
  </head>
  <body>
    <h2> Famous Proverbs</h2>
    <hr/>
    <ul type="disc">
      <li>A friend in need is a friend indeed</li>
      <li>Cleanliness is next to Godliness</li>
      <li>All is fair in love and war</li>
      <li>An apple a day keeps the doctor away</li>
      <li>God helps them who help themselves</li>
```

```

<br/> <br/>
<em>
  -From My Collection.
</em>
</body>
</html>

```

Output



As we have defined the universal selector that sets the green color. Hence the text that is appearing on the above web page is in green color.

1.23.6 Attribute Selector

- CSS allows authors to specify rules that match elements which have certain attributes defined in the source document.
- The syntax is

[att] - Match when the element sets the "att" attribute, whatever the value of the attribute.

[att=val] - Match when the element's "att" attribute value is exactly "val".

- Example : In the following example all the <a> elements get selected with a target attribute.

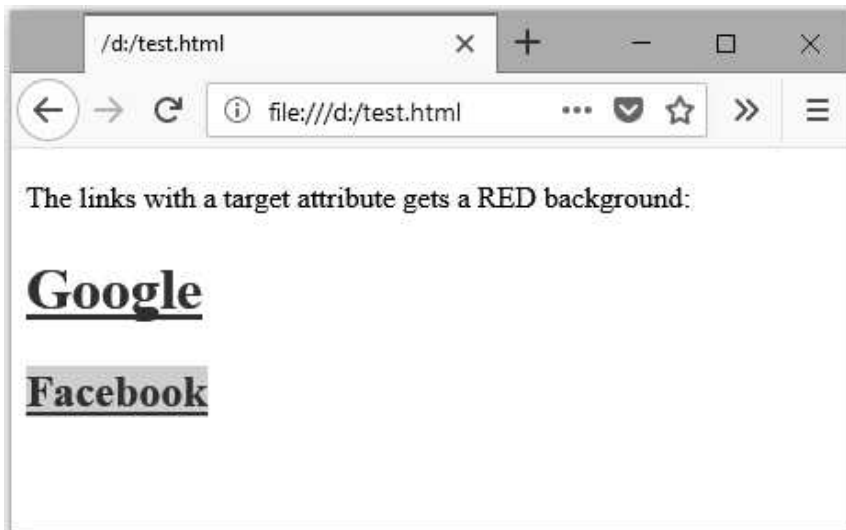
```

<!DOCTYPE html>
<html>
<head>
<style>
  a[target] {
    background-color: red;
  }
</style>

```

```
</head>
<body>
<p>The links with a target attribute gets a RED background:</p>
<h1><a href="https://www.google.com">Google</a> </h1>
<h2><a href="http://www.facebook.com" target="_blank">Facebook</a> </h2>
</body>
</html>
```

Output



1.23.7 Pseudo Classes

Using pseudo classes we can give special effects on the selectors. There some pseudo classes which are more commonly used and those are -

- Focus
- Hover
- Hyperlink

In the following XHTML documents we have used these pseudo classes.

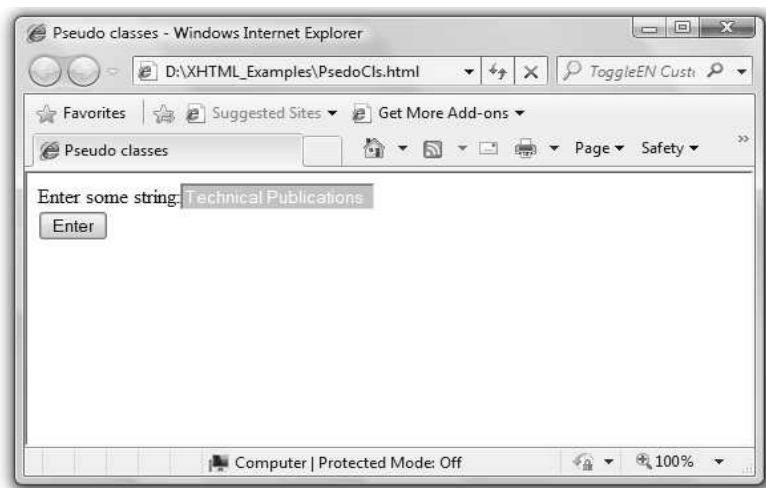
HTML Document[PseudoClasses.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title>Pseudo classes</title>
    <style type="text/css">
      input:focus
      {
        background-color:skyblue;
```

```
color:white;
}
</style>
</head>

<body>
  <form>
    Enter some string:<input type="text" name="my_txt"/>
    <br/>
    <input type="submit" value="Enter"/>
  </form>
</body>
</html>
```

Output



Let us now see one more illustration in which the pseudo class is used.

HTML Document[PseudoCls1.html]

```
<!DOCTYPE html>
<html>
<head>
  <title>Pseudo classes for hyperlinks</title>
  <style type="text/css">
    a.ColoredTxt:link {color: red}
    a.ColoredTxt:visited {color: blue}
    a.ColoredTxt:hover {color: green}

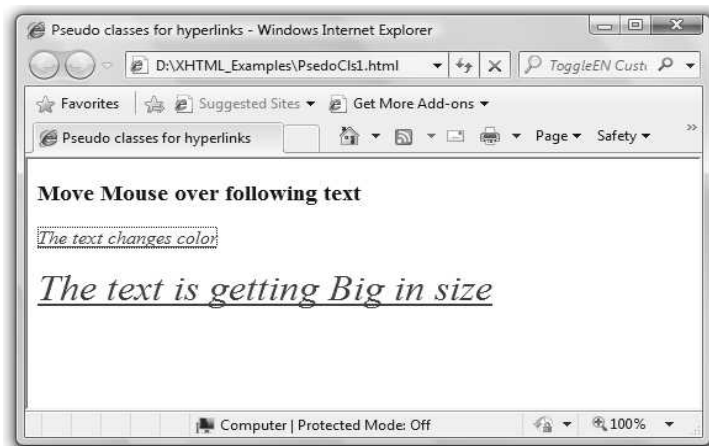
    a.BigTxt:link {color: red}
    a.BigTxt:visited {color: blue}

    a.BigTxt:hover {font-size: 200%}
```

We have defined two pseudo classes named ColoredTxt and BigTxt

```
</style>
</head>
<body>
  <h3>Move Mouse over following text</h3>
  <p>
    <em>
      <a class="ColoredTxt" href="mypage.html" target="_blank">The text
        changes color
      </a>
    </em>
  </p>
  <p>
    <em>
      <a class="BigTxt" href="mypage.html" target="_blank">The text is getting
        Big in size
      </a>
    </em>
  </p>
</body>
```

Output



1.23.8 Contextual Selector

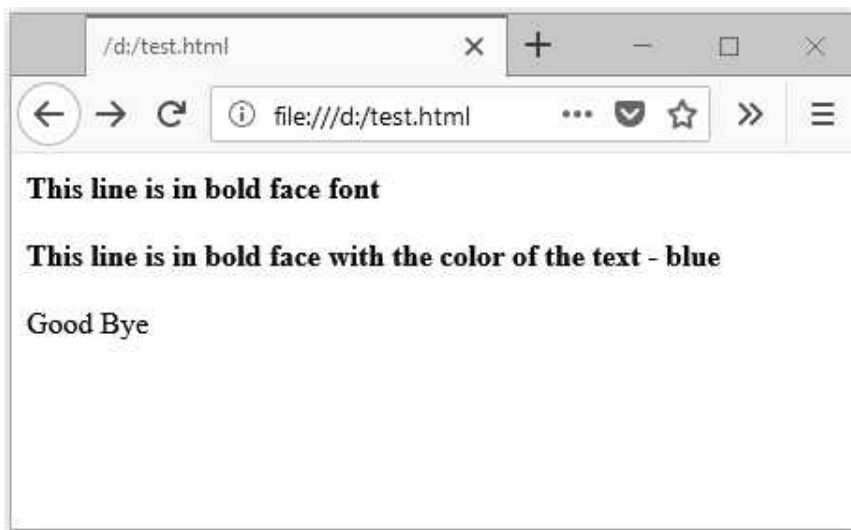
- "Contextual selectors" in CSS allow you to specify different styles for different parts of your document.
- You can assign styles directly to specific HTML tags, or you can create independent classes and assign them to tags in the HTML.

For example -

```
<!DOCTYPE html>
<html>
```

```
<head>
<style>
  p b {
    font-family: Times, serif; /* Font family */
    font-weight: bold; /* heavy faced type*/
    color: blue; /* blue colour of the text */
  }
</style>
</head>
<body>
<div><b>This line is in bold face font</b></div>
<p><b>This line is in bold face with the color of the text - blue </b></p>
<p> Good Bye</p>
</body>
</html>
```

Output



1.24 Style Sheets and HTML

There are three levels of cascading style sheets -

- Inline style sheet
- Document level style sheet
- External level style sheet

1.24.1 Inline Style Sheet

The inline cascading style sheet is a kind of style sheet in which the styles can be applied to HTML tags. This tag can be applied using following rule -

```
Tag
{
property: value
}
```

For example :

```
<p style="font-family: Arial;color:red" >
```

Here for the tag **p** two properties are used such as **font-family** and **color** and those are associated with the values such as arial and red respectively.

Note that if we want to use **more than one property then** we have to use separator such as **semicolon**. In the following HTML document we have used cascading style sheet-

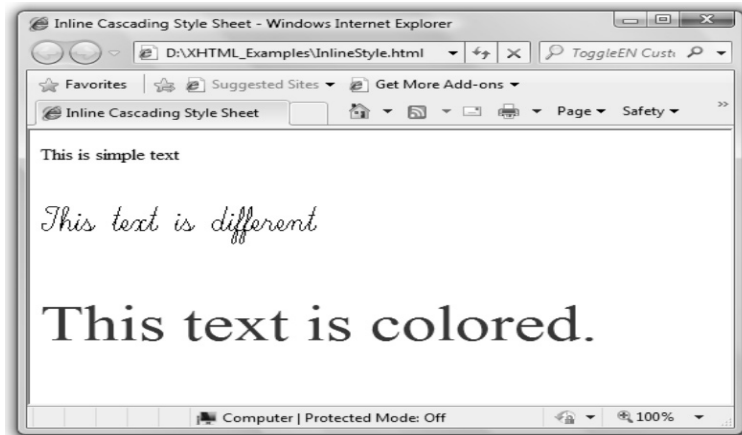
HTML Document [InlineStyle.html]

```
<!DOCTYPE html >
<html>
  <head>
    <title>Inline Cascading Style Sheet</title>
  </head>
  <body>
    <p>This is simple text</p>
    <p style="font-size: 30pt ;font-family:Script">This text is different </p>
    <p style="font-size: 40pt ;color:#ff0000">This text is colored.</p>
  </body>
</html>
```

Script explanation

- 1) In this document, in the body section the style sheets are created.
- 2) In this section first of all we have displayed a simple sentence This is simple text. There is no style for this sentence.
- 3) In the next line, we have applied style in which **font-size** is set to the size of 30 point and **font-family** is set by the font name "Script".
- 4) Then colored text will be displayed by **color:#ff0000**". The first two digits ff denote the red color, there is no green and blue color as the values next to ff are 00. Hence the text will be displayed in red.

This can be very well illustrated in output.

Output**1.24.2 Document Level Style Sheet**

- This type style sheet appears only in the head section and in the body section newly defined Selector tags are used with the actual contents.
- For example : In the following HTML script we have defined **h1**, **h2**, **h3** and **p** selectors. For each of these **selectors** different property and values are set. Such setting will help us to represent our web page in some decorative form.
- The most important thing while writing document level style sheet is that we should mention the

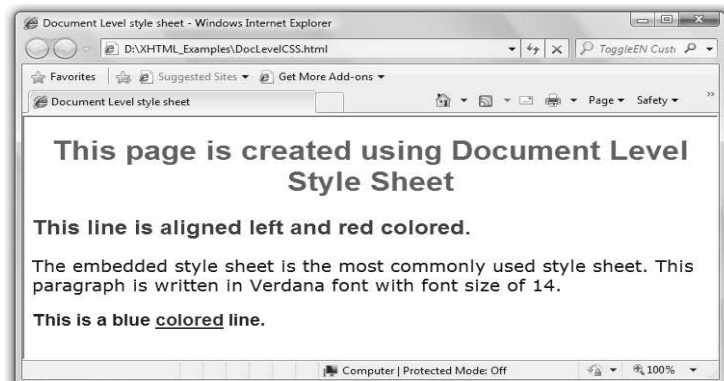
style type="text/css" in the head section. By this the browser will come to know that the program is making use of cascading style sheet.

HTML Document [DocLevelCSS.html]

```
<!DOCTYPE html>
<html >
<head>
  <title>Document Level style sheet</title>
  <style type="text/css">
    h1
    {
      font-family:Arial;
      color:green
    }
    h2
    {
      font-family:Arial;
      color:red;
```

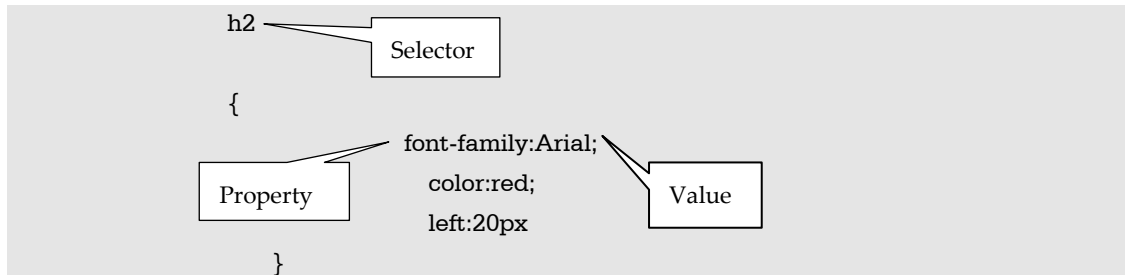
```
left:20px
}
h3
{
  font-family:arial;
  color:blue;
}
p
{
  font-size:14pt;
  font-family:verdana
}
</style>
</head>
<body>
  <h1>
<center>
  This page is created using Document Level Style Sheet
  </center>
  </h1>
  <h2>
  This line is aligned left and red colored.
  </h2>
  <p>
  The embedded style sheet is the most commonly used style sheet.
  This paragraph is written in Verdana font with font size of 14.
  </p>
  <h3>
  This is a blue <a href="colormame.html">colored</a> line.
  </h3>
</body>
</html>
```

Output



Script explanation

- 1) In above program, we have defined all the selectors in the head sections only. And these HTML elements are then used along with the web page contents.
- 2) The setting defined in the selectors will affect the web page contents. For example we have defined the selector h2 as,



and then in the body section we have written -

<h2> This line is aligned left and red colored.</h2>

- 3) Now as h2 defines font to be "Arial" with color as "red" having left alignment of 20 pixels, the sentence "This line is aligned left and red colored." will be displayed in Arial font, which is red colored and aligned from left by 20 pixels. Surely we can see this effect on our web browser.

1.24.3 External Stylesheet

- Sometimes we need to apply particular style to more than one web documents in such cases external style sheets can be used.
- The central idea in this type of style sheet is that the desired style is stored in one .css file. And the name of that file has to be mentioned in our web pages.
- Then the styles defined in .css file will be applied to all these web pages.
- Here is a sample program in which external style sheet is used.

Step 1 : Create an HTML document

HTML Document [ExtCSS.html]

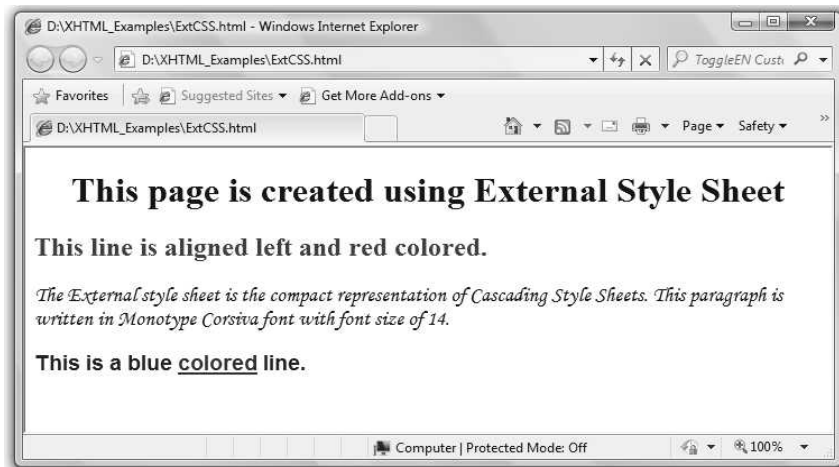
```
<!DOCTYPE html>
<html >
  <head>
    <link rel="stylesheet" type="text/css" href="ex1.css" />
  </head>
  <body>
    <h1 class="special"> <center> This page is created using External Style
    Sheet</center> </h1>
    <h2>
```

```
        This line is aligned left and red colored.
    </h2>
    <p>
    The External style sheet is the compact representation of Cascading Style Sheets.
    This paragraph is written in Monotype Corsiva font with font size of 14.
    </p>
    <h3>
        This is a blue <a href="colormame.html">colored</a> line.
    </h3>
</body>
</html>
```

Step 2 : Create a css file which contains the styles that can be applied to different HTML elements present in the above HTML document.

The cascading style sheet ex1.css can be

```
<!-- The file name ex1.css and can be opened in notepad.-->
h1
{
font-family:Arial
}
h2
{
font-family:times new roman;
color:red;
left:20px
}
h3
{
font-family:arial;
color:blue;
}
p
{
font-size:14pt;
font-family:Monotype Corsiva
}
```



Script explanation

When we want to link the external style sheet then we have to use `<link>` tag which is to be written in the head section.

- **link** tells the browser some file must be linked to the page.
- **rel=stylesheet** tells the browser that this hyperlink is a style sheet.
- **href=" "** denotes the path name of style sheet file.
- **type="text/css"** tells the browser that what it is reading is text which is affected by the cascading style sheet.

1.25 Style Rule Cascading and Inheritance

1.25.1 Style Rule Cascading

- When there are two or more values for the same property then conflict occurs. There are various levels of style sheets such as embedded CSS and external CSS. Out of which **embedded CSS** have **more precedence over the external CSS**.
- There are some sorting order rules that are used to decide the precedence. These are based on two things, importance (important or normal) of the property and origin (author origin, user origin and user agent origin). Let us first understand the terms important and origins.
- The keyword **!important** can be applied to specify that particular property is important.

For example :

```
p.myclass {font-family:Arial !important ;font-style:bold;}
```

specifies that the font being Arial is one property but font-style being bold is a important property.

- When the specification is marked by !important then it actually specifies the weight. The weight can be either important or normal.
- The origin can be author origin, user origin and user agent origin. These terms can be defined as follows -
 - **Author** is a person who writes the XHTML document and associated style sheets.
 - **User** is a person who interacts with user agent and use the web document along with the associated style sheets.
 - **User agent** is a program that interprets the web document written in scripting language and applies the style sheet according to the specification. For example, HTML is an user agent.
- The sorting order can be determined by applying following rules with highest precedence to lowest -
 1. Important declaration having user origin.
 2. Important declaration having author origin.
 3. Normal declaration having author origin.
 4. Normal declaration having user origin.
 5. Any declaration (either important or normal) with user agent (i.e. browser) origin.
- Following rules are also be used to decide the sorting order of the elements from highest precedence to lowest -
 1. **id** selectors have the highest precedence.
 2. **class** and **pseudo class** selectors
 3. Contextual selectors
 4. Universal selectors
- If the conflict still occurs then it can be resolved writing the properties in the specification. The rules that are defined later in the document tree have the highest precedence than those defined earlier.
- **Style rule cascading** can be defined as the process which is used to resolve the conflicts in the style specification. Fig. 1.25.1 shows the steps in CSS cascade.

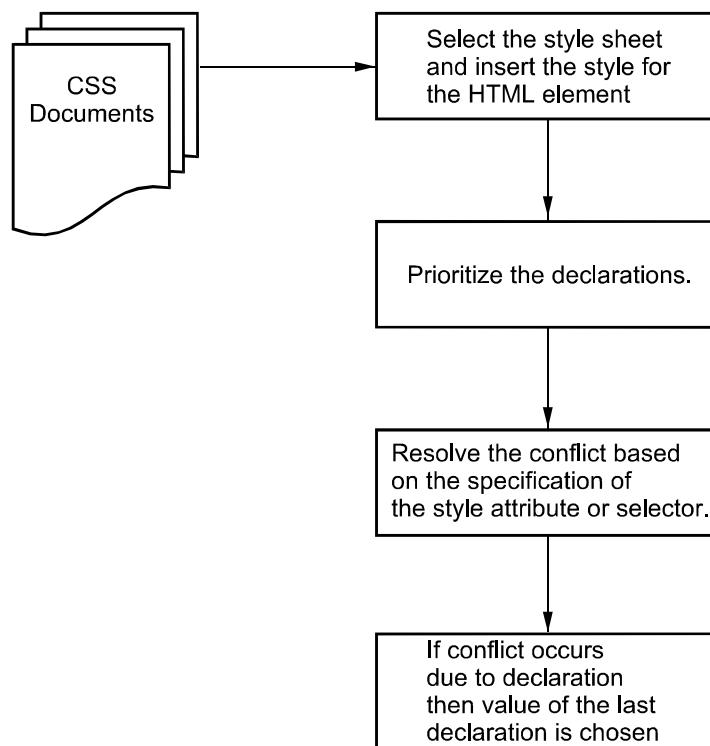


Fig. 1.25.1 Steps in CSS cascade

1.25.2 Inheritance

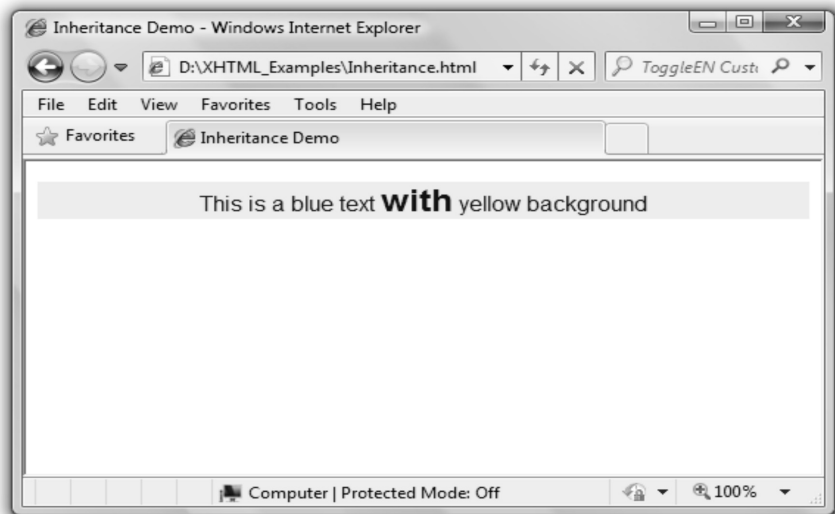
- In CSS, if a property of an element has no associated declarations then using inheritance technique the value for this property can be obtained.
- In an HTML document, the tags have parent-child relationships. For example, <title> tag is always inside the <head> tag, so the <head> tag is the parent of the <title> tag. Similarly <p> tag is defined inside the <body> tag. Then <body> becomes parent of <p> tag.
- In cascading style sheets, inheritance is based on **tree structure** of the document itself.
- An element inherits the value for its properties by checking the parent element. If the parent has the value for the property demanded by the child then this property value can be inherited. If the parent does not have this value, then its parent (i.e. grandparent) is checked. If this parent has no such property value then its parent is checked. This process is continued until a root element is obtained.
- Thus the search will be made in upward direction from the parent of the element to the root of the tree or to the <html> tag of the document.

- Inheritance can be defined as a concept in which child always inherits the properties of parent tag along with its own properties.
- Following is an HML document in which inheritance is demonstrated.

HTML Document

```
<!DOCTYPE html>
<html>
<head>
  <title>Inheritance Demo</title>
  <style type="text/css">
    p
    {
      font-family:Arial;
      color:blue;
      background-color:yellow;
    }
    strong
    {
      font-size:24px;
    }
  </style>
</head>
<body>
  <center>
    <p>This is a blue text <strong>with</strong> yellow background</p>
  </center>
</body>
</html>
```

Output



Script explanation :

In above, script <p> tag has some properties as **Arial** font, blue font color and **yellow** background. The tag has only one property can that is font-size which is assigned by the value **24px**. But in the body section, tag is the child of <p> tag. And note that inherits the properties of <p> tag. Hence the text defined under tag is also blue colored with yellow background.

Now just modify by the above script as follows and observe the sample output.

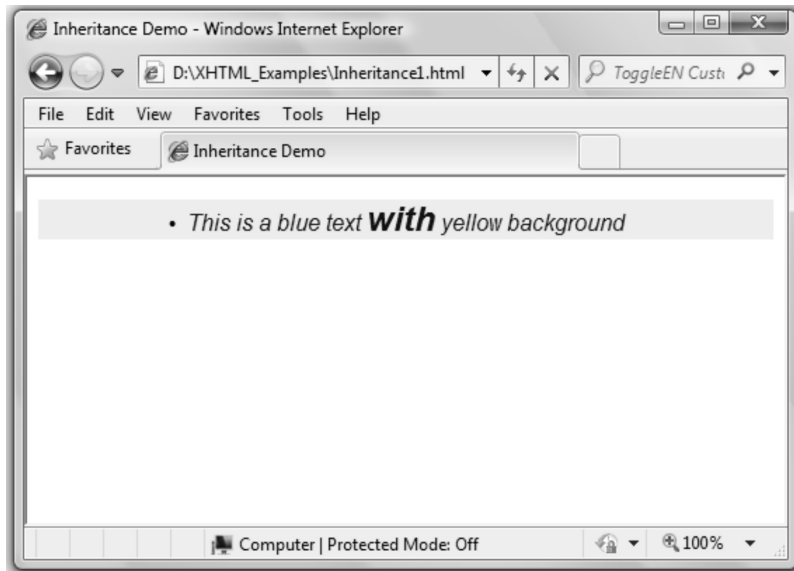
HTML Document [inheritance1.html]

```
<!DOCTYPE html>
<html>
<head>
  <title>Inheritance Demo</title>
  <style type="text/css">
    li
    {
      font-style:italic
    }
    p
    {
      font-family:Arial;
      color:blue;
      background-color:yellow
    }
    strong
    {
      font-size:24px
    }
  </style>
</head>

<body>
  <center>
    <li>
      <p>This is a blue text <strong>with</strong> yellow background</p>
    </li>
  </center>
</body>
</html>
```



Added tag as grand parent for tag.

Output

Note that the above style sheet contains the nested tags -

```
<li>
.....<p>
.....<strong>
.....</strong>
.....</p>
</li>
```

The tag contains the italic property. Hence the text inside the tag becomes italic. That means tag has inherited the italic property from its grandchild.

1.26 Text Properties

SPPU : March-20, Marks 5

The font properties can be setting different types of fonts, styles and sizes.

1.26.1 Font Families

The **font-family** denotes different types of fonts such as Arial, Times New Roman, Script, monospace and so on. Following are some examples of generic fonts

Generic Font Name	Example
sans-serif	Arial, Helvetica, Futura
cursive	Zapf-chancery
fantasy	Critter, Cottonwood
monospace	Courier, Prestige
serif	Times New Roman, Garamond

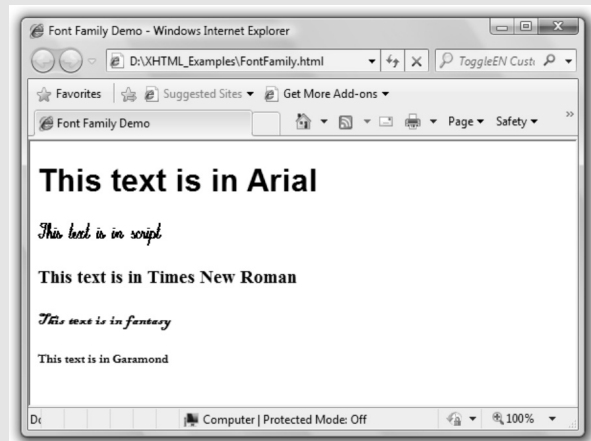
Let us see usage of **font-family** property in HTML document.

HTML Document[FontFamily.html]

```
<!DOCTYPE html>
<html>
<head>
<title>Font Family Demo</title>
<style type="text/css">
  h1
  {
    font-family:Arial;
  }
  h2
  {
    font-family:script;
  }
  h3
  {
    font-family:'Times New Roman';
  }
  h4
  {
    font-family:fantasy;
  }
  h5
  {
    font-family:Garamond;
  }
</style>
</head>
<body>
<h1>This text is in Arial </h1>
<h2>This text is in script </h2>
<h3>This text is in Times New Roman</h3>
<h4> This text is in fantasy</h4>
<h5> This text is in Garamond</h5>

</body>
</html>
```

Output



If the font name has more than one word then the name should be enclosed by the single quotes. Although quotes are not mandatory, it is a good practice to use quotes for specifying the font name.

1.26.2 Font Sizes

This property is used to specify the size of the font. One can specify the font size in points (pts), pixels (px) or in percentage (%). We can also specify the font size using the relative values such as **small**, **medium**, **large**. Use of font size in such a way is relative. And the disadvantage of using such relative sizes is that one cannot have the strict control over the font-size. Different browsers may have different font size values.

In the following HTML document we will use the font-size property.

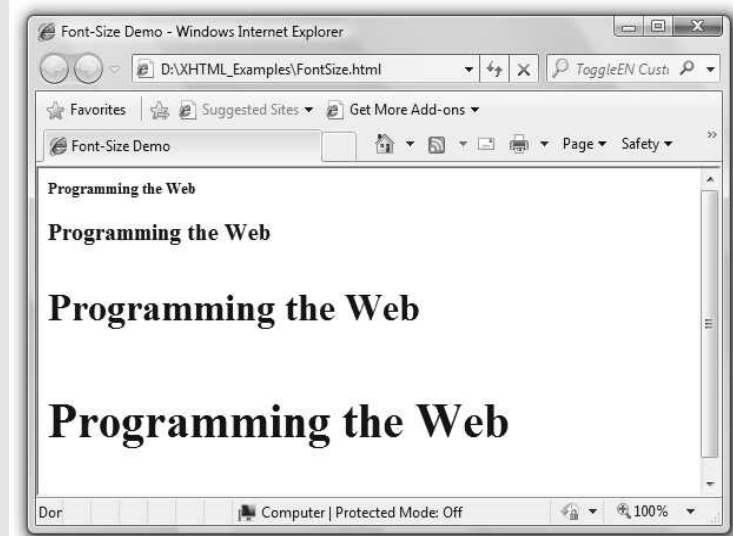
HTML Document[FontSize.html]

```
<!DOCTYPE html>
<html>
  <head>

<title>Font-Size Demo</title>
  <style type="text/css">
    h1
    {
      font-size:10pt;
    }
    h2
    {
      font-size:20px;
    }
    h3
    {
      font-size:xx-large;
    }
    h4
    {
      font-size:250%;
    }

  </style>
</head>
<body>
<h1>Programming the Web</h1>
  <h2>Programming the Web</h2>
  <h3>Programming the Web</h3>
  <h4>Programming the Web</h4>
</body>
</html>
```

Output



1.26.3 Font Variants

The font variation can be achieved by making setting the font in upper case or in lower case.

We use **font-variant** property for this purpose.

Value	Meaning
normal	The font can be displayed in normal form.
small-caps	The font can be displayed in small capital letters.

HTML Document [FontVariant.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title>Font-Variant Demo</title>
    <style type="text/css">
      h1
    {
      font-variant:small-caps;
    }
  </style>
</head>
<body>
  <h1>Programming the Web</h1>
</body>
</html>
```

Output



1.26.4 Font Styles

Various font styles are -

- normal
- italic
- oblique

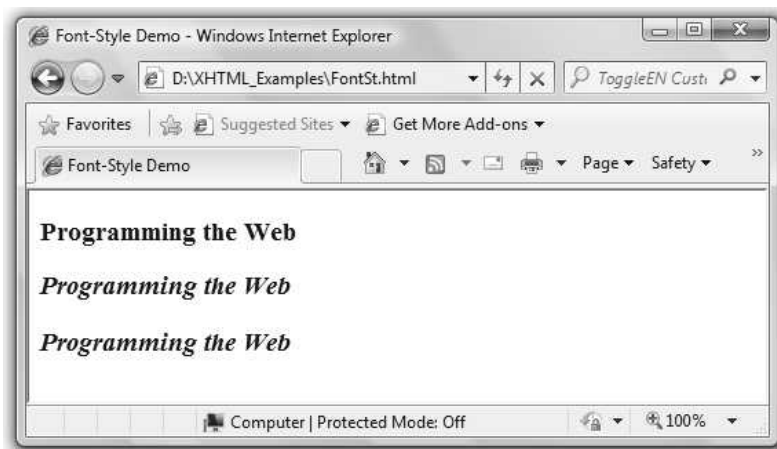
The illustration is as given below -

HTML Document[FontSt.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title>Font-Style Demo</title>
```

```
<style type="text/css">
  h3.normal { font-style:normal;}
  h3.italic { font-style:italic;}
  h3.oblique { font-style:oblique;}
</style>
</head>
<body>
  <h3 class="normal">Programming the Web</h3>
  <h3 class="italic">Programming the Web</h3>
  <h3 class="oblique">Programming the Web</h3>
</body>
</html>
```

Output



1.26.5 Font Weights

Various font styles are -

- Normal (by default)
- Bold
- Bolder
- Lighter

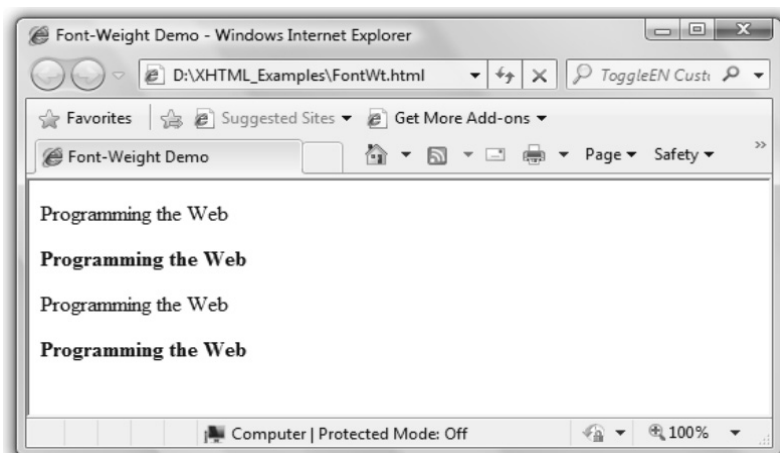
The illustration is as given below –

HTML Document[FontWt.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title>Font-Weight Demo</title>
    <style type="text/css">
```

```
p.normal { font-weight:normal;}
p.bold { font-weight:bold;}
p.bolder { font-weight:bolder;}
p.lighter { font-weight:lighter;}
</style>
</head>
<body>
  <p class="normal">Programming the Web</p>
  <p class="bold">Programming the Web</p>
  <p class="lighter">Programming the Web</p>
  <p class="bolder">Programming the Web</p>
</body>
</html>
```

Output



1.26.6 Font Shorthands

- We can specify more than one font properties in a list.

For example

HTML Document[Fontshrthnd.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title>Font Properties</title>
    <style type="text/css">
      p.myfont {font-family:Courier New;
                font-style:italic;
                font-weight:bold;
```

```
font-size:28pt;
}
</style>
</head>
<body>
  <p class="myfont">I Love my Country!!!</p>
</body>
</html>
```

Output



1.26.7 Text Decoration

Using text decoration property we can include special features in the text. Various properties of text decoration are,

- underline
- overline
- line-through

HTML Document[TxtDecor.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title>Text Decoration</title>
    <style type="text/css">
      p.linethrough {text-decoration:line-through;}
      p.underline {text-decoration:underline;}
      p.overline {text-decoration:overline;}
    </style>
  </head>
  <body>
    <p class="linethrough">I Love my Country!!!</p>
    <p class="underline">I Love my Country!!!</p>
    <p class="overline">I Love my Country!!!</p>
  </body>
</html>
```

```

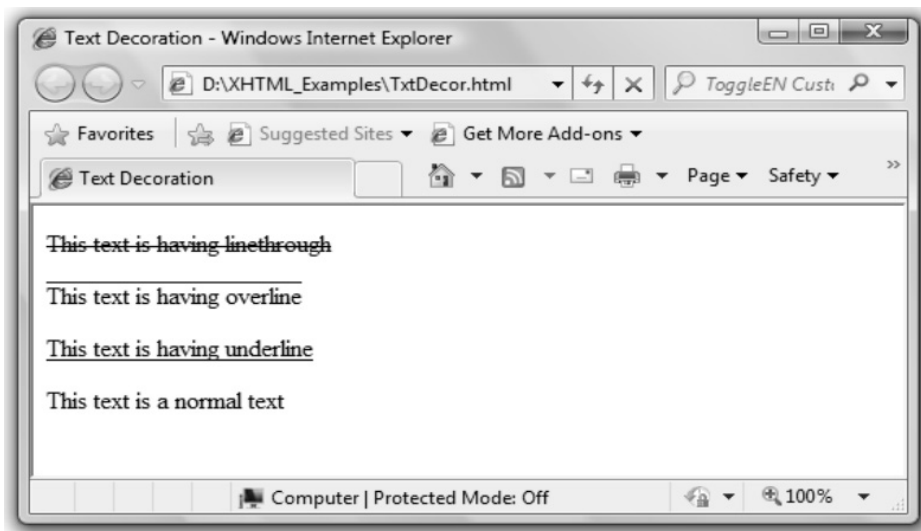
    p.normal {text-decoration:normal;}

</style>
</head>
<body>
    <p class="linethrough"> This text is having linethrough </p>
    <p class="overline"> This text is having overline </p>
    <p class="underline"> This text is having underline </p>
    <p class="normal"> This text is a normal text </p>

</body>
</html>

```

Output



1.26.8 Alignment of Text

Using cascading style sheets we can align the text. This alignment can be done using following properties such as -

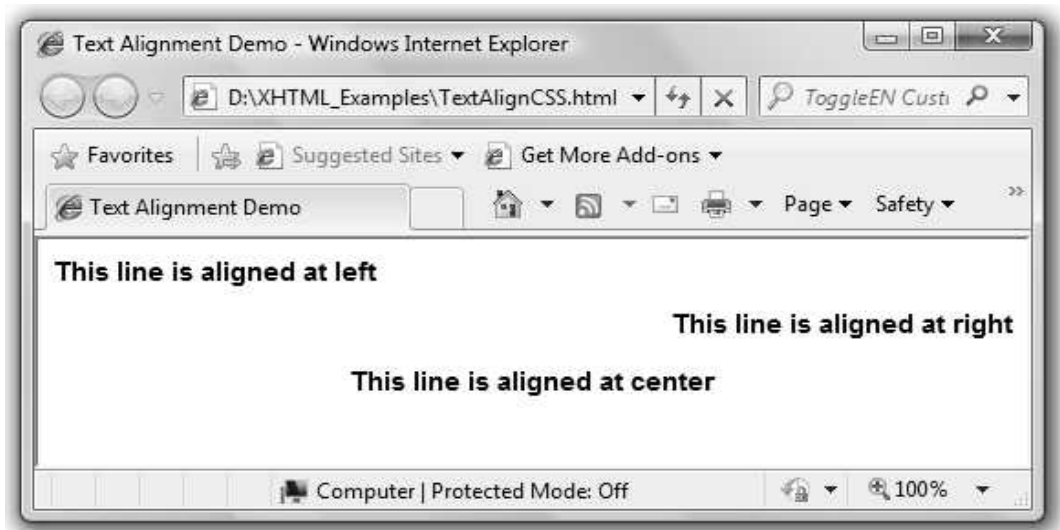
Property	Value	Meaning
text-align	left	Aligning the text to the left.
text-align	right	Aligning the text to the right
text-align	center	Alignment of text at the centre.
text-indent	value in inches	Desired indentation of the text can be applied.

Following is script which uses various text alignment properties.

HTML Document [TextAlignCSS.html]

```
<!DOCTYPE html>
<html>
  <head>
    <title>Text Alignment Demo</title>
    <style type="text/css">
      h1
      {
        font-family:Arial;
        font-size:15px;
        text-align:left;
      }
      h2
      {
        font-family:Arial;
        font-size:15px;
        text-align:right;
      }
      h3
      {
        font-family:Arial;
        font-size:15px;
        text-align:center;
      }
    </style>
  </head>

  <body>
    <h1>This line is aligned at left</h1>
    <h2>This line is aligned at right</h2>
    <h3>This line is aligned at center</h3>
  </body>
</html>
```

Output

Example 1.26.1 *How to you create a multi colour text on a web page using CSS ?*

SPPU : March-20, Marks 5

Solution :

```
<!DOCTYPE html>
<html>
<head>
  <title>Rainbow Effect</title>
</head>
<body>
<h1>
  <span style="color:violet">P</span>
  <span style="color:indigo">r</span>
  <span style="color:green">o</span>
  <span style="color:blue">g</span>
  <span style="color:orange">r</span>
  <span style="color:yellow">a</span>
  <span style="color:red">m</span>
</h1>
</body>
</html>
```

1.27 Bootstrap

SPPU : May-18, Dec.-19, Marks 9

- Bootstrap is open source, front end web framework. It is used for designing web-site and web applications.
- Bootstrap helps the developer to create responsive designs. Responsive web design means creating the web site which automatically adjusts itself to look good on all the devices. These devices range from small phones to large desktops screens.
- Bootstrap includes HTML, CSS and JavaScript based design templates which includes forms, buttons, tables, navigation, modals and so on.

Features of Bootstrap

Setting up Environment

There are two ways to use Bootstrap in a webpage.

- 1) The first way is to use a CDN or **C**ontent **D**elivery **N**etwork. Using bootstrap CDN means that we will not download and store the bootstrap files in our server or local machine. We will just include the bootstrap CSS and JavaScript links on our page.
- 2) The second way is downloading and storing a copy Bootstrap files in our web server or local machine. We can download the Bootstrap from <https://getbootstrap.com/>

I have created a folder named **bootstrap-3.3.7** and downloaded my Bootstrap in this folder. After getting downloaded I can see three folders in it namely css, fonts and js

- If we want to **use the CDN** then we will include Bootstrap CSS as follows -

```
<link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">
```

- If we want to use our **own downloaded copy** then we will use

```
<link rel="stylesheet"
href="bootstrap-3.3.7/css/bootstrap.min.css">
```

- Similarly the jquery can be used as

```
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.0.0/jquery.min.js"> </script>
Or
<script src="jquery-3.0.0.min.js"> </script>
```

Components of Bootstrap

1. **HTML 5** : The Bootstrap uses the HTML elements and CSS properties, hence the code is contained within the <doctype>.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
... ..
        Script declarations
</head>
<body>
...
...
</body>
</html>
```

2. **The mobile-first** : Bootstrap 3 is designed to be responsive to mobile devices. Mobile-first styles are part of the core framework.

For proper rendering the <meta> tag inside the <head> section is specified. It is specified as follows -

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

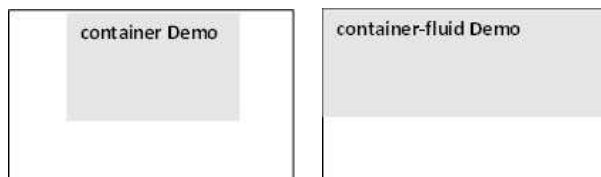
The width=device-width part sets the width of page which is based on the screen-width of the device.

The initial-scale = 1 part sets the initial zoom level when the page is loaded on the browser.

3. Container

The container element is for wrapping the site contents. There are two classes for the container -

- i) **Container** : It provides a responsive fixed width container.
- ii) **Container-fluid** : It provides full width container occupying the entire width of the viewport.

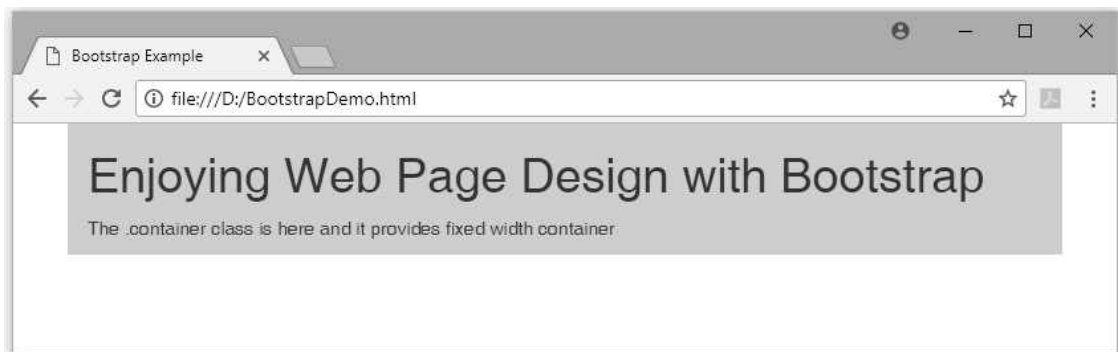


How to write first Bootstrap Code ?

Step 1 : Open a Notepad and type the following code. Save it using the filename extension .html

BootstrapDemo.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"></script>
  <script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"></script>
</head>
<body>
<div class="container" style="background-color:grey;">
  <h1 >Enjoying Web Page Design with Bootstrap</h1>
  <p>The .container class is here and it provides fixed width container</p>
</div>
</body>
</html>
```

Output

Similarly if we change the <div> element class to **container-fluid** we will get the content to be occupied the entire web page. It can be demonstrated by modifying the above script as follows -

BootstrapDemo.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
```

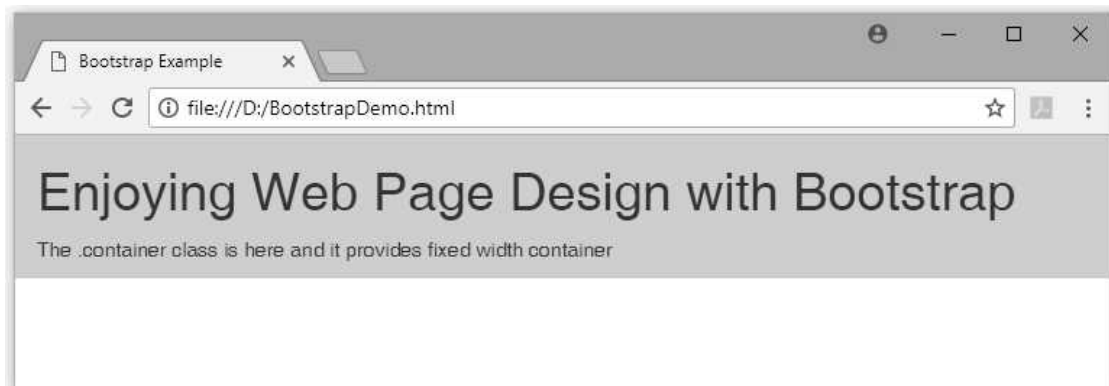
```
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">

<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"></script>
<script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"></script>
</head>
<body>

<div class="container-fluid" style="background-color:grey;">
  <h1 >Enjoying Web Page Design with Bootstrap</h1>
  <p>The .container class is here and it provides fixed width container</p>
</div>

</body>
</html>
```

Output



1.27.1 Grid System

- In Bootstrap there are 12 columns across the page. It is not always necessary to have 12 columns we can group the columns and wider columns can be created.
- The Bootstrap's Grid system is responsive. That means as we resize the browser window, the columns arrangement gets changed as per the screen size.
- There are four classes of Bootstrap's Grid system and those are -

Class	Purpose
xs	It is used for phones
sm	It is used for tablet screen
md	It is used for small laptops
lg	It is for laptops and desktops.

The above classes can be combined to create more dynamic and flexible display.

Structure of Bootstrap Grid

The typical structure of Bootstrap Grid is as follows -

```
<div class="row">
  <div class="col-*-*"></div>
  <div class="col-*-*"></div>
</div>
<div class="row">
  <div class="col-*-*"></div>
  <div class="col-*-*"></div>
  <div class="col-*-*"></div>
</div>
...
...
```

In above structure we adding the row by

```
<div class="row">
```

And then adding the column for that row using

```
<div class="col-*-*"></div>
```

Here first * indicates the class as xs or sm or md or so on. The second * indicates the number of columns. We can add at the most 12 columns.

Let us understand how to use Grid system with a simple example

GridDemo.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">

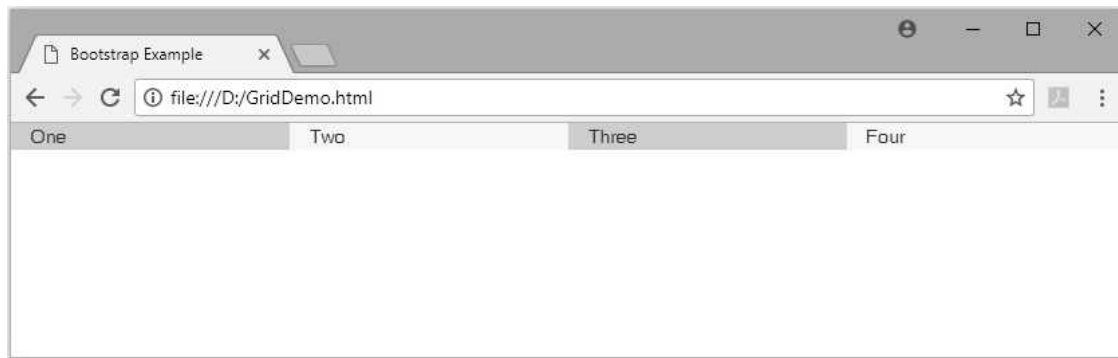
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"></script>
  <script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"></script>

</head>
<body>

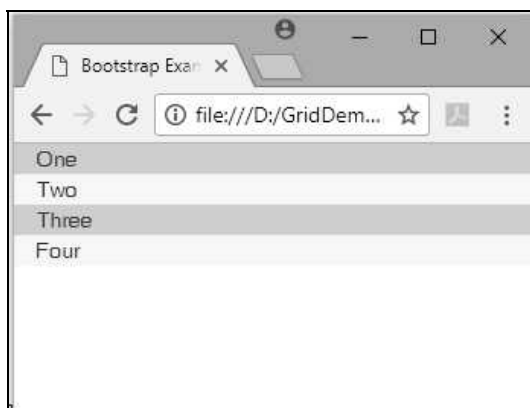
<div class="container-fluid">
<div class="row">
  <div class="col-sm-3" style="background-color:grey;">One</div>
  <div class="col-sm-3" style="background-color:aqua;">Two</div>
```

```
<div class="col-sm-3" style="background-color:grey;">Three</div>
<div class="col-sm-3" style="background-color:aqua;">Four</div>
</div>
</div>
</body>
</html>
```

Output



If we resize the browser window we get the columns to be stacked one upon other



Similarly we can create the columns of unequal width, if we change the **<col-sm>**. For example -

```
<div class="col-sm-8" style="background-color:grey;">One</div>
<div class="col-sm-4" style="background-color:aqua;">Two</div>
```

1.27.2 Typography

- In this section we will learn how to display the text in different style using Bootstrap elements.
- By default font size is 14 px.
- The HTML elements will be displayed with different style in Bootstrap.

Heading

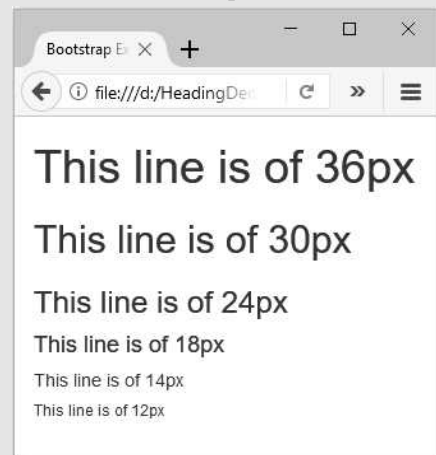
The tags <h1> to <h6> are used to display the heading. For example

HeadingDemo.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"> </script>
  <script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"> </script>

</head>
<body>
<div class="container">
  <h1> This line is of 36px </h1>
  <h2> This line is of 30px </h2>
  <h3> This line is of 24px </h3>
  <h4> This line is of 18px </h4>
  <h5> This line is of 14px </h5>
  <h6> This line is of 12px </h6>
</div>
</body>
</html>
```

Output



The small element

The <small> element is used to create the small light text in heading. For example

HeadingDemo.html

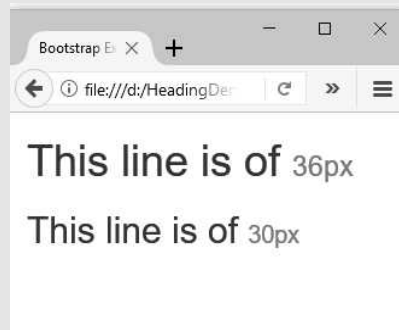
```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">
```

```

<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"> </script>
<script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"> </script>

</head>
<body>
<div class="container">
    <h1> This line is of <small>36px</small> </h1>
    <h2> This line is of <small>30px</small> </h2>
</div>
</body>
</html>

```

Output**The mark element**

The <mark> element can be used to highlight the important text present in the line. Here is the simple illustration

MarkDemo.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <title>Bootstrap Example</title>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">

    <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"> </script>
    <script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"> </script>

</head>
<body>

<div class="container">
    <p> This line is an <mark>important</mark> message </p>
</div>
</body>
</html>

```

Output**The <pre> element**

For displaying the text on multiple lines we can use <pre> tag. Here is the simple example of this tag.

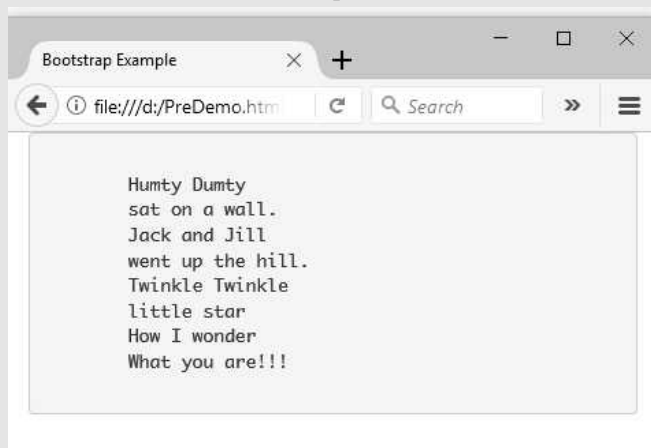
PreDemo.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">

  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"> </script>
  <script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"> </script>

</head>
<body>

<div class="container">
  <pre>
Humty Dumty
sat on a wall.
Jack and Jill
went up the hill.
Twinkle Twinkle
little star
How I wonder
What you are!!!
  </pre>
</div>
</body>
</html>
```

Output

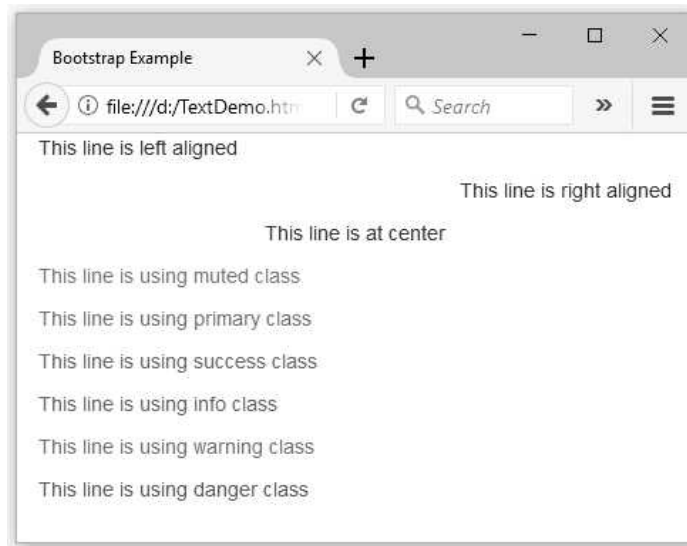
Colors and Background

- There are some contextual classes using which specific meaning can be represented through the colors. The classes for text colors are -
.text-muted, .text-primary, .text-success, .text-info, .text-warning, .text-danger
- Similarly we can align the text to left, right or center using text-left, text-right and text-center.
- These are called contextual classes which provides the meaning through text.
- Let us use these classes in following demo application

TextDemo.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">

  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"> </script>
  <script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"> </script>
</head>
<body>
  <div class="container">
    <p class="text-left">This line is left aligned</p>
    <p class="text-right">This line is right aligned</p>
    <p class="text-center">This line is at center </p>
    <p class="text-muted">This line is using muted class</p>
    <p class="text-primary">This line is using primary class</p>
    <p class="text-success">This line is using success class</p>
    <p class="text-info">This line is using info class</p>
    <p class="text-warning">This line is using warning class</p>
    <p class="text-danger">This line is using danger class</p>
  </div>
</body>
</html>
```

Output

Similarly we can set background color for these classes as follows :

TextBGDemo.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">

  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"> </script>
  <script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"> </script>

</head>
<body>
<div class="container">
  <p class="bg-primary">This line is using primary class</p>
  <p class="bg-success">This line is using success class</p>
  <p class="bg-info">This line is using info class</p>
  <p class="bg-warning">This line is using warning class</p>
  <p class="bg-danger">This line is using danger class</p>
</div>
</body>
</html>
```

Output**1.27.3 Tables**

The `<table>` tag is used to create the table. The `<tr>` tag is used for rows and `<td>` tag is used for columns. The heading in the table can be denoted within `<thead><th>` `</th>` `</thead>` tag while remaining contents of the table are specified in `<tbody>` tag.

The following example demonstrates the use of table tag.

TableDemo1.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">

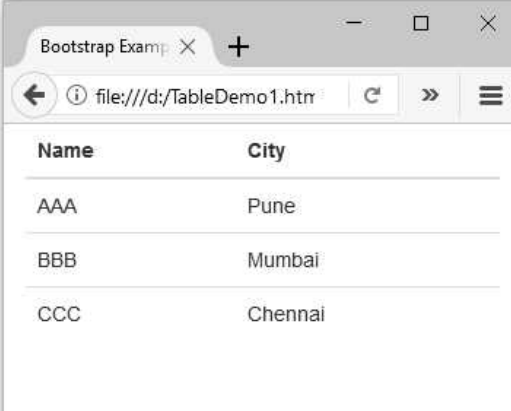
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"> </script>
  <script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"> </script>
</head>
<body>
<div class="container">
  <table class="table">
    <thead>
      <tr><th> Name </th><th> City </th></tr>
    </thead>
    <tbody>
      <tr>
        <td>AAA</td>
        <td>Pune</td>
      </tr>
      <tr>
```

```

      <td>BBB</td>
      <td>Mumbai</td>
    </tr>
    <tr>
      <td>CCC</td>
      <td>Chennai</td>
    </tr>
  </tbody>
</div>
</body>
</html>

```

Output



Name	City
AAA	Pune
BBB	Mumbai
CCC	Chennai

For having the grids to the table, we simply change the

```
<table class="table">
```

to

```
<table class="table table-bordered">
```

1.27.4 Images

- For displaying the images the `<img src="">` tag is used. You have to specify the image file name within the double quotes.
- There are three classes that are used to display the image. These classes are -
 1. **img-rounded** : The image is displayed in a border-circle of radius 6px.
 2. **img-circle** : The image is displayed in circle of radius 500px
 3. **img-thumbnail** : The image is displayed in a rectangle having padding and grey border.

Here is illustrative example of displaying images.

ImgDemo.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"> </script>
  <script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"> </script>

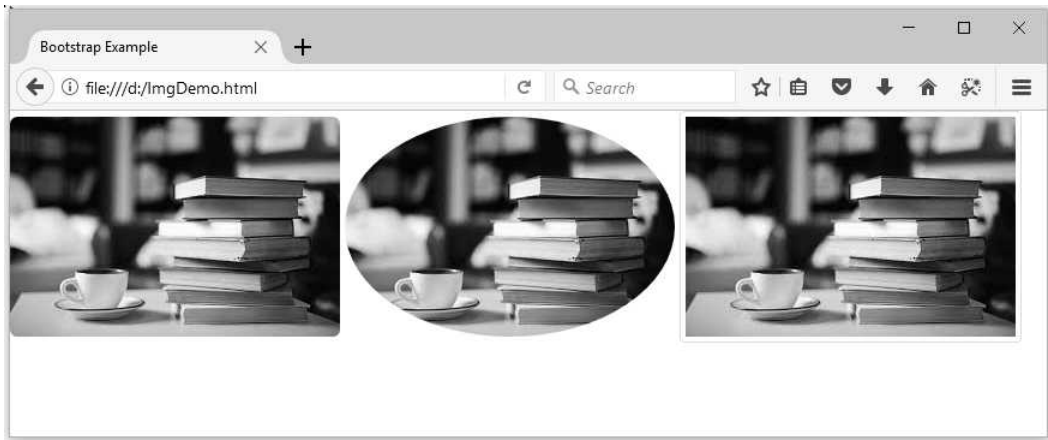
```

```

</head>
<body>
<img src = "bookImg.jpg" class = "img-rounded">
<img src = "bookImg.jpg" class = "img-circle">
<img src = "bookImg.jpg" class = "img-thumbnail">
</body>
</html>

```

Output



1.27.5 Button

- In Bootstrap we can create the buttons of different styles. Various button classes are used as follows -
 .btn, .btn-default, .btn-primary, .btn-success, .btn-info, .btn-warning,
 .btn-danger, .btn-link
- The example code is as follows -

ButtonDemo.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
    href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"></script>
  <script
    src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"></script>
</head>
<body>

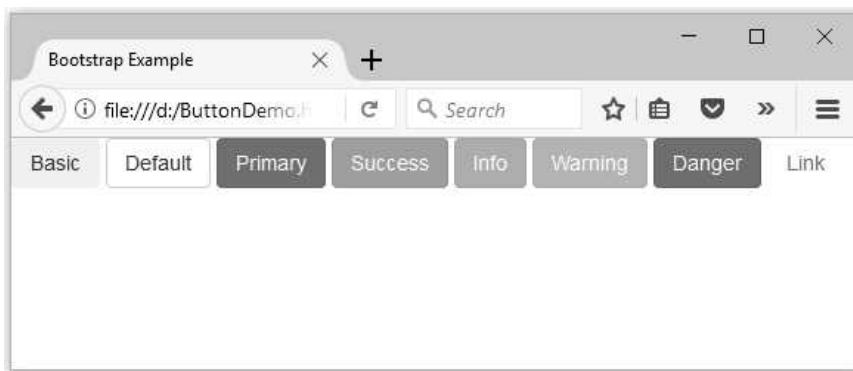
```

```

<button type="button" class="btn">Basic</button>
<button type="button" class="btn btn-default">Default</button>
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-link">Link</button>
</body>
</html>

```

Output



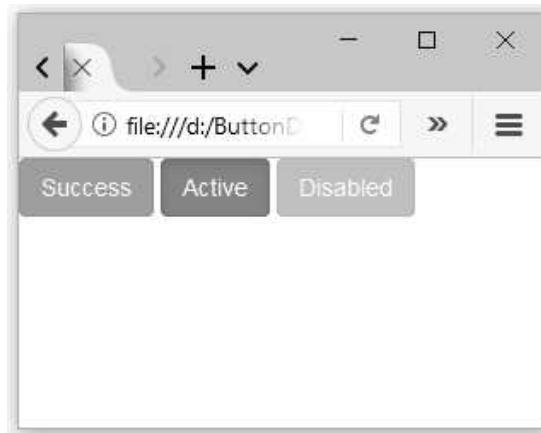
We can also create active or disabled buttons using active and disabled class. The illustration is as follows -

ButtonDemo1.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"> </script>
  <script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"> </script>
</head>
<body>
  <button type="button" class="btn btn-success">Success</button>
  <button type="button" class="btn btn-success active">Active</button>
  <button type="button" class="btn btn-success disabled">Disabled</button>
</body>
</html>

```

Output**1.27.6 Form**

- In Bootstrap there are three types of form layouts.
 1. Vertical Form
 2. Horizontal Form
 3. Inline Form
- For using the form we must add the <div> element as **<div class="form-group">**. The required controls such as <input>, <select>, <textarea> and so on can be added within this <div> class.
- Let us discuss and learn how to create the above mentioned types of forms –

1. Vertical or Basic Form

To create the vertical or basic form use following steps –

Step 1 : Add a role to form element as -

```
<form role="form">
```

Step 2 : Inside the <div class="form-group"> wrap all the controls

Step 3 : While adding any textual control such as <text>, <textarea> and <select> add the class **"form-control"**

Here is example code –

FormDemo1.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
```

```
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">

<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"></script>
<script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"></script>

</head>
<body>
<form role = "form">

    <div class = "form-group">

    <label for = "name">User Name</label>

    <input type = "text" class = "form-control" id = "name">
    </div>

    <div class = "form-group">

    <label for = "pwd">Password</label>

    <input type = "password" class = "form-control" id = "pwd">

    </div>

    <div class = "checkbox">

    <label><input type = "checkbox"> Show Me Always</label>

    </div>

    <button type = "submit" class = "btn btn-default">Submit</button>

</form>
</body>
</html>
```

Output

The screenshot shows a web browser window with the title 'Bootstrap Example'. The address bar shows a file path: 'file:///d:/FormDe'. The browser's search bar contains the word 'Search'. The form itself is styled with Bootstrap classes. It features a 'User Name' label above a text input field, a 'Password' label above a password input field, a checkbox labeled 'Show Me Always', and a 'Submit' button.

2. Horizontal Form

To create the vertical or basic form use following steps –

Step 1 : Add a role to form element as -

```
<form role="form"> and form class="form-horizontal"
```

Step 2 : Inside the <div class="form-group"> wrap all the labels and controls

Step 3 : While adding any textual control such as <text>, <textarea> and <select> add the class "form-control"

Step 4 : Add a class of "control-label" to the labels.

Here is example code –

FormDemo2.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"></script>
  <script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"></script>
</head>
<body>
  <form class ="form-horizontal" role = "form">
    <div class = "form-group">
      <label class="control-label col-sm-4" for = "name">User Name</label>
```

```
<div class="control-label col-sm-8">
  <input type = "text" class = "form-control" id = "name">
</div>
</div>
<div class = "form-group">
<label class="control-label col-sm-4" for = "pwd">Password</label>
<div class="control-label col-sm-8">
  <input type = "password" class = "form-control" id = "pwd">
</div>
</div>
<div class="form-group">
<div class="col-sm-offset-4 col-sm-8">
<div class = "checkbox">
<label><input type = "checkbox"> Show Me Always</label>
</div>
</div>
</div>
<div class="form-group">
<div class="col-sm-offset-4 col-sm-8">
<button type = "submit" class = "btn btn-default">Submit</button>
</div>
</div>
</form>
</body>
</html>
```

3. Inline Form

- In an inline form, all of the elements are inline, left-aligned, and the labels are alongside.
- To create the inline form use following steps –

Step 1 : Add a role to form element as -

```
<form role="form">
```

And the class as “form-inline” to the <form> element.

Step 2 : Inside the <div class=“from-group”> wrap all the controls

Step 3 : While adding any textual control such as <text>, <textarea> and <select> add the class “**form-control**”

Here is example code –

FormDemo3.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<title>Bootstrap Example</title>
```

```
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">

<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.2.1/jquery.min.js"> </script>
<script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"> </script>

</head>
<body>
<form class="form-inline" role = "form">
  <div class = "form-group">
    <label for = "name">User Name</label>
    <input type = "text" class = "form-control" id = "name">
  </div>

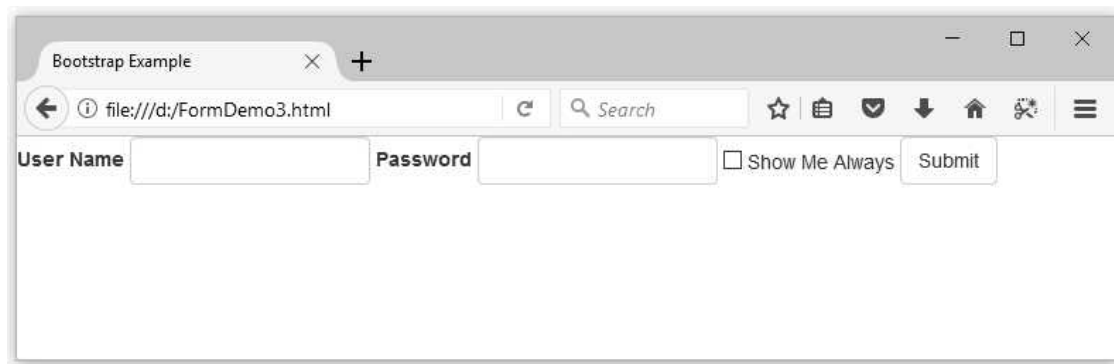
  <div class = "form-group">
    <label for = "pwd">Password</label>
    <input type = "password" class = "form-control" id = "pwd">
  </div>

  <div class = "checkbox">
    <label><input type = "checkbox"> Show Me Always</label>
  </div>

  <button type = "submit" class = "btn btn-default">Submit</button>

</form>
</body>
</html>
```

Output



Review Questions

1. Write short note on Bootstrap.

SPPU : May-18, Marks 4

2. Write short note on Bootstrap ? Write a program to demonstrate button styles (warning, danger, default, primary, link, success) ?

SPPU : Dec.-19, Marks 9

1.28 Multiple Choice Questions

Q.1 A collection of hyperlinked documents on the internet forms the ____.

- | | |
|---|--|
| ☐ a World Wide Web (WWW) | ☐ b e-mail system |
| ☐ c mailing list | ☐ d hypertext markup language |

Q.2 Which of the following is a popularly used internet protocol ?

- | | |
|---------------------------------|--|
| ☐ a HTML | ☐ b TCP/IP |
| ☐ c IPX | ☐ d None of these |

Q.3 FTP stands for ____.

- | | |
|---|---|
| ☐ a File Transmission Protocol | ☐ b File Transfer Program |
| ☐ c File Transfer Protection | ☐ d File Transfer Protocol |

Q.4 Which layer of the TCP / IP stack corresponds to the OSI model transport layer ?

- | | |
|---|---|
| ☐ a Host to host | ☐ b Application |
| ☐ c Internet | ☐ d Network access |

Q.5 Which of the following is not a web browser ?

- | | |
|----------------------------------|---|
| ☐ a Opera | ☐ b Microsoft edge |
| ☐ c www | ☐ d Google chrome |

Q.6 What services are not a provided by Webservers ?

- | | |
|--|---|
| ☐ a Serving web pages | ☐ b Running gateway programs |
| ☐ c Resource sharing | ☐ d Server side scripting |

Q.7 HTML is stand for ____.

- | | |
|--|---|
| ☐ a Higher Text Modeling Language | ☐ b Huge Text Modeling Library |
| ☐ c Hyper Text Modeling Library | ☐ d Hyper Text Markup Language |

Q.8 HTML is a subset of ____.

- | | |
|---------------------------------|--|
| ☐ a HTTP | ☐ b SGML |
| ☐ c XML | ☐ d None of these |

Q.9 All the HTML tags are enclosed within ____.

☐ a < ? >

☐ b < >

☐ c ##

☐ d { }

**Q.10 What does
 tag do ?**

☐ a Long break

☐ b Paragraph break

☐ c Line break

☐ d None of these

Q.11 Which tag tells the browser where the page starts and stops ?

☐ a <html>

☐ b <body>

☐ c <head>

☐ d <title>

Q.12 The main container for <TR>, <TD> and <TH> is ____.

☐ a <list>

☐ b <frame>

☐ c <table>

☐ d none of the above

Q.13 Which of the following is not a newly added element in HTML5 ?

☐ a Article

☐ b Audio

☐ c Nav

☐ d Frameset

Q.14 In HTML5, which element may be used within content to represent material that is tangential ____.

☐ a aside

☐ b cite

☐ c article

☐ d class

Q.15 CSS stands for ____.

☐ a Color and Style Specification

☐ b Cascading Style Sheets

☐ c Concurrent Style Specification

☐ d None of these

Q.16 The property in CSS used to change the background color of an element is ____.

☐ a bgcolor

☐ b color

☐ c background-color

☐ d all of the above

Q.17 The property in CSS used to change the color of a text is ____.

☐ a bgcolor

☐ b color

☐ c background-color

☐ d all of the above

Q.18 Which of the following attributes is used to specify elements to bind style rules to ____.

☐ a id

☐ b class

☐ c tag

☐ d all of the above

Answer Keys for Multiple Choice Questions :

Q.1	a	Q.2	b	Q.3	d	Q.4	a	Q.5	c
Q.6	c	Q.7	d	Q.8	b	Q.9	b	Q.10	c
Q.11	a	Q.12	c	Q.13	d	Q.14	a	Q.15	b
Q.16	c	Q.17	b	Q.18	d				



Notes

[illegible]